



Degree Project in Heat and Power Technology
Second cycle, 30 credits

Numerical Investigation of Steady-State Thermo-Fluid Performance of a Reducer for a High-Temperature *Dynamic* Pressure Sensor Calibration Rig

Minimizing Heat Loss and Pressure Loss in a Combustion-
Dynamics Calibration Rig

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Abstract

High-temperature laboratory test rigs are essential for evaluating components and measurement systems intended for gas-turbine-related applications. In Siemens Energy's Pulsatorn dynamic pressure calibration rig, the reducer connecting the air heater to the downstream sensor section was identified as a likely source of both aerodynamic losses and unwanted heat dissipation. The legacy reducer is a commercially available hydraulic fitting that produces two abrupt contractions (35.05 mm $\rightarrow$ 11.25 mm $\rightarrow$ 6.0 mm) and was selected primarily for mechanical compatibility rather than thermo-fluid performance. A redesigned reducer was therefore developed using a quintic C^2 -continuous contraction profile over a solid length of 150 mm, providing a smooth transition from the 35.05 mm heater bore to the 6.0 mm outlet.

The project was initially intended to include experimental high-temperature testing on the Pulsatorn rig. However, heater hardware failure prevented sustained testing within the available timeframe. With approval from both the industrial supervisor at Siemens Energy AB and the academic supervisor at KTH Royal Institute of Technology, the scope was revised to a comparative steady-state numerical investigation under controlled and repeatable boundary conditions.

The work comprised two studies. Study 1 established mesh independence for both geometries using compressible adiabatic CFD across three refinement levels. The selected baseline meshes contained 829,621 fluid cells for the legacy reducer and 913,301 for the redesigned reducer. At the most demanding pressure condition, the legacy reducer consistently exhibited a localised supersonic vena-contracta pocket ($Ma = 1.010$ – 1.016), whereas the redesigned reducer operated near-choked ($Ma = 0.989$ – 0.990), confirming that the flow-regime difference was not a numerical artefact.

Study 2 comprised six steady-state conjugate heat transfer simulations for both geometries at outlet gauge pressures of 80,000, 40,000, and 0 Pa, with an inlet condition of 1 bar(g) and 673 K. Under uninsulated conditions, the legacy reducer delivered slightly higher outlet static temperatures, by 7.80 K, 4.75 K, and 3.36 K across the three cases, because its external surface area was 3.35 times smaller and therefore lost less heat to the surroundings. However, the redesigned reducer consistently showed 9.5–14.3% lower external heat flux per unit area, indicating reduced local thermal loss intensity associated with the smooth contraction. At the supercritical pressure-ratio condition, the redesigned reducer reached $Ma = 0.990$ while the legacy reducer reached $Ma = 1.006$, confirming a clear flow-regime asymmetry between the two designs. Additional insulated reference simulations at the 0 Pa case reduced the outlet-temperature difference from 3.36 K to 1.24 K and reversed it in favour of the redesigned reducer, showing that the uninsulated thermal deficit of the redesigned geometry arose predominantly from its larger external surface area rather than inferior internal flow behaviour.

The results show that the redesigned reducer provides the superior internal flow field and lower heat-loss intensity per unit external area, but that its longer body introduces a thermal penalty under uninsulated operation. External insulation is therefore identified as the key practical measure required for the redesigned geometry to realise its full thermo-fluid advantage in the Pulsatorn rig.

Keywords: CFD, conjugate heat transfer, SST k - ω , compressible flow, reducer geometry, Pulsatorn, choking flow, quintic contraction, dynamic pressure calibration

Sammanfattning

Högtemperaturriggar i laboratoriemiljö är viktiga för att utvärdera komponenter och mätsystem avsedda för gasturbinrelaterade tillämpningar. I Siemens Energys Pulsatorn-rigg för kalibrering av dynamiska trycksensorer identifierades reduktionsstycket mellan luftvärmaren och den nedströms belägna sensorsektionen som en sannolik källa till både aerodynamiska förluster och oönskade värmeförluster. Den ursprungliga reducern är en kommersiellt tillgänglig hydraulisk koppling som ger upphov till två abrupta kontraktioner ($35,05 \text{ mm} \rightarrow 11,25 \text{ mm} \rightarrow 6,0 \text{ mm}$) och valdes i första hand utifrån mekanisk kompatibilitet snarare än termofluidisk prestanda. En omkonstruerad reducer utvecklades därför med en kvintisk C^2 -kontinuerlig kontraktionsprofil över en solid längd på 150 mm, vilket ger en mjuk övergång från värmarens diameter på 35,05 mm till utloppets diameter på 6,0 mm.

Projektet var initialt planerat att omfatta experimentella högtemperaturtester i Pulsatorn-riggen. Haveri i värmarutrustningen förhindrade dock långvarig provning inom den tillgängliga tidsramen. Med godkännande från både industriell handledare vid Siemens Energy AB och akademisk handledare vid Kungliga Tekniska högskolan reviderades därför projektets omfattning till en jämförande stationär numerisk studie under kontrollerade och reproducerbara randvillkor.

Arbetet omfattade två delstudier. Delstudie 1 fastställde nätoberoende för båda geometrierna med hjälp av kompressibel adiabatisk CFD vid tre nätförfiningsnivåer. De valda basnäten innehöll 829 621 fluidceller för den ursprungliga reducern och 913 301 för den omkonstruerade reducern. Vid det mest krävande tryckfallet uppvisade den ursprungliga reducern konsekvent en lokaliserad supersonisk *vena-contracta*-zon ($Ma = 1,010\text{--}1,016$), medan den omkonstruerade reducern arbetade nära strypt flöde ($Ma = 0,989\text{--}0,990$), vilket bekräftar att skillnaden i strömningsregim inte är en numerisk artefakt.

Delstudie 2 omfattade sex stationära simuleringar av konjugerad värmeöverföring för båda geometrierna vid utloppstrycken 80 000, 40 000 och 0 Pa övertryck, med ett inloppsvillkor på 1 bar(g) och 673 K. Under oisolerade förhållanden gav den ursprungliga reducern något högre massmedelvärdesbildad statisk utloppstemperatur, med 7,80 K, 4,75 K respektive 3,36 K över den omkonstruerade reducern i de tre fallen, eftersom dess yttre area var 3,35 gånger mindre och därmed avled mindre värme till omgivningen. Den omkonstruerade reducern uppvisade dock genomgående 9,5–14,3 % lägre externt värmeflöde per areaenhet, vilket visar att den mjuka kontraktionen reducerade de lokala termiska förluster som är kopplade till den abrupta äldre geometrin. Vid fallet med överkritisk tryckkvot nådde den omkonstruerade reducern $Ma = 0,990$ medan den ursprungliga reducern nådde $Ma = 1,006$, vilket bekräftar en tydlig asymmetri i strömningsregim mellan de två konstruktionerna. Ytterligare isolerade referenssimuleringar vid 0 Pa-fallet minskade temperaturskillnaden vid utloppet från 3,36 K till 1,24 K och vände fördelen till den omkonstruerade reducerns favör, vilket visar att dess termiska nackdel i oisolerat tillstånd huvudsakligen orsakas av den större yttre arean och inte av sämre invändigt strömningsbeteende.

Resultaten visar att den omkonstruerade reducern ger ett bättre invändigt strömningsfält och lägre värmeförlustintensitet per areaenhet, men att dess längre

kropp medför en termisk nackdel under oisolerad drift. Extern isolering identifieras därför som den viktigaste praktiska åtgärden för att den omkonstruerade geometrin fullt ut ska kunna realisera sina termofluidiska fördelar i Pulsatorn-riggen.

Nyckelord: CFD, konjugerad värmeöverföring, SST $k-\omega$, kompressibelt flöde, reducergeometri, Pulsatorn, strypt flöde, kvintisk kontraktion, kalibrering av dynamiska trycksensorer

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Table of Contents

Abstract	3
Sammanfattning	4
Acknowledgements	6
List of Symbols	13
List of Abbreviations	16
List of Tables	17
List of Figures	19
1. Introduction	21
1.1 The Pulsatorn Test Rig at Siemens Energy	21
1.2 Background and Motivation.....	23
1.3 Problem Statement	23
1.4 Project Scope and Delimitations	24
1.4.1 Delimitations	24
1.5 Research Aim and Research Questions	24
1.6 Thesis Outline	25
2. Literature Review	26
2.1 High-Temperature Dynamic Pressure Sensing and Calibration	26
2.1.1 The Instrumentation Gap in High-Temperature Gas Turbine Operation ..	26
2.1.2 The Requirement for Temperature-Traceable Calibration.....	26
2.1.3 European Research Programmes on High-Temperature Instrumentation	27
2.1.4 The Pulsatorn Rig: Prior Characterisation Work.....	27
2.2 Heat-Transfer Mechanisms in Heated Gas Delivery Systems	27
2.2.1 The Coupled Heat-Transfer Chain	27
2.2.2 Internal Forced Convection and Separation-Zone Heat Transfer.....	27
2.2.3 External Convection and the Thermal Resistance Framework.....	28
2.3 Flow Conditioning and Pressure Loss in Reducer Geometries.....	28
2.3.1 The Vena Contracta and Borda–Carnot Mechanism.....	28
2.3.2 Smooth Contraction Design and C^2 Continuity	29
2.3.3 Compressible Flow and Choking in Contracting Ducts	30
2.4 CFD and Conjugate Heat-Transfer Modelling.....	30
2.4.1 The Role of CFD in Comparative Design Assessment.....	30
2.4.2 The SST $k-\omega$ Turbulence Model.....	31
2.5 Mesh Resolution and Near-Wall Treatment.....	31

2.6 Identified Research Gap and Thesis Contribution	31
3. Methodology and Numerical Model Setup.....	33
3.1 Overview of the Numerical Investigation.....	33
3.2 Geometry Description and Computational Domain.....	33
3.2.1 Coordinate System and Symmetry Assumptions	33
3.2.2 Legacy Reducer Geometry	34
3.2.3 Redesigned Reducer Geometry.....	35
3.2.4 Geometric Simplifications.....	37
3.3 First-Order Engineering Pre-Analysis	37
3.3.1 Continuity and Velocity Scaling	37
3.3.2 Pressure Loss and Critical Pressure Ratio.....	38
3.3.3 First-Order Thermal Energy Balance	38
3.4 Simulation Study Design and Case Definition.....	38
3.4.1 Study 1: Mesh Sensitivity Assessment.....	38
3.4.2 Study 2: Conjugate Heat Transfer Campaign.....	39
3.5 Governing Equations and Physical Models	40
3.5.1 Conservation Equations for Compressible Turbulent Flow	40
3.5.2 Equation of State and Thermophysical Properties of Air.....	40
3.5.3 Solid Domain Energy Equation and Conjugate Coupling	40
3.6 Turbulence Modelling	41
3.6.1 Selection Rationale	41
3.6.2 The SST $k-\omega$ Transport Equations	41
3.6.3 Turbulence Inlet Boundary Specification	41
3.7 Boundary Conditions	41
3.7.1 Inlet Boundary Conditions	41
3.7.2 Outlet Boundary Conditions	42
3.7.3 Fluid Wall Boundary Conditions	42
3.7.4 External Solid Surface Boundary Condition	42
3.7.5 Symmetry Conditions	42
3.8 Mesh Strategy and Near-Wall Treatment	42
3.8.1 Mesh Topology and Generation Approach.....	42
3.8.2 Inflation Layer Parameters	43
3.8.3 Solid Domain Meshing	43
3.8.4 Resulting Mesh Statistics: CHT Simulations	43
3.8.5 Near-Wall y^+ Performance	45
3.9 Mesh Sensitivity Assessment	45
3.9.1 Rationale and Validation Strategy	45

3.9.2 Mesh Refinement Strategy.....	46
3.9.3 Acceptance Criterion.....	46
3.9.4 Sensitivity Results Summary.....	46
3.10 Solver Settings and Numerical Discretisation	47
3.10.1 Solver Type and Formulation	47
3.10.2 Spatial Discretisation Schemes.....	47
3.10.3 Initialisation Strategy	47
3.10.4 Convergence Monitoring	47
3.10.5 Convergence Criteria for CHT Simulations	48
3.11 Post-Processing and Output Quantities.....	48
3.11.1 Primary Performance Indicators.....	49
3.11.2 Quarter-Symmetry Correction	49
3.11.3 Derived Quantities.....	49
3.12 Methodological Limitations	49
4. Results	51
4.1 Overview.....	51
4.2 Mesh Sensitivity Assessment	51
4.2.1 Simulation Configuration and Acceptance Criterion	51
4.2.2 Legacy Reducer: Mesh Sensitivity Results	51
4.2.3 Redesigned Reducer: Mesh Sensitivity Results.....	53
4.2.4 Preliminary Flow Regime Observation from Study 1	54
4.3 Conjugate Heat Transfer Simulation Overview.....	55
4.3.1 Simulation Configuration and Convergence	55
4.3.2 Consolidated CHT Results.....	56
4.4 Downstream Thermal Delivery.....	57
4.4.1 Mass-Weighted Outlet Temperature	57
4.5 External Heat Loss and Normalised Heat Flux	58
4.5.1 Absolute External Heat Loss	58
4.5.2 Near-Invariance of External Heat Loss with Pressure Condition.....	59
4.5.3 Normalised External Heat Flux Density.....	60
4.6 Peak Mach Number and Flow Regime Classification	61
4.6.1 Subsonic Cases A and B	61
4.6.2 Redesigned Reducer: Near-Choked Flow at Case C.....	61
4.6.3 Legacy Reducer: Supersonic Vena Contracta Pocket at Case C.....	62
4.7 The Choking Asymmetry: Flow Regime Analysis at Case C	63
4.7.1 Statement of the Asymmetry	63
4.7.2 Physical Mechanism	63

4.7.3 Operational Significance	65
4.8 Insulated Reference Simulations	65
4.8.1 Purpose and Configuration.....	66
4.8.2 Results	66
4.8.3 Interpretation	66
4.9 Summary of Principal Findings	67
5. Discussion	70
5.1 Overview.....	70
5.2 Answers to the Research Questions.....	70
5.2.1 RQ1: Bulk Thermal Delivery.....	70
5.2.2 RQ2: Normalised External Heat Flux	71
5.2.3 RQ3: Internal Flow Regime and Choked Flow	71
5.3 Physical Mechanisms and Contextualisation	72
5.3.1 The External Resistance Paradox.....	72
5.3.2 The Choking Asymmetry in Context.....	73
5.3.3 Total Pressure Loss: An Apparent Contradiction	74
5.4 Implications for the Pulsatorn Rig	74
5.4.1 Operational Recommendation.....	74
5.4.2 Target Operating Temperature.....	75
5.4.3 Flow Quality for Calibration	75
5.5 Limitations and Their Effect on Conclusions	76
5.5.1 Uniform External Convective Coefficient.....	76
5.5.2 Neglect of Thermal Radiation.....	76
5.5.3 Absence of Experimental Validation.....	76
5.5.4 Steady-State Assumption.....	77
5.6 Sustainability and Societal Implications	77
5.6.1 Energy Efficiency in Gas Turbine Development Infrastructure	77
5.6.2 Broader Application of Results	77
5.6.3 Limitations of Purely Numerical Work	78
6. Conclusions and Future Work	79
6.1 Summary of the Investigation	79
6.2 Principal Conclusions	79
6.3 Limitations of the Present Study	82
6.4 Recommendations for Future Work.....	83
6.4.1 Experimental Validation: Highest Priority	83
6.4.2 Inclusion of Thermal Radiation.....	83
6.4.3 CHT-Specific Mesh Independence Study	83

6.4.4 Unsteady Simulation of Near-Choked Flow Stability	83
6.4.5 Parametric Optimisation of the Quintic Profile.....	84
6.4.6 Extended System-Level CHT Model	84
6.4.7 Quantification of External Insulation Benefit.....	84
6.5 Final Remarks.....	84
References.....	86
Appendix A: CAD Geometry and Reducer Design Details	88
A.1 Purpose and Scope.....	88
A.2 Geometry Overview.....	88
A.2.1 Legacy Reducer Configuration	88
A.2.2 Redesigned Reducer Configuration.....	90
A.3 Geometric Simplifications.....	91
Appendix B: Mesh Sensitivity Study	92
B.1 Purpose and Scope.....	92
B.2 Legacy Reducer: Mesh Settings and Quality	92
B.3 Redesigned Reducer: Mesh Settings and Quality	92
B.4 Confirmed Mesh Sensitivity Results	93
Appendix C: Full Simulation Configuration	95
C.1 Boundary Conditions.....	95
C.2 Solver Configuration.....	95
C.3 Convergence Criteria	96
C.4 Air Thermophysical Properties	97
Appendix D: Risk Analysis	98
D.1 Purpose and Scope.....	98
D.2 Risk Management Framework	98
D.3 Identified Hazard Categories.....	98
D.4 Key Risks and Mitigation Measures	98
D.5 Relation to the Present Thesis	99
Appendix E: Project Scope Revision	100
E.1 Initial Project Scope.....	100
E.2 Hardware Failure and Scope Revision	100
E.3 Justification for Scope Revision.....	100
E.4 Implications for the Thesis.....	100
Appendix F: Software Tools and Computing Environment.....	101
F.1 CAD and Geometry Preparation.....	101
F.2 Numerical Simulation Software.....	101
F.3 Post-Processing and Documentation	101

Appendix G: Project Planning and Milestones	102
G.1 Initial Project Plan	102
G.2 Project Milestones.....	102
G.3 Supervision Structure.....	102
G.4 Adaptation and Project Management.....	103

List of Symbols

Roman Symbols

Symbol	Definition	Units
A	Cross-sectional flow area	m ²
A_ext	External convective surface area of solid body	mm ² or m ²
a	Local speed of sound: $a = \sqrt{\gamma R_s T}$	m/s
a ₁	SST k – ω model constant = 0.31	-
Bi	Biot Number: $Bi = h \cdot L_c / \lambda_s$	-
C_c	Vena contracta contraction coefficient ≈ 0.62	
C_p	Specific heat capacity of air at constant pressure	J/kg·K
C_ps	Specific heat capacity of solid (structural steel)	J/kg·K
F_1, F_2	SST k – ω blending functions	-
H	Specific total enthalpy: $H = h + \frac{1}{2} \mathbf{u} ^2$	J/kg
h_ext	External convective heat transfer coefficient = 10 W/m ² ·K	W/m ² ·K
K	Borda–Carnot loss coefficient: $K = (1/C_c - 1)^2$	-
k	Turbulence kinetic energy	m ² /s ²
L	Quintic contraction body length	m
L_c	Characteristic length for Biot number: $L_c = V/A_{ext}$	m
Ma	Mach number: $Ma = u/a$	-
Nu	Nusselt number	-
n	Outward normal direction at fluid–solid interface	-
P	Static pressure	Pa
P ₀	Total (stagnation) pressure	Pa
PR	Absolute pressure ratio: $P_{0,in} / P_{outlet}$	-
PR_crit	Critical pressure ratio for air = 1.893 ($\gamma = 1.4$)	-
Pr	Prandtl number	-
Pr_t	Turbulent Prandtl number ≈ 0.85	-
$\tilde{P}_k$	Production rate of turbulence kinetic energy (with limiter)	W/m ³
Q_ext	Total external heat loss, full geometry (quarter $\times$ 4)	W
q_ext	External heat flux density: Q_{ext} / A_{ext}	W/m ²
R	Internal bore radius profile (redesigned reducer)	m

R_{ext}	External thermal resistance: $1/(h \cdot A_{ext})$	K/W
Ra	Rayleigh number	-
Re	Reynolds number	-
R_s	Specific gas constant for dry air = 287 J/kg·K	J/kg·K
S	Magnitude of mean strain-rate tensor	s ⁻¹
T	Static temperature	K
T_{amb}	Ambient (laboratory) temperature = 300K	K
T_{in}	Mass-weighted inlet total temperature = 673K	K
T_{out}	Mass-weighted outlet static temperature	K
T_s	Solid body temperature	K
$T_{wall,ext}$	Area-weighted external wall temperature	K
u	Mean axial velocity	m/s
u_{vc}	Velocity at vena contracta	m/s
u_d	Mean downstream velocity after reattachment	m/s
u_τ	Friction velocity: $u_\tau = \sqrt{\tau_w/\rho}$	m/s
V	Solid body volume, quarter model	m ³
y_1	Wall-normal distance to first mesh cell centre	m
y^+	Non-dimensional wall distance: $\rho u_\tau y_1/\mu$	-
z	Axial coordinate (primary flow direction)	m
z_0	Axial position of quintic contraction inlet junction	m
$\dot{m}$	Mass flow rate	kg/s

Greek Symbols

Symbol	Definition	Units
Symbol	Definition	Units
α	SST $k-\omega$ model coefficient	—
β, β^*	SST $k-\omega$ dissipation coefficients ($\beta^* = 0.09$)	—
γ	Ratio of specific heats for air = 1.4	—
δ_{ij}	Kronecker delta	—
ΔP_0	Total pressure drop: $P_{0,inlet} - P_{0,outlet}$	Pa
ΔT	Temperature drop from inlet: $T_{in} - T_{out}$	K

ε	Surface emissivity of oxidised stainless steel ≈ 0.8	—
η	Thermal efficiency: $(T_{\text{out}} - T_{\text{amb}})/(T_{\text{in}} - T_{\text{amb}}) \times 100\%$	%
λ	Thermal conductivity of air	W/m·K
λ_s	Thermal conductivity of solid (structural steel) = 16.27 W/m·K	W/m·K
μ	Dynamic viscosity of air	Pa·s
μ_t	Turbulent eddy viscosity	Pa·s
ξ	Normalised axial coordinate: $(z - z_0)/L$	—
ρ	Fluid density	kg/m ³
ρ_s	Solid density (structural steel) = 8,030 kg/m ³	kg/m ³
σ	Stefan–Boltzmann constant = 5.67×10^{-8} W/m ² ·K ⁴	W/m ² ·K ⁴
σ_k, σ_ω	SST k – ω diffusion coefficients (blended)	—
τ_w	Wall shear stress	Pa
φ	Quintic shape function: $\varphi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3$	—
ω	Specific dissipation rate	s ⁻¹

Subscripts

Subscript	Meaning
amb	Ambient ($T = 300$ K)
ext	External (outer solid surface)
f	Fluid domain
in	Inlet boundary
max	Maximum over domain
out	Outlet boundary
s	Solid domain
t	Turbulent
vc	Vena contracta
wall	Wall surface
∞	Free-stream / ambient condition

List of Abbreviations

Abbreviation	Definition
ANSYS	Analysis System (commercial simulation software suite)
ASME	American Society of Mechanical Engineers
AWA	Area-Weighted Average
BSP	British Standard Pipe (thread standard)
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CHT	Conjugate Heat Transfer
DO	Discrete Ordinates (radiation model)
FP6	EU Framework Programme 6
FP7	EU Framework Programme 7
GaPO ₄	Gallium orthophosphate (piezoelectric material)
HEATTOP	High-Temperature Engine Aero-Thermal Measurements for Optimised Performance (EU FP6 project)
KPI	Key Performance Indicator
KTH	KTH Royal Institute of Technology
LES	Large Eddy Simulation
MEMS	Micro-Electromechanical Systems
MWA	Mass-Weighted Average
NIST	National Institute of Standards and Technology
NPT	National Pipe Thread (thread standard)
PID	Proportional–Integral–Derivative (control system)
PR	Pressure Ratio
RANS	Reynolds-Averaged Navier–Stokes
SiC	Silicon Carbide
SST	Shear Stress Transport (turbulence model)
STARGATE	Sensors Towards Advanced Monitoring and Control of Gas Turbine Engines (EU FP7 project)
URANS	Unsteady Reynolds-Averaged Navier–Stokes
V&V	Verification and Validation

List of Tables

Number	Caption
3.1	Confirmed geometric parameters for both reducer configurations.
3.2	Study 2 CHT simulation case matrix.
3.3	CHT baseline mesh statistics for both reducer geometries (Study 2).
3.4	Study 1 (adiabatic CFD) mesh levels and average quality metrics.
4.1	Mesh quality metrics for the legacy reducer at three Study 1 refinement levels.
4.2	Mesh sensitivity KPIs for the legacy reducer at Case C (PR = 1.987).
4.3	Mesh quality metrics for the redesigned reducer at three Study 1 refinement levels.
4.4	Mesh sensitivity KPIs for the redesigned reducer at Case C (PR = 1.987).
4.5	Maximum Mach number across all Study 1 mesh refinement levels for both geometries.
4.6	Consolidated CHT results for all six pressure cases and both reducer geometries.
4.7	Comparison of thermal KPIs under CHT and insulated ($q_{\text{ext}} = 0$) conditions at Case C.
4.8	Summary of principal findings across all CHT and insulated simulation results.
6.1	Comparative performance summary for both reducer geometries across all principal metrics.
A.1	Confirmed geometric parameters of the legacy reducer.
A.2	Confirmed geometric parameters of the redesigned reducer.
B.1	Legacy reducer mesh generation settings and quality metrics for three Study 1 refinement levels.
B.2	Redesigned reducer mesh generation settings and quality metrics for three Study 1 refinement levels.
B.3	Study 1 mesh sensitivity KPIs at Case C for both geometries across all refinement levels.
B.4	Percentage changes in KPIs between consecutive mesh refinement levels.
C.1	Boundary conditions for Study 1 (adiabatic CFD) and Study 2 (CHT) simulations.
C.2	Fluent solver settings applied to all Study 1 and Study 2 simulations.

C.3	Convergence criteria for Study 2 (CHT) simulations.
C.4	Air piecewise-linear thermophysical properties (C_p and thermal conductivity k).
D.1	Risk scoring criteria (severity, probability, risk level).
D.2	Key risks, risk scores, and mitigation measures for the experimental campaign.
G.1	Key project milestones showing planned versus actual completion dates.

List of Figures

Number	Caption
1.1	Pulsatorn high-temperature dynamic pressure calibration rig at Siemens Energy AB, Finspång
2.1	Borda–Carnot mechanism in abrupt contractions
2.2	Quintic profile shape function
3.1	Quarter-symmetry 3D CAD geometry of the legacy reducer
3.2	Dimensioned cross-section of the legacy reducer internal geometry
3.3	Legacy reducer computational domain with inlet and outlet extensions
3.4	Redesigned reducer computational domain with inlet and outlet extensions
3.5	Quarter-symmetry 3D CAD geometry of the redesigned reducer
3.6	Dimensioned cross-section of the redesigned reducer internal geometry
3.7	Detailed Dimensions of the redesigned reducer geometry
3.8	ANSYS Workbench project schematic showing Study 1 and Study 2 simulation systems with cascade initialisation links.
3.9	Redesigned Reducer CHT baseline mesh throat section
3.10	Redesigned reducer CHT baseline mesh cross-section at the throat (XZ plane)
3.11	Legacy reducer CHT baseline mesh cross-section at the first abrupt contraction step
3.12	Redesigned reducer CHT baseline mesh, symmetry-plane cross-section
3.13	Legacy reducer CHT baseline mesh, symmetry-plane cross-section
3.14	Residual convergence history for a representative CHT case (redesigned reducer, Case A, 80 kPa gauge outlet) — continuity, momentum, energy, k , ω
3.15	Physical convergence monitor
4.1	Legacy Reducer y^+
4.2	Redesigned Reducer y^+
4.3	Thermal efficiency for both reducer geometries across Cases A–C
4.4	Redesigned Reducer Total Heat Flux
4.5	Legacy Reducer Total Heat Flux
4.6	Total external heat loss for both reducer geometries across Cases A–C,
4.7	Legacy Reducer Wall Temperature

4.8	Redesigned Reducer Wall Temperature
4.9	Redesigned Reducer Mach Number
4.10	Legacy Reducer Mach Number
4.11	Centerline Mach Number plot for the Legacy reducer
4.12	Centerline Mach Number plot for the redesigned reducer
4.13	Outlet-temperature gap between the two geometries at Case C
5.1	Redesigned Reducer Streamlines
5.2	Legacy Reducer Streamlines

1. Introduction

Gas turbines are central to modern energy systems, powering electricity generation, industrial processes, and aircraft propulsion worldwide. Their development depends on the ability to measure, with high fidelity, the physical quantities that govern their internal behaviour, chief among them, pressure. Gas turbines are inherently unsteady machines: the pressure oscillations generated by combustion, thermoacoustic instabilities, and rotor–stator interaction carry diagnostic information that static measurements cannot capture. Sensors capable of resolving these rapid, high-amplitude fluctuations, dynamic pressure sensors, are therefore essential tools in both turbine development and ongoing operational monitoring.

A dynamic pressure sensor that performs accurately at ambient conditions may behave very differently when exposed to the elevated temperatures and mechanical stresses of real turbine operation. Before such a sensor can be trusted in the field, it must be calibrated under controlled conditions that replicate the thermal and flow environment it will encounter in service. Delivering those controlled conditions, stable temperature, defined pressure, and repeatable pulsating flow is the fundamental challenge of high-temperature dynamic pressure sensor calibration.

This thesis is concerned with one specific and critical component in the thermal delivery chain of such a calibration rig: the reducer geometry connecting the upstream air heater to the downstream test section. The reducer may appear to be a minor detail, but it is in fact a governing design element. Pressure and heat losses through the reducer degrade the thermal and flow conditions delivered to the sensor under test, reducing both the achievable operating temperature and the repeatability of the calibration result. The present work quantifies this degradation for a legacy reducer geometry and evaluates a redesigned geometry intended to reduce it, using steady-state computational fluid dynamics (CFD) and conjugate heat transfer (CHT) simulations.

1.1 The Pulsatorn Test Rig at Siemens Energy

Siemens Energy AB in Finspång, Sweden, operates the Pulsatorn test rig to support the verification and calibration of dynamic pressure sensors used in gas turbine development and service. The rig delivers a controlled stream of hot, pressurised, pulsating air to a sensor under test, supporting systematic characterisation of sensor response across a defined range of temperatures, mean pressures, and dynamic pressure amplitudes.

The rig architecture consists of a compressed air supply drawn from the site facility network, conditioned to the required inlet pressure, then heated by an inline resistance heater. The heated air flows through the heater-to-downstream interface, the reducer that is the subject of the present thesis, before reaching the instrumented test section where the sensor under calibration is located. A gear-wheel pulsation generator downstream of the test section imposes controlled pressure oscillations on the flow, creating the dynamic pressure environment required for sensor characterisation (Bartkevičius, 2023).

The Pulsatorn rig has been the subject of two preceding master's theses conducted at Siemens Energy under the supervision of Jonas Deurell. Bartkevičius (2023)

designed and built the high-temperature variant of the rig, developing its thermal management system, including the inline resistance heater and PID-controlled heating and characterising the sinusoidal pressure wave generation capability of the gear-wheel mechanism down to 1 Hz. Järleby (2025) extended this work to include heat radiation effects and to evaluate the piezoelectric signal conditioning chain, while also conducting the first systematic thermocouple measurements of the thermal delivery chain during rig operation.

Järleby's measurements provided a critical quantified observation: with the inline heater stabilised at 250 °C, the temperature at the pulsator housing separated from the heater by the connecting hardware reached a maximum of only approximately 150 °C, representing a loss of approximately 100 K across the interface. Järleby identified the heater-to-pipe connector as the dominant thermal loss location, and explicitly noted that even with additional insulation, the target operating temperature of 500 °C could not be reached (Järleby, 2025). The present thesis directly addresses this finding by numerically characterising the thermo-fluid performance of the reducer at this interface and evaluating a redesigned geometry intended to reduce these losses.

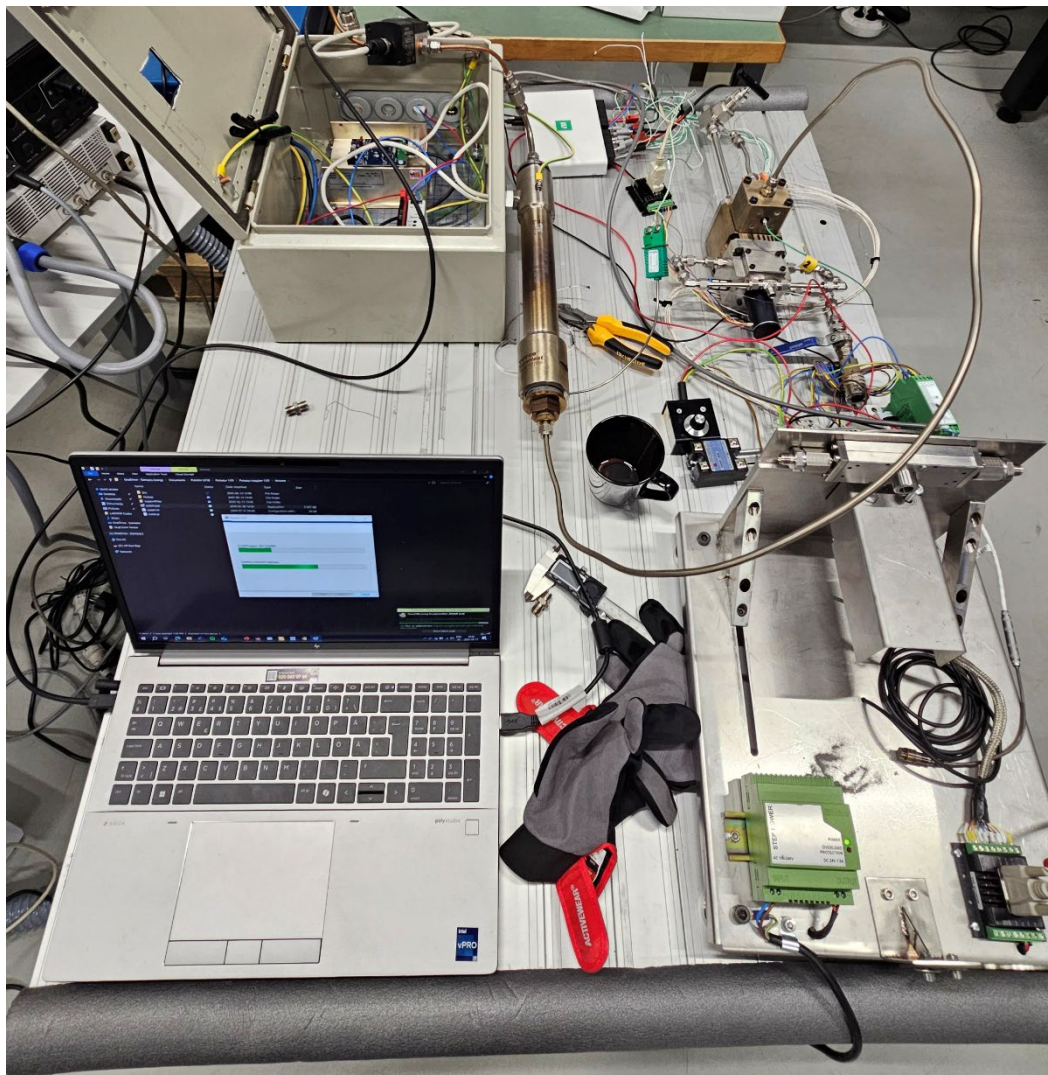


Figure 1.1. Pulsatorn high-temperature dynamic pressure calibration rig at Siemens Energy AB, Finspång, showing the inline heater assembly (centre), control electronics (left), and pulsator mechanism (right).

The societal relevance of reliable dynamic pressure sensor calibration extends beyond the immediate industrial application. Gas turbines operating in combined-cycle power generation supply a significant fraction of European baseload electricity and increasingly serve as the primary dispatchable backup for intermittent wind and solar generation in the transition to a low-carbon energy system. In both roles, the safety-critical monitoring of combustion dynamics made possible by well-calibrated dynamic pressure sensors is not merely an industrial efficiency concern but a matter of public safety and energy security. The Pulsatorn calibration rig and the heater-to-downstream interface that this thesis investigates form part of the technical infrastructure that supports this monitoring capability.

1.2 Background and Motivation

High-temperature laboratory test rigs play a critical role in the development and verification of components and measurement systems for gas-turbine-related applications. In such environments, it is essential to deliver hot compressed air to a downstream test section with well-defined thermal and flow conditions. Excessive pressure loss or unintended heat loss in upstream components can limit the achievable operating range and introduce uncertainty in downstream measurements.

In the Pulsatorn rig, the reducer connecting the heater outlet to the downstream section is particularly important for two coupled reasons. First, it is the component with the largest cross-sectional area change in the flow path, making it the dominant source of aerodynamic minor loss. Second, it is fabricated from metal and operates across a large temperature difference, hot gas at several hundred degrees Celsius on its inner surface and ambient air at approximately 20 °C on its outer surface, which drives significant conductive and convective heat loss through the reducer body. The result is a measurable drop in air temperature between the heater outlet and the downstream test section, directly compromising calibration conditions.

The legacy reducer used in the rig was designed primarily for mechanical compatibility between dissimilar components. Its internal geometry includes a strong, abrupt contraction and a relatively long metallic section, both of which promote flow separation and elevated surface heat loss. Practical experience during rig operation and the experimental evidence reported by Järleby (2025) confirmed that this component is a dominant contributor to thermal degradation in the delivery chain. A redesigned reducer geometry was subsequently developed to provide a smoother internal contraction, reduce aerodynamic losses, and improve thermal efficiency. Numerically quantifying the performance difference between these two geometries is the central engineering problem addressed by this thesis. As the results demonstrate, these objectives are achieved aerodynamically, but the thermal outcome under uninsulated conditions is counterintuitive and constitutes a principal finding

1.3 Problem Statement

The legacy reducer in the Pulsatorn rig was not designed with thermo-fluid performance as a primary criterion. Its geometry imposes a strong internal contraction, leading to flow separation, elevated peak velocities, and significant total pressure loss.

At the same time, although its compact 38 mm body limits external surface area, the absence of insulation allows heat loss through the metallic wall, reducing the temperature of the air delivered to the downstream test section relative to the heater outlet condition

To address these limitations, a new reducer geometry was developed to provide a smoother internal contraction, a more controlled flow path, and reduced heat loss. The design employs a quintic C^2 -continuous polynomial profile that ensures zero slope and zero curvature at the inlet and outlet junctions, eliminating the geometric discontinuities that cause flow separation in the legacy geometry. The specific engineering problem this thesis addresses is numerically quantifying the performance improvement achievable by this geometric change in terms of pressure loss, peak Mach number, downstream outlet temperature, and external heat flux

1.4 Project Scope and Delimitations

The project was initially planned to include an experimental campaign involving high-temperature operation of the Pulsatorn rig, combined with thermocouple and pressure measurements downstream of the heater. During the course of the work, the heater assembly sustained damage, preventing sustained high-temperature experimental operation within the available project timeframe. Following discussions with both the industrial supervisor at Siemens Energy and the academic supervisor at KTH, the project scope was formally revised with the approval of both supervisors to a controlled numerical investigation. This revision allowed the original group of engineering questions to be addressed under repeatable boundary conditions while assuring timely completion. A detailed account of the original experimental scope and the circumstances of the scope revision is provided in Appendix E

The final thesis scope is defined as follows: (i) a compressible adiabatic CFD mesh sensitivity assessment for both reducer geometries across three mesh refinement levels, to establish spatial discretisation adequacy; and (ii) six steady-state conjugate heat transfer simulations for both geometries at three outlet pressure conditions to compare thermo-fluid performance under identical boundary conditions. The primary output is a quantified comparative assessment of the two geometries across subsonic and supercritical operating regimes.

1.4.1 Delimitations

The following are explicit boundaries of the present work. Steady-state simulations only include transient heat-up behaviour, start-up dynamics, and the pulsating flow introduced by the gear-wheel mechanism are not modelled. Thermal radiation is neglected; external heat loss is represented using a uniform convection coefficient ($h = 10 \text{ W/m}^2\cdot\text{K}$) applied to all outer solid surfaces. No experimental validation is performed, the redesigned reducer has not been manufactured, and prior temperature measurements by Järleby (2025) were obtained under different operating conditions and serve only for qualitative contextualisation. Only the legacy and one redesigned geometry are evaluated; parametric optimisation is outside the scope.

1.5 Research Aim and Research Questions

The aim of this thesis is to evaluate and quantify the improvement in steady-state thermo-fluid performance achievable by replacing the legacy reducer in the Pulsatorn

test rig with a redesigned quintic contraction geometry, using controlled CFD and CHT simulations under identical boundary conditions as the primary analytical method.

The specific research questions addressed are:

RQ1: Does the redesigned reducer improve bulk thermal delivery to the downstream outlet section, and by how much, across the range of outlet pressure conditions investigated?

RQ2: Does the redesign reduce the normalised external heat loss per unit surface area, and is this reduction consistent across all tested operating conditions?

RQ3: Does the redesign alter the internal compressible flow regime specifically the ability to achieve choked flow at supercritical pressure ratios and what are the implications for the calibration environment?

1.6 Thesis Outline

Chapter 2 reviews relevant literature on high-temperature dynamic pressure sensing and calibration, heat-transfer mechanisms in heated gas delivery systems, flow losses in reducer and contraction geometries, and numerical methods for conjugate heat transfer modelling. Chapter 3 describes the two reducer geometries, the numerical methodology, boundary conditions, mesh strategy, and solver settings. Chapter 4 presents the mesh sensitivity results and the six-case CHT comparative dataset, including the choking asymmetry finding. Chapter 5 discusses the physical mechanisms behind the results, addresses modelling limitations, answers the three research questions, and considers sustainability and societal implications. Chapter 6 states the conclusions and outlines recommendations for future work.

Supplementary documentation covering CAD geometry, mesh sensitivity details, solver configuration, risk analysis, project scope evolution, software tools, and project planning is provided in Appendices A-G.

2. Literature Review

This chapter reviews the scientific and engineering literature relevant to the present work across five thematic areas. Section 2.1 establishes the operational context of high-temperature dynamic pressure sensing and calibration in gas turbine applications, situating the Pulsatorn rig within the broader landscape of European research on high-temperature instrumentation. Section 2.2 describes the coupled heat-transfer mechanisms governing thermal losses in heated gas delivery systems. Section 2.3 addresses the aerodynamic loss mechanisms specific to reducer and contraction geometries. Section 2.4 reviews the numerical methods used to model these phenomena. Section 2.5 discusses near-wall mesh resolution requirements. Section 2.6 identifies the specific research gap that motivates the present study and states the thesis contribution.

2.1 High-Temperature Dynamic Pressure Sensing and Calibration

2.1.1 The Instrumentation Gap in High-Temperature Gas Turbine Operation

Gas turbines are fundamentally unsteady machines. The pressure oscillations generated by combustion heat release, thermoacoustic coupling between the flame and the acoustic modes of the combustor volume, and the periodic rotor–stator interaction loading of turbine blades all produce time-varying pressure signals whose amplitude, frequency content, and phase relationships carry diagnostic information inaccessible to static or time-averaged measurements. Dynamic pressure sensors capable of resolving these signals under the thermal and mechanical conditions of gas turbine operation are therefore essential tools in turbine development, condition monitoring, and safety certification.

The fundamental challenge is that the materials capable of transducing pressure with high bandwidth and sensitivity have temperature-dependent properties that degrade, drift, or fail permanently at temperatures well below those present in the gas turbine hot section. Standard alpha-quartz piezoelectric crystals maintain their piezoelectric response up to approximately 400–500 °C before a displacive phase transition destroys the transduction mechanism. Gallium orthophosphate (GaPO_4) extends reliable sensing to approximately 780 °C. At the material research frontier, silicon carbide (SiC) MEMS sensors have been demonstrated in combustor environments at temperatures up to 600 °C by Okojie et al. (2014) at NASA Glenn Research Center, successfully detecting thermoacoustic instabilities with pressure amplitudes as small as 0.5 psi. However, commercial SiC sensors with full dynamic characterisation across their operating range remain at an advanced research stage.

2.1.2 The Requirement for Temperature-Traceable Calibration

The temperature dependence of sensor sensitivity is not merely a failure mode to be avoided: it is an inherent property that must be characterised and compensated throughout the intended operating temperature range. Salminen et al. (2020) characterised a capacitive dynamic pressure sensor for high-temperature industrial applications and reported a sensitivity variation of approximately 1% per 100 °C. At this rate, a sensor calibrated at ambient temperature and deployed at 300 °C would exhibit an uncorrected sensitivity error of approximately 3%. At the time of their publication, only two national metrology institutes worldwide offered traceable dynamic

pressure calibration services, reflecting both the technical difficulty of the problem and the inadequacy of existing calibration infrastructure.

2.1.3 European Research Programmes on High-Temperature Instrumentation

The gap between the temperatures at which gas turbines operate and those at which reliable sensing is achievable has motivated successive European Commission research programmes. The HEATTOP project (EU Framework Programme 6, 2006–2010) investigated sensor materials, packaging technologies, cooling strategies, and calibration methodologies capable of extending reliable operation toward the gas path temperature (European Commission, 2010). The STARGATE project (EU Framework Programme 7, 2012–2016) extended this work targeting dynamic pressure sensing capability at gas-path temperatures substantially exceeding 1,000 °C, concluding that the temperature limits of available sensing materials and the accuracy limitations of high-temperature calibration constitute the binding constraints on how close to their thermal design ceiling gas turbines can be operated with instrumented confidence (European Commission, 2016).

2.1.4 The Pulsatorn Rig: Prior Characterisation Work

Bartkevičius (2023) designed and constructed the high-temperature variant of the Pulsatorn system and demonstrated stable sinusoidal pressure wave generation at frequencies from 1 Hz upward. He explicitly acknowledged that the thermal delivery chain from heater to test section had not been characterised under high-temperature conditions, identifying it as a probable source of thermal losses whose magnitude remained unquantified. Järleby (2025) addressed this through systematic thermocouple measurements, measuring a temperature deficit of approximately 100 K between the heater outlet and the pulsator housing, and identifying the reducer at the heater-to-pipe interface as the dominant single loss location. These two findings define the specific engineering problem that the present thesis addresses.

2.2 Heat-Transfer Mechanisms in Heated Gas Delivery Systems

2.2.1 The Coupled Heat-Transfer Chain

The temperature of a heated gas stream at a downstream measurement location is determined by a chain of coupled heat-transfer mechanisms. Internal forced convection transfers thermal energy from the hot gas to the solid wall, governed by the local Nusselt number and the temperature difference between the bulk gas and the wall surface. Conduction through the solid wall transports this energy to the outer surface. External convection and radiation from the outer surface then dissipate this energy to the ambient. For metallic components such as the stainless steel reducers in the Pulsatorn rig, the thermal conductivity of the solid (approximately 16 W/m·K for structural steel) is approximately 600 times higher than that of the internal gas, meaning that the solid wall contributes negligibly to the total thermal resistance. The dominant resistances are therefore the internal convective resistance and the external convective-radiative resistance.

2.2.2 Internal Forced Convection and Separation-Zone Heat Transfer

For fully developed turbulent pipe flow, the Dittus–Boelter correlation gives $Nu = 0.023 Re^{0.8} Pr^{0.4}$, where Re is the Reynolds number and Pr the Prandtl number (Incropera and DeWitt, 2007; Cengel and Cimbala, 2018). In regions of flow separation and turbulent reattachment, the local Nusselt number can substantially exceed this fully-developed value. The reattachment of a turbulent shear layer produces locally elevated heat transfer coefficients several times the undisturbed pipe-flow value, owing to the thinning of the thermal boundary layer at the reattachment point. Martin (1977) demonstrated that for impinging jets, local Nusselt numbers at the stagnation point can reach 2–5 times the corresponding fully-developed pipe flow value at comparable Reynolds numbers. In the legacy reducer, the annular flat face at the inlet step presents exactly such a stagnation-point geometry, constituting a dominant source of enhanced thermal loss absent from the redesigned geometry.

2.2.3 External Convection and the Thermal Resistance Framework

For engineering estimates of natural convection from metallic surfaces, the Churchill–Chu correlation for horizontal cylinders gives h values in the range 5–20 W/m²·K for the relevant temperature levels and geometric scales (Churchill and Chu, 1975). The value $h_{ext} = 10$ W/m²·K adopted in the present CHT simulations represents a representative mid-range estimate. For the reducer geometries in this study, a first-order thermal resistance analysis based on the external convective resistance $R_{ext} = 1/(h_{ext} \times A_{ext})$ and the Biot numbers of the solid bodies ($Bi = 0.003$ – 0.004 , confirming internal conduction resistance is negligible) suggests that R_{ext} is likely to dominate the thermal circuit; this is subsequently confirmed by the CHT simulations in Chapter 4. At outer wall temperatures of 600–630 K and emissivity $\epsilon \approx 0.8$ for oxidised stainless steel (Incropera and DeWitt, 2007), the Stefan–Boltzmann radiative flux is approximately 5,000–7,000 W/m², substantially exceeding the convective flux of 2,768–3,279 W/m² predicted by the CHT simulations. Neglect of radiation is therefore the most significant absolute limitation of the present study, as discussed in Section 5.5.2.

2.3 Flow Conditioning and Pressure Loss in Reducer Geometries

2.3.1 The Vena Contracta and Borda–Carnot Mechanism

Pressure losses in internal flow systems are conventionally divided into major (frictional) losses along straight duct lengths and minor (form) losses associated with geometric changes (White, 2011; Idelchik, 2008). For an abrupt contraction, the dominant loss arises from the vena contracta formed downstream of the sharp inlet edge, where flow streamlines cannot follow the abrupt change in wall direction. For a thin-edged orifice at high area ratio, the contraction coefficient $C_c \approx 0.62$ (White, 2011), meaning the contracted jet cross-section is approximately 62% of the downstream bore area. The subsequent irreversible expansion of this jet back to the full bore gives the Borda–Carnot pressure loss:

$$\Delta P_{BC} = \frac{1}{2}\rho (u_{vc} - u_d)^2 = \frac{1}{2}\rho u_d^2 (1/C_c - 1)^2 \rightarrow K \approx 0.376 \text{ for } C_c = 0.62$$

This is a substantial loss coefficient. Both reducer geometries share the same overall area ratio of 34.13:1 (35.05 mm inlet to 6.0 mm outlet), but the legacy reducer achieves this through two sequential abrupt contraction steps, while the redesigned reducer achieves it through a single smooth quintic contraction. The cumulative Borda–Carnot losses at the two abrupt steps of the legacy reducer degrade the effective total

pressure at the vena contracta plane. At supercritical overall pressure ratios, this total pressure deficit prevents choking at the geometric throat, even when the nominal supply-to-outlet ratio exceeds the critical value of 1.893 for air ($\gamma = 1.4$). This is the physical mechanism behind the central aerodynamic finding of the present study.

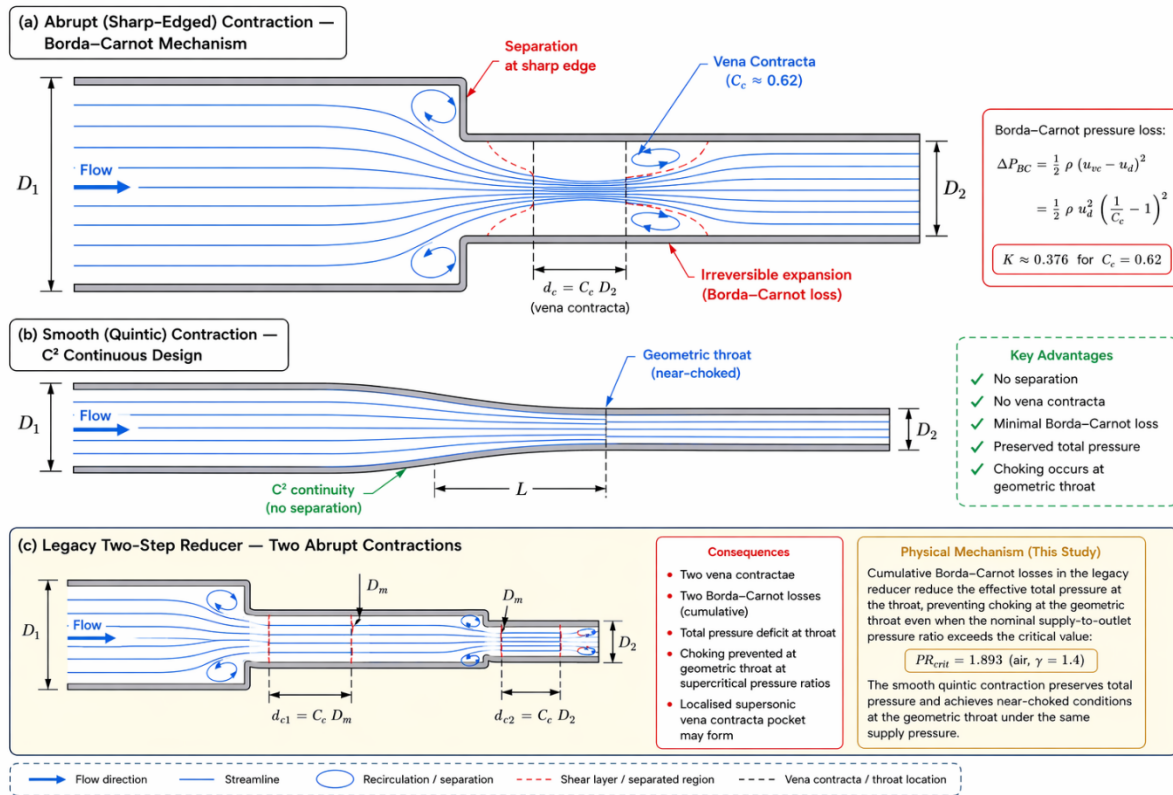


Figure 2.1: Borda–Carnot mechanism in abrupt contractions. (a) Single abrupt contraction showing vena contracta formation and irreversible pressure loss. (b) Smooth C^2 -continuous quintic contraction eliminating separation and preserving total pressure. (c) Legacy two-step reducer geometry showing cumulative Borda–Carnot losses that prevent choked flow at the geometric throat despite supercritical supply pressure ratios.

2.3.2 Smooth Contraction Design and C^2 Continuity

Morel (1975) demonstrated that a contraction profile with continuous slope and curvature at the junctions with the upstream and downstream pipe sections (C^2 continuity) suppresses flow separation by avoiding the adverse pressure gradient perturbations associated with geometric discontinuities. A quintic Hermite polynomial $\phi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3$ satisfies $\phi(0) = 0$, $\phi(1) = 1$, $\phi'(0) = \phi'(1) = 0$, and $\phi''(0) = \phi''(1) = 0$, providing C^2 continuity at both junctions with zero slope and zero curvature. The resulting wall radius profile $R(z) = R_{in} + (R_{out} - R_{in}) \times \phi((z - z_0)/L)$ is the standard choice for smooth nozzle and contraction design (Anderson, 2003), and is the profile adopted for the redesigned reducer in this study.

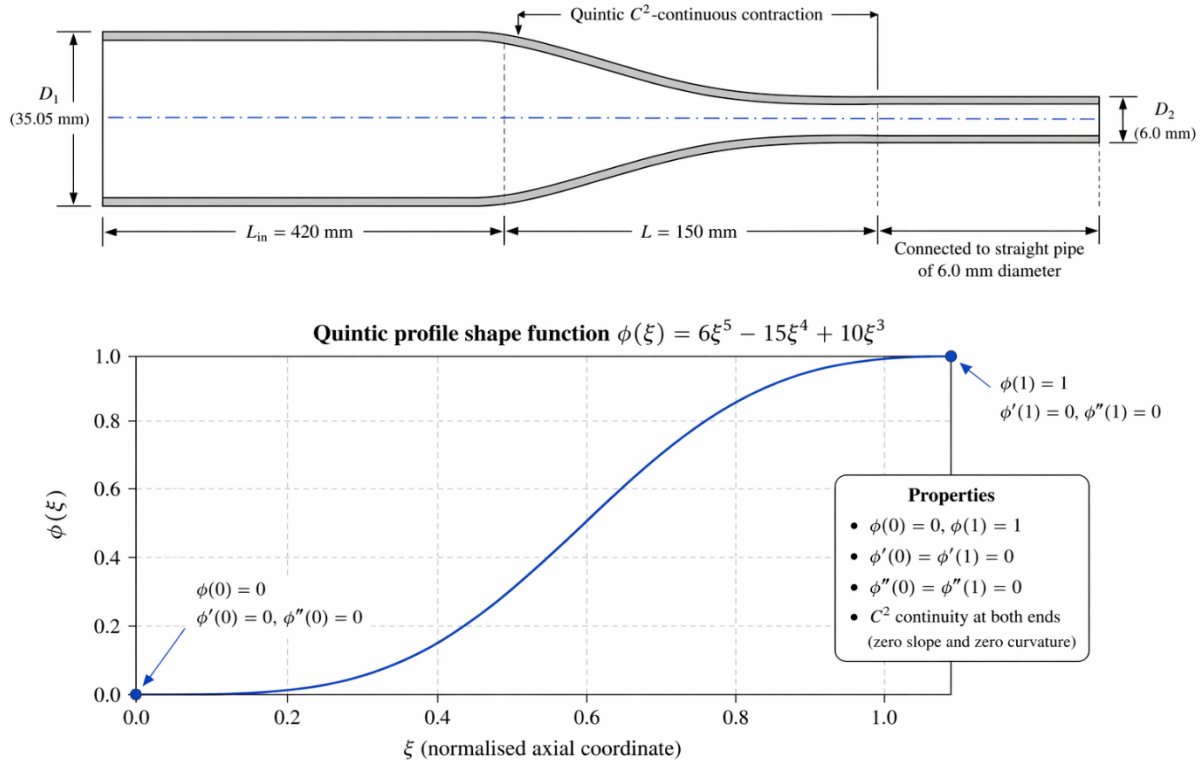


Figure 2.2 Quintic profile shape function $\phi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3$ plotted against normalised axial coordinate $\xi \in [0, 1]$

2.3.3 Compressible Flow and Choking in Contracting Ducts

For isentropic flow through a converging duct, choking at the geometric throat occurs when the supply-to-outlet pressure ratio exceeds the critical value $PR_{crit} = ((1+(\gamma-1)/2))^{\gamma/(\gamma-1)} = 1.893$ for air (Anderson, 2003; Shapiro, 1953; Cengel and Boles, 2015). In practice, total pressure losses upstream of the throat reduce the effective stagnation pressure available at the throat, requiring a higher supply pressure to achieve the same critical condition. For the legacy reducer, the Borda–Carnot losses at the two abrupt steps prevent choked flow at the geometric throat despite a supercritical nominal pressure ratio, producing instead a localised supersonic vena contracta pocket. The redesigned smooth contraction, by distributing total pressure losses along the smooth contraction body rather than concentrating them at abrupt steps upstream of the throat, achieves near-choked conditions. This qualitative flow regime divergence is the central aerodynamic result of the present study and is discussed in detail in Sections 4.6 and 4.7.

2.4 CFD and Conjugate Heat-Transfer Modelling

2.4.1 The Role of CFD in Comparative Design Assessment

Computational fluid dynamics combined with conjugate heat transfer (CHT) provides a rigorous framework for comparative design assessment when experimental access is limited. For comparative studies, where the primary objective is quantifying the performance difference between two geometries rather than predicting absolute performance, the CHT approach is particularly powerful because identical boundary

conditions for both geometries ensure that any difference in predicted output is attributable solely to geometry. Guo et al. (2021) performed a combined experimental and CHT study of turbulent hot-air flow through a heated metallic pipe with an external coefficient $h_{\text{out}} = 14 \text{ W/m}^2\cdot\text{K}$, demonstrating good agreement between CHT predictions and thermocouple measurements and confirming that CHT with an imposed outer convection coefficient can reproduce the dominant heat-transfer behaviour in heated metallic flow paths. The methodology and boundary condition approach adopted in the present study is methodologically similar to that of Guo et al. (2021), with the external convection coefficient applied uniformly to the outer solid surfaces of both reducer geometries.

2.4.2 The SST $k-\omega$ Turbulence Model

The SST (Shear Stress Transport) $k-\omega$ model of Menter (1994) is widely considered the most suitable two-equation closure for engineering internal flow applications with adverse pressure gradients, incipient separation, and wall heat transfer. It combines the near-wall accuracy of the Wilcox $k-\omega$ formulation (Wilcox, 2006), which correctly captures adverse pressure gradient boundary layers and the onset of separation, with the free-stream independence of the $k-\epsilon$ formulation. An eddy-viscosity limiter in the SST formulation prevents over-prediction of turbulent viscosity in strongly strained flows, giving better performance than standard $k-\epsilon$ for flows with adverse pressure gradients and separated zones. This is the critical advantage for the legacy reducer's vena contracta flow, where the separated and reattaching shear layer presents precisely the strongly strained, adverse pressure gradient conditions for which the SST model is specifically designed. Published benchmark studies confirm that SST with $y^+ \leq 5$ achieves substantially improved wall heat flux predictions for turbulent pipe flow with heating (Menter and Esch, 2001).

2.5 Mesh Resolution and Near-Wall Treatment

The non-dimensional wall distance $y^+ = \rho u_\tau y_1 / \mu$ characterises the placement of the first mesh node relative to the boundary layer structure, where $u_\tau = \sqrt{(\tau_w / \rho)}$ is the friction velocity and y_1 is the wall-normal distance to the first cell centre. For the SST $k-\omega$ model with automatic near-wall treatment in ANSYS Fluent, a maximum $y^+ \leq 5$ is required for direct resolution of the viscous sublayer; $y^+ \leq 1$ is recommended for the highest-accuracy wall heat flux prediction. The automatic wall treatment blends continuously between a low- y^+ wall-resolved formulation and a wall-function treatment depending on the local y^+ of the first wall cell, without requiring a user-specified switch between the two approaches. Published benchmarks indicate that wall heat flux accuracy degrades measurably when $y^+ \approx 10$ compared to $y^+ \approx 1$ (Menter and Esch, 2001). Inflation layer best-practice parameters, specifically first cell height targeting $y^+ \leq 5$ at worst-case conditions, 15–25 layers, and growth rate 1.1–1.2, guided the mesh generation strategy in this study. The CHT meshes use a first cell height of $1 \times 10^{-6} \text{ m}$ with 15 layers and growth ratio 1.2, achieving maximum wall y^+ below 3.0 for both geometries across all three pressure cases (Section 3.8.5).

2.6 Identified Research Gap and Thesis Contribution

The reviewed literature establishes the following coherent narrative. High-temperature dynamic pressure sensor calibration is a recognised strategic need confirmed by HEATTOP (2010) and STARGATE (2016). The Pulsatorn rig was established by

Bartkevičius (2023) and its thermal delivery problem quantified by Järleby (2025), who identified the reducer as the dominant thermal loss location. The heat-transfer literature (Section 2.2) provides the physical framework for understanding this loss, including the stagnation-point impingement mechanism at the annular step face (Martin, 1977) and the dominance of the external convective resistance. The hydraulic resistance literature (Section 2.3) provides the aerodynamic loss context via the Borda–Carnot formulation (White, 2011; Idelchik, 2008), and the Morel (1975) C^2 continuity criterion provides the design basis for the quintic contraction. One-dimensional compressible flow theory (Anderson, 2003) establishes the critical pressure ratio framework against which the numerical Mach number results are benchmarked. The CFD and CHT literature provides validated numerical tools, with Guo et al. (2021) providing precedent that imposed outer convection coefficients can reproduce dominant heat-transfer behaviour in heated metallic flow paths, and Menter and Esch (2001) establishing the y^+ requirements for accurate wall heat flux prediction with the SST $k-\omega$ model.

What is absent from the existing literature is a quantified, geometry-to-geometry comparison of the specific reducer at the Pulsatorn heater-to-downstream interface, across a representative range of operating back-pressures, under controlled numerical boundary conditions. No previous study has identified the choking asymmetry, specifically the failure of the legacy geometry to achieve near-choked flow at supercritical pressure ratios where the redesigned geometry reaches $Ma = 0.990$, as a consequence of the abrupt contraction’s Borda–Carnot total pressure losses. No previous study has isolated the role of external surface area as the primary determinant of thermal delivery performance in this class of heated gas delivery component, using insulated reference simulations to decouple aerodynamic and thermal contributions. This thesis addresses both gaps.

3. Methodology and Numerical Model Setup

3.1 Overview of the Numerical Investigation

This chapter describes the numerical methodology used to evaluate the comparative thermo-fluid performance of the legacy and redesigned reducer geometries in the Pulsatorn high-temperature dynamic pressure calibration rig. The investigation is organised as two complementary studies, each designed to address a distinct aspect of numerical credibility and physical characterisation.

Study 1 is a compressible, adiabatic CFD sensitivity study in which both reducer geometries are simulated across three systematically refined mesh levels under the most aerodynamically demanding operating condition. Its purpose is to establish mesh independence for the flow field, thereby providing a rigorous numerical basis for the conjugate heat transfer simulations that follow.

Study 2 is a steady-state conjugate heat transfer (CHT) campaign in which both geometries are simulated at three outlet pressure conditions chosen to span subsonic, high-subsonic, and supercritical pressure-ratio regimes. Its purpose is to generate the primary quantitative dataset for comparative assessment of thermal delivery, external heat loss, and internal flow regime.

All simulations were performed using ANSYS Workbench 2025 R1 with ANSYS Fluent as the flow and heat-transfer solver, in double-precision parallel mode on four processing cores constrained by the KTH Royal Institute of Technology academic licence. Geometry preparation was performed in ANSYS SpaceClaim / Discovery Premium; meshing was carried out in ANSYS Meshing. A quarter-symmetry (1/4) model was used for both geometries, exploiting two perpendicular planes of geometric and flow symmetry to reduce computational cost by a factor of four. All integral quantities reported in this thesis are corrected to full-geometry values by multiplication by four unless otherwise stated. A complete summary of all software versions, hardware specifications, and licence details is provided in Appendix F. Full solver configuration parameters and material property tables are provided in Appendix C.

3.2 Geometry Description and Computational Domain

3.2.1 Coordinate System and Symmetry Assumptions

All geometries were defined using a right-handed Cartesian coordinate system in which the primary flow direction is aligned with the positive z-axis, the radial directions are x and y, and the origin is located at the centre of the inlet boundary face. Exploiting two perpendicular planes of geometric symmetry ($x = 0$ and $y = 0$ for a circular cross-section) reduces the computational domain to one quarter of the full three-dimensional assembly, with symmetry boundary conditions applied at both cut planes. This approach is valid provided the flow remains axisymmetric, which is appropriate for the steady-state geometries modelled here.

The computational domain for each geometry consists of three regions in series: a straight inlet extension, the reducer body (fluid zone only in Study 1, fluid plus solid zone in Study 2), and a straight outlet extension. The extensions serve to decouple the numerical solution from boundary-condition artefacts at the inlet and outlet faces.

3.2.2 Legacy Reducer Geometry

The legacy reducer is a commercially available threaded stainless-steel fitting originally selected for mechanical compatibility between the heater outlet thread (NPT) and the downstream hydraulic connector (BSP), without thermo-fluid performance as a design criterion. Its internal geometry creates two sequential abrupt coaxial area reductions. The first contraction occurs at the flat annular inlet face, reducing the 35.05 mm heater bore diameter to an 11.25 mm through-bore (area ratio 9.71:1). Hot gas impinges on this annular face before being forced through the 11.25 mm bore. The through-bore continues for approximately 25 mm axially before the second abrupt contraction reduces the bore from 11.25 mm to 6.0 mm (area ratio 3.52:1) at the outlet face. The total solid body length is 38 mm. Both contractions are geometrically abrupt, with zero streamwise length at each step face.

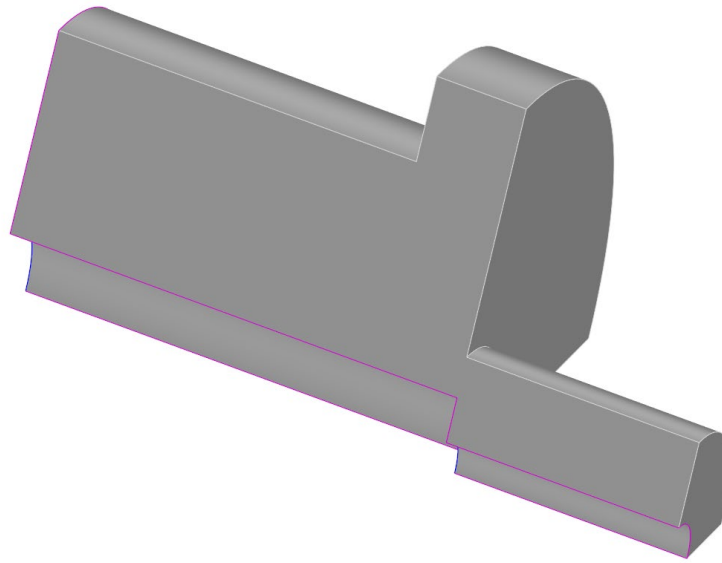


Figure 3.1. Quarter-symmetry 3D CAD geometry of the legacy reducer.

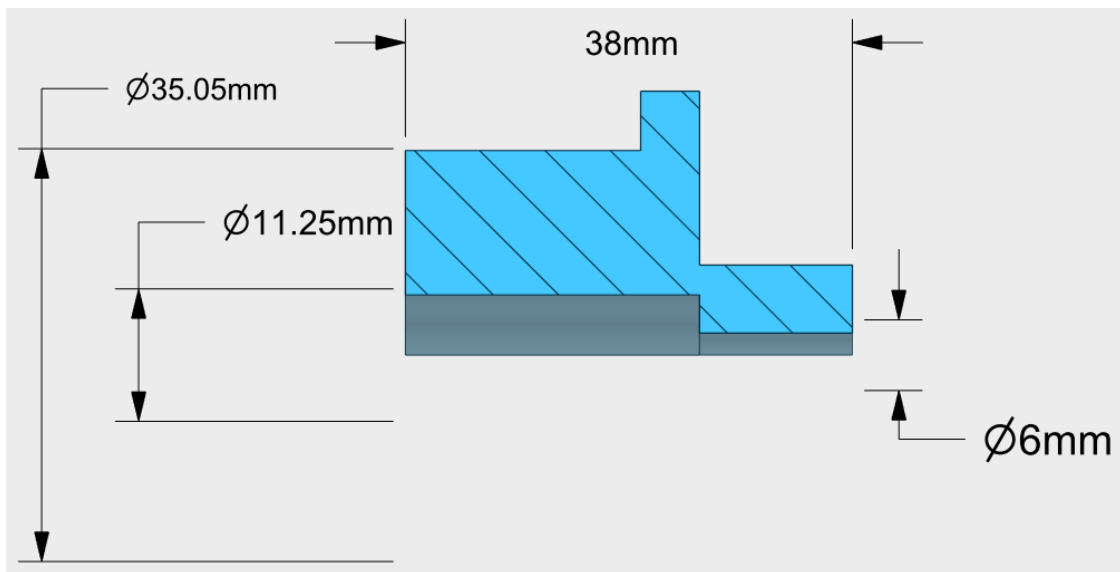


Figure 3.2. Dimensioned cross-section of the legacy reducer internal geometry.

For the numerical model, a 200 mm straight inlet extension (35.05 mm bore) was added upstream of the reducer body, and a 100 mm straight outlet extension (6.0 mm bore) was added downstream. The total modelled axial length is 338 mm (200 + 38 + 100). Internal threads and external features of the physical fitting were excluded from the computational domain. The solid domain for CHT simulations comprises the reducer body only; the inlet and outlet extensions are adiabatic fluid-only zones.

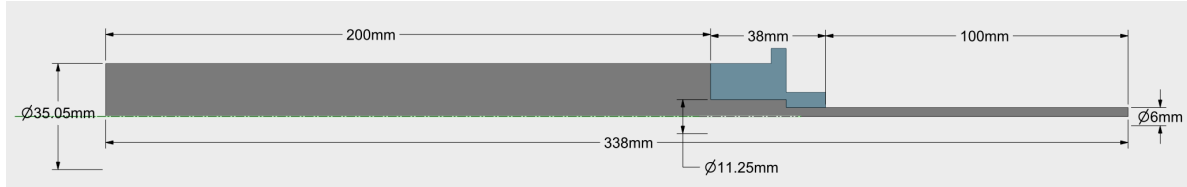


Figure 3.3. Legacy reducer computational domain with inlet and outlet extensions. Total domain length: 338 mm.

The external convective surface area of the solid body, comprising the annular inlet flange face, the hex body outer surface, and the outlet spigot outer surface, was verified by integrating mesh face areas over all faces carrying the external convection boundary condition in Fluent, yielding 5,758.15 mm² for the full geometry (1,439.54 mm² in the quarter model). The solid body volume is 6.754×10⁻⁶ m³ (quarter model), giving a Biot number $Bi = h \cdot L_c / \lambda_s = 0.00288$, where $L_c = V/A_{ext} = 4.69$ mm, confirming that internal conduction resistance is negligible relative to the external convective resistance.

3.2.3 Redesigned Reducer Geometry

The redesigned reducer employs a fifth-order quintic Hermite polynomial to define the internal bore radius R as a function of axial position z :

$$R(z) = R_{in} + (R_{out} - R_{in}) \times \varphi(\xi),$$

$$\varphi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3,$$

$$\xi = (z - z_0)/L$$

where $R_{in} = 17.525$ mm (35.05 mm inlet bore), $R_{out} = 3.0$ mm (6.0 mm outlet bore), and $L = 150$ mm. The polynomial satisfies $\varphi(0) = 0$, $\varphi(1) = 1$, $\varphi'(0) = \varphi'(1) = 0$, and $\varphi''(0) = \varphi''(1) = 0$, ensuring C^2 continuity of position, slope, and curvature at both junctions with the straight bore sections. This guarantees no geometric discontinuity is presented to the approaching boundary layer, eliminating separation at the inlet and outlet junctions (Morel, 1975). The overall area ratio is $A_{in}/A_{out} = (17.525/3.0)^2 = 34.13:1$. The 51 coordinate points defining the quintic profile were generated analytically and imported as a spline in ANSYS SpaceClaim.

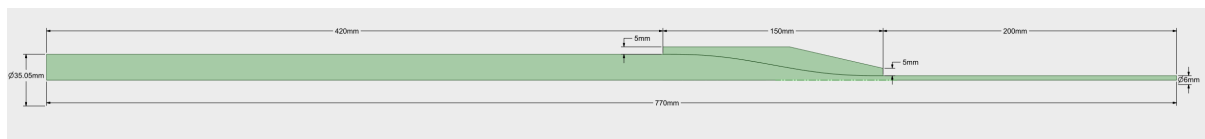


Figure 3.4. Redesigned reducer computational domain with inlet and outlet extensions. Total domain length: 770 mm.

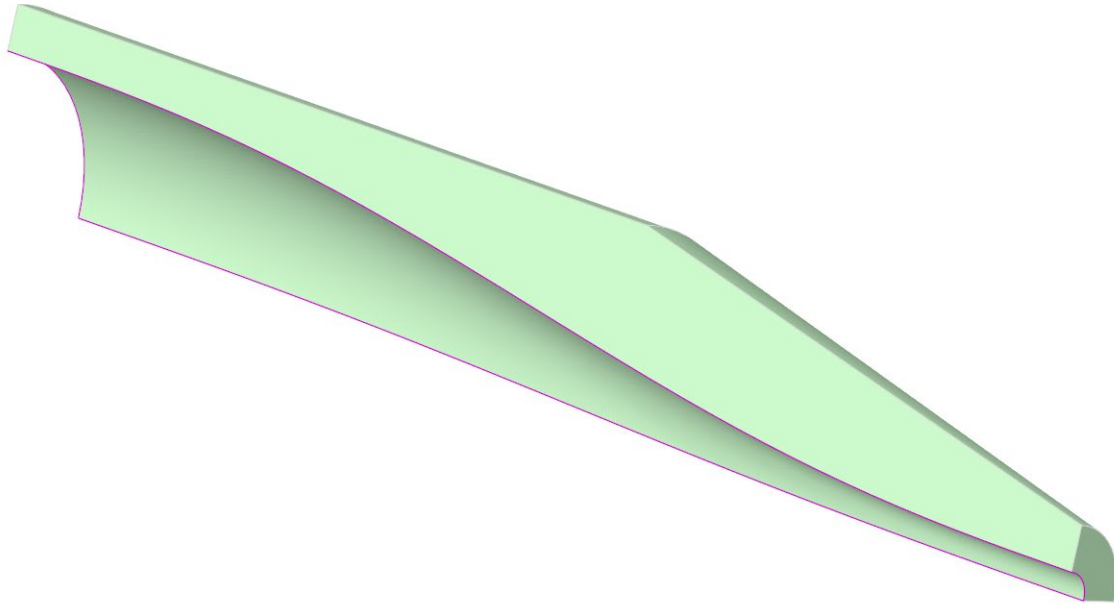


Figure 3.5. Quarter-symmetry 3D CAD geometry of the redesigned reducer.

A 420 mm inlet extension (35.05 mm bore) was added upstream of the reducer body, providing a hydrodynamically undisturbed inlet condition upstream of the smooth contraction. A 200 mm outlet extension (6.0 mm bore) was added downstream. The total modelled domain length is 770 mm (420 + 150 + 200). The solid domain for CHT simulations spans the 150 mm contraction body only; inlet and outlet extensions are adiabatic fluid-only zones.

The external convective surface area was verified by integration in Fluent, yielding 19,282.45 mm² for the full geometry (4,820.61 mm² in the quarter model), which is 3.35 times larger than the legacy reducer. This area difference is the primary driver of the thermal performance distinction between the two geometries under uninsulated conditions. The solid body volume is 3.103×10^{-5} m³ (quarter model), giving $Bi = 0.00396$ ($L_c = 6.44$ mm), consistent with external resistance dominance for both geometries. Table 3.1 summarises the confirmed geometric parameters of both reducer configurations.

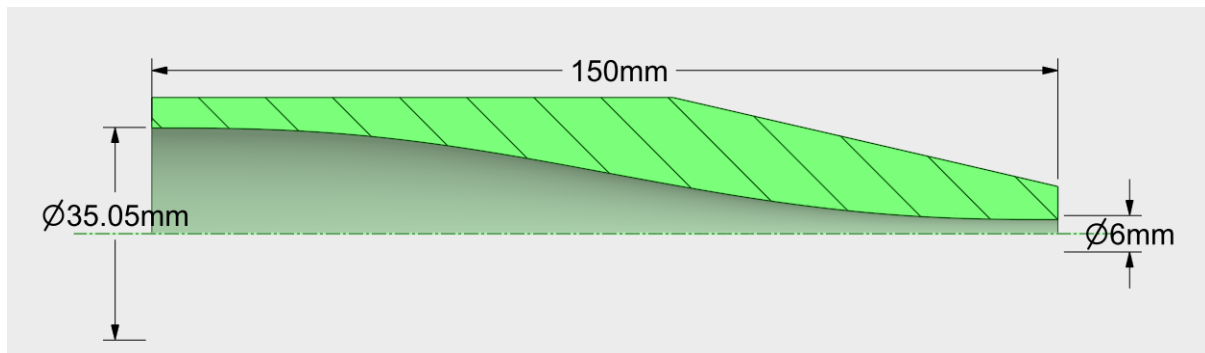


Figure 3.6. Dimensioned cross-section of the redesigned reducer internal geometry.

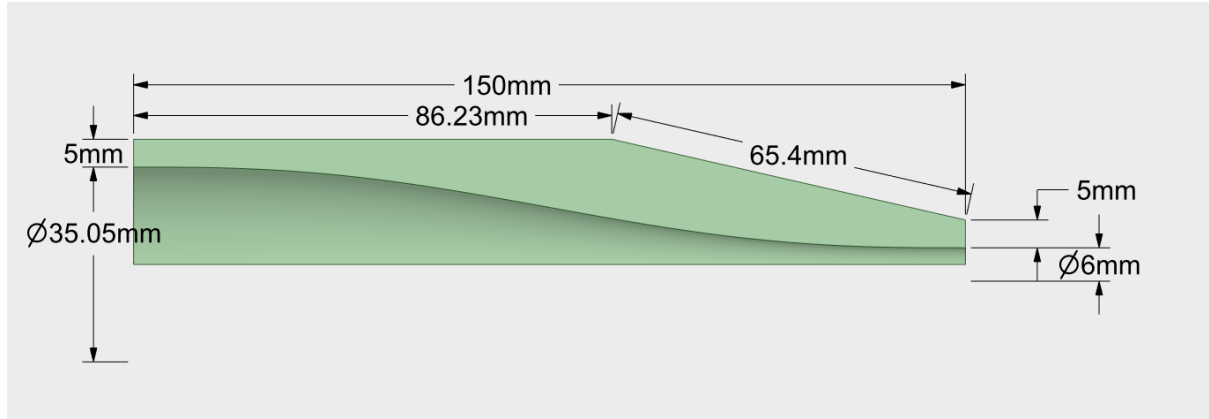


Figure 3.7. Detailed Dimensions of the redesigned reducer geometry

Parameter	Legacy reducer	Redesigned reducer
Inlet bore diameter	Ø35.05 mm	Ø35.05 mm
Outlet bore diameter	Ø6.0 mm	Ø6.0 mm
Internal profile	Two abrupt Borda–Carnot steps 35.05→11.25→6.0 mm	Quintic C ² -continuous $\phi(\xi)$ $= 6\xi^5 - 15\xi^4 + 10\xi^3$
Solid body length	38 mm	150 mm
Overall area ratio A_{in}/A_{out}	34.13:1	34.13:1
Domain length (inlet + body + outlet)	338 mm (200+38+100)	770 mm (420+150+200)
A_{ext} quarter model (mm ²)	1,439.54	4,820.61
A_{ext} full geometry (mm ²)	5,758.15	19,282.45
A_{ext} ratio (new/legacy)	-	3.349× larger
Solid volume V (quarter, m ³)	6.754×10^{-6}	3.103×10^{-5}
Biot number Bi	0.00288	0.00396
<i>A_{ext} cross-validated between SpaceClaim measurement and Fluent face-area integration to within 0.01%. Biot number: $Bi = h \cdot (V/A_{ext})/\lambda_s$ with $h = 10 \text{ W/m}^2\text{K}$, $\lambda_s = 16.27 \text{ W/m}\cdot\text{K}$.</i>		

Table 3.1. Confirmed geometric parameters for both reducer configurations.

3.2.4 Geometric Simplifications

Internal threads were excluded from the fluid domain and treated as smooth cylindrical bore surfaces. Minor external chamfers, fastener recesses, and surface features not contributing to the primary heat conduction path were omitted from the solid domain. These simplifications were applied consistently to both geometries, preserving the dominant thermo-fluid behaviour and introducing no systematic bias in the comparative analysis.

3.3 First-Order Engineering Pre-Analysis

3.3.1 Continuity and Velocity Scaling

Before constructing numerical models, first-order analytical estimates were applied to establish physical expectations and provide order-of-magnitude checks against which

simulation outputs could be assessed for plausibility. For steady compressible flow through a duct of varying cross-section, conservation of mass requires $\dot{m} = \rho Au$, where $\dot{m}$ is mass flow rate, ρ is local fluid density, A is cross-sectional area, and u is mean axial velocity. At the operating conditions of this study (inlet temperature 673 K, gauge pressure 100,000 Pa), the inlet density from the ideal gas law is $\rho = (101,325 + 100,000)/(287 \times 673) \approx 1.042 \text{ kg/m}^3$. At the 35.05 mm inlet bore, the mean inlet velocity is approximately 1.3–1.5 m/s (pre-analysis estimate; full CHT simulations yield 4–7 m/s), well subsonic. The strong area reduction (34.13:1) drives the flow in the 6 mm outlet bore toward the local speed of sound, which at 673 K is $a = \sqrt{\gamma R_s T} = \sqrt{(1.4 \times 287 \times 673)} \approx 519 \text{ m/s}$.

3.3.2 Pressure Loss and Critical Pressure Ratio

For the legacy abrupt contraction, geometric losses dominate and are estimated using the Borda–Carnot formula for the irreversible loss in the reattachment zone downstream of the vena contracta: $\Delta P_{BC} = \frac{1}{2}\rho_{vc}(u_{vc} - u_d)^2$, where u_{vc} is the velocity at the vena contracta and u_d is the fully-recovered downstream velocity. For a sharp-edged contraction with $C_c \approx 0.62$, the loss coefficient $K = (1/C_c - 1)^2 \approx 0.376$ per step (White, 2011). The redesigned quintic contraction replaces this geometric loss with the comparatively modest frictional loss along the contraction wall.

The critical pressure ratio for choked flow in air ($\gamma = 1.4$) is $PR_{crit} = ((1.4+1)/2)^{1.4/(1.4-1)} = (1.2)^{3.5} = 1.893$. With the inlet absolute pressure of 201,325 Pa and atmospheric outlet pressure of 101,325 Pa, the overall pressure ratio is $201,325/101,325 = 1.987 > 1.893$. Isentropic flow theory therefore predicts choked flow at the geometric throat; however, the legacy reducer's abrupt contractions dissipate total pressure, potentially preventing choking. This hypothesis was confirmed by the numerical results.

3.3.3 First-Order Thermal Energy Balance

A direct estimate of external heat loss cannot be obtained from $\dot{m}C_p(T_{in} - T_{out})$ in the present study because the inlet boundary is specified in terms of total temperature, while T_{out} is reported as a mass-weighted outlet static temperature. In compressible accelerating flow, static temperature decreases even in the absence of external heat loss. This is demonstrated by the insulated Case C simulations, for which $q_{ext} = 0$ while T_{out} remains 588.94 K for the redesigned reducer and 587.70 K for the legacy reducer. Accordingly, external heat loss is quantified in this thesis using the CHT wall-flux integral Q_{ext} , while T_{out} is interpreted as a bulk thermal-delivery metric rather than a direct measure of heat loss.

3.4 Simulation Study Design and Case Definition

3.4.1 Study 1: Mesh Sensitivity Assessment

Study 1 comprises six compressible, adiabatic CFD simulations: both reducer geometries evaluated at three mesh refinement levels each. The outlet pressure was fixed at 0 Pa gauge ($PR = 1.987 > 1.893$) for all Study 1 runs, as this condition produces the highest flow velocities and steepest velocity gradients, the most demanding regime for spatial discretisation. No solid domain was included. All fluid walls were treated as adiabatic (zero heat flux). The three refinement levels are denoted Coarse, Baseline, and Fine, with refinement ratios of approximately 2–3× in total cell count between adjacent levels.

3.4.2 Study 2: Conjugate Heat Transfer Campaign

Study 2 comprises six steady-state CHT simulations: both reducer geometries at three outlet pressure conditions spanning qualitatively distinct flow regimes.

Case	Outlet BC (Pa gauge)	PR_{abs}	Flow regime	Physical motivation
A	80,000	1.110	Subsonic	Moderate downstream restriction; lowest velocity condition
B	40,000	1.424	Subsonic, high Ma	Significant compressibility; below choking threshold
C	0	1.987	Supercritical ($PR > 1.893$)	Maximum flow rate; tests choked flow capability; most differentiating condition

$PR_{abs} = \text{absolute pressure ratio } P_{o,in}/P_{outlet,static} = 201,325 \text{ Pa} / P_{outlet,abs}$.
Critical pressure ratio $PR_{crit} = 1.893$ for $\gamma = 1.4$.

Table 3.2. Study 2 CHT simulation case matrix.

Cases B and C were initialised using the cascade approach: the converged solution from the preceding case (both .cas.h5 and .dat.h5 files) was loaded as the initial condition, with only the outlet pressure boundary modified. This approach preserves the converged solid thermal state from the preceding case, substantially reducing the iterations required for convergence at the new operating point. The correctness of each cascade initialisation was verified by running ten diagnostic iterations at the original boundary condition and confirming that residuals begin at low values (continuity $\sim 10^{-6}$, energy $\sim 10^{-10}$) rather than spiking to unity.

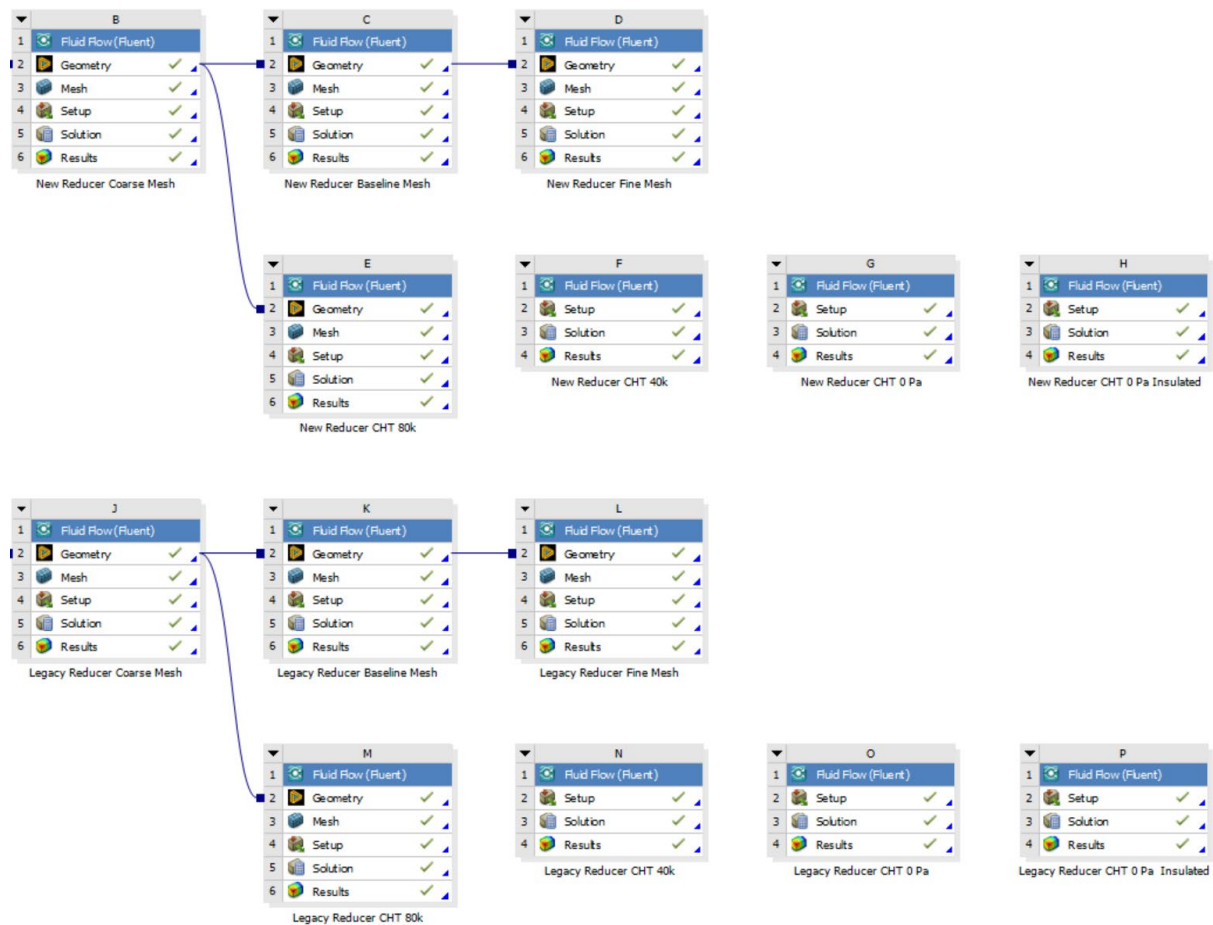


Figure 3.8. ANSYS Workbench project schematic showing Study 1 and Study 2 simulation systems with cascade initialisation links.

3.5 Governing Equations and Physical Models

3.5.1 Conservation Equations for Compressible Turbulent Flow

The flow was modelled as compressible, viscous, and turbulent under steady-state conditions. The governing equations are the Reynolds-Averaged Navier–Stokes (RANS) equations for conservation of mass, momentum, and energy in the fluid domain (Versteeg and Malalasekera, 2007).

Conservation of mass:

$$\partial(\rho u_j)/\partial x_j = 0$$

Conservation of momentum (RANS):

$$\rho u_j \partial u_i / \partial x_j = -\partial p / \partial x_i + \partial / \partial x_j [\mu_t^d (\partial u_i / \partial x_j + \partial u_j / \partial x_i) - (2/3) \mu_t^d (\partial u_k / \partial x_k) \delta_{ij}]$$

where $\mu_t^d = \mu + \mu_t$ is the sum of molecular and turbulent viscosity. The Reynolds stress tensor is closed through the Boussinesq hypothesis using μ_t from the SST $k-\omega$ model.

Conservation of energy (total enthalpy formulation, ANSYS Fluent):

$$\partial(\rho u_j H) / \partial x_j = \partial / \partial x_j [(\lambda / C_p + \mu_t / Pr_t) \partial H / \partial x_j]$$

where $H = h + \frac{1}{2}|u|^2$ is specific total enthalpy, λ is fluid thermal conductivity, C_p is specific heat, and $Pr_t \approx 0.85$ is the turbulent Prandtl number. The full total-enthalpy formulation includes pressure-work and viscous dissipation contributions at compressible conditions (ANSYS Inc., 2025).

3.5.2 Equation of State and Thermophysical Properties of Air

Air was treated as a compressible ideal gas: $\rho = p/(R_s T)$ with $R_s = 287 \text{ J/kg}\cdot\text{K}$. Dynamic viscosity was computed using Sutherland's law: $\mu(T) = \mu_{\text{ref}}(T/T_{\text{ref}})$

$$\mu(T) = \mu_{\text{ref}} \times (T/T_{\text{ref}})^{3/4} \times (T_{\text{ref}} + S)/(T + S), \mu_{\text{ref}} = 1.716 \times 10^{-5} \text{ Pa}\cdot\text{s}, T_{\text{ref}} = 273.15 \text{ K}, S = 110.4 \text{ K}$$

Specific heat C_p and thermal conductivity λ of air were specified as piecewise-linear functions of temperature based on NIST reference data for dry air, defined at thirteen temperature points in the range 300–1000 K (Table C.4, Appendix C) (NIST, 2023). At the heater inlet temperature of 673 K, $C_p = 1,069.4 \text{ J/kg}\cdot\text{K}$ (confirmed from the Fluent material panel). The piecewise-linear specification is necessary because neither C_p nor λ varies linearly with temperature over the 400 K range between ambient and the inlet condition.

3.5.3 Solid Domain Energy Equation and Conjugate Coupling

In Study 2 simulations, the solid domain is governed by the steady-state heat conduction equation: $\partial / \partial x_i (\lambda_s \partial T_s / \partial x_i) = 0$, where λ_s is solid thermal conductivity and T_s is solid temperature. Isotropic conductivity is assumed. The solid material properties are those of structural steel (316L) from the ANSYS Fluent material database: $\rho_s = 8,030 \text{ kg/m}^3$, $C_{p,s} = 502.5 \text{ J/kg}\cdot\text{K}$, $\lambda_s = 16.27 \text{ W/m}\cdot\text{K}$.

Conjugate coupling between fluid and solid is enforced at the shared interface by requiring continuity of temperature and heat flux: $T_{\text{fluid}} = T_{\text{solid}}$ and $\lambda_f \partial T_f / \partial n = \lambda_s \partial T_s / \partial n$, where n is the interface outward normal. These conditions are applied simultaneously at every interface face at every iteration, enabling fully coupled solution without prescribing wall temperature or wall heat flux a priori.

3.6 Turbulence Modelling

3.6.1 Selection Rationale

The SST (Shear Stress Transport) k – ω turbulence model (Menter, 1994) was selected for all simulations. In the legacy reducer, the abrupt area changes induce flow separation and turbulent reattachment with an adverse pressure gradient, conditions for which the SST model is specifically formulated through its eddy-viscosity limiter in regions of strong strain, a feature absent from the k – ϵ family. In the redesigned reducer, the attached boundary layer through the smooth contraction is a more benign regime well served by any wall-resolving two-equation model; the SST model provides consistent performance in both geometries.

The SST model employs an automatic near-wall treatment that blends continuously between a wall-resolved low- y^+ formulation (viscous sublayer resolved directly) and a wall-function treatment based on the local y^+ of the first wall cell. This flexibility is particularly relevant for the present study, in which the two geometries produce different local shear stresses and therefore different y^+ values for the same mesh inflation settings, as discussed in Section 3.8.5.

3.6.2 The SST k – ω Transport Equations

The SST model solves transport equations for turbulence kinetic energy k [m^2/s^2] and specific dissipation rate ω [s^{-1}]:

$$\partial(\rho u_i k) / \partial x_i = \tilde{P}_k - \beta^* \rho \omega k + \partial / \partial x_j [(\mu + \sigma_k \mu_t) \partial k / \partial x_j]$$

$$\partial(\rho u_i \omega) / \partial x_i = (\alpha / \nu_t) \tilde{P}_k - \beta \rho \omega^2 + \partial / \partial x_j [(\mu + \sigma_\omega \mu_t) \partial \omega / \partial x_j] + 2(1 - F_1) \rho \sigma_{\omega 2} (1 / \omega) (\partial k / \partial x_j) (\partial \omega / \partial x_j)$$

The eddy viscosity is computed with the SST limiter: $\mu_t = \rho a_1 k / \max(a_1 \omega, S F_2)$, where $a_1 = 0.31$ is the SST model constant, S is the mean strain-rate magnitude, and F_2 is a blending function that activates the limiter within boundary layers. This limiter prevents over-prediction of turbulent viscosity in strongly strained flows. Full model coefficients and blending function definitions are given in Menter (1994) and ANSYS Inc. (2025).

3.6.3 Turbulence Inlet Boundary Specification

At the inlet boundary, the turbulence intensity and hydraulic diameter method was used. A turbulence intensity of 5% was applied, consistent with fully developed turbulent pipe flow from the heater bore. The hydraulic diameter was set to 35.05 mm (the common inlet bore diameter) for both geometries. This specification method was used in preference to explicit turbulent length scale or viscosity ratio specification to avoid sensitivity to user-defined constants at the inlet.

3.7 Boundary Conditions

3.7.1 Inlet Boundary Conditions

The inlet boundary was specified as a pressure inlet in both Study 1 and Study 2. The inlet total (stagnation) gauge pressure was set to 100,000 Pa (absolute: 201,325 Pa) and the inlet total temperature was 673 K in all simulations. The pressure-inlet formulation in ANSYS Fluent specifies total pressure and total temperature at the boundary, with static pressure determined internally from the flow solution. The operating pressure was set to 101,325 Pa; all reported gauge pressures are referenced to this value.

3.7.2 Outlet Boundary Conditions

The outlet boundary was specified as a pressure outlet prescribing gauge static pressure at the outlet face. For subsonic outlets (Cases A and B), this is the backpressure seen at the domain exit. For Case C of the redesigned reducer, where the outlet is near-sonic, the pressure outlet specification becomes inactive and the pressure is determined by the internal flow field, allowing the near-sonic solution to pass through without artificial reflection. The three gauge outlet pressures are 80,000 Pa, 40,000 Pa, and 0 Pa (Cases A, B, C respectively).

3.7.3 Fluid Wall Boundary Conditions

In Study 1, all internal fluid wall faces were specified as no-slip, adiabatic walls (zero heat flux). In Study 2, the internal fluid wall faces coincide with the fluid–solid interface and are treated as fully coupled conjugate boundaries with no-slip velocity condition; temperature and heat flux continuity are enforced by the conjugate formulation described in Section 3.5.3.

3.7.4 External Solid Surface Boundary Condition

In Study 2, the external faces of the solid domain were assigned a convective heat transfer boundary condition: $q_{\text{ext}} = h_{\text{ext}} \times (T_{\text{wall,ext}} - T_{\infty})$ with $h_{\text{ext}} = 10 \text{ W/m}^2\text{K}$ and $T_{\infty} = 300 \text{ K}$. The value $h_{\text{ext}} = 10 \text{ W/m}^2\text{K}$ is representative of natural convection from a metallic surface in still ambient air, consistent with the Churchill–Chu correlation for horizontal cylinders at the relevant temperature differences and geometric scales (Churchill and Chu, 1975). The same h_{ext} was applied uniformly to all exposed outer solid surfaces in both geometries, ensuring that differences in CHT results are attributable solely to geometric differences rather than boundary condition differences.

Thermal radiation from the external surfaces was neglected in all simulations. At the operating outer wall temperatures of 577–630 K, the Stefan–Boltzmann radiative flux with $\varepsilon \approx 0.8$ (oxidised stainless steel) is approximately 5,000–7,000 W/m², which substantially exceeds the convective flux. This represents the most significant absolute limitation of the present study and is discussed fully in Section 5.5.2.

3.7.5 Symmetry Conditions

Symmetry boundary conditions were applied at the two cut planes ($x = 0$ and $y = 0$). The symmetry condition enforces zero normal velocity and zero normal gradient of all scalar quantities (pressure, temperature, k , ω), equivalent to no flux or force perpendicular to the plane of symmetry.

3.8 Mesh Strategy and Near-Wall Treatment

3.8.1 Mesh Topology and Generation Approach

Unstructured computational meshes were generated in ANSYS Meshing using the Patch Conforming method. The core flow region of the fluid domain was discretised with tetrahedral elements, providing the geometric flexibility required to resolve the continuously varying wall curvature of the redesigned reducer contraction and the internal radius transitions of the legacy geometry. Structured prismatic inflation layers were generated adjacent to all no-slip wall faces to resolve the near-wall velocity and thermal boundary layers.

The solid domain was meshed independently using tetrahedral elements with body sizing controls, and the mesh was made conforming (node-compatible) at the fluid–solid interface to avoid interpolation error in the coupled heat flux calculation.

3.8.2 Inflation Layer Parameters

Identical inflation layer settings were applied to all no-slip wall faces of both geometries to ensure comparative validity: any difference in observed heat transfer must be attributable to geometry, not meshing differences.

For Study 1 (adiabatic CFD, mesh sensitivity), the inflation settings were: first cell height 5×10^{-5} m, 10 layers, growth ratio 1.2. These settings yield a maximum wall y^+ of approximately 10.5–11.2 across both geometries at Case C conditions.

For Study 2 (CHT), the inflation settings were: first cell height 1×10^{-6} m, 15 layers, growth ratio 1.2. The first cell height of 1 μ m was selected to achieve full viscous sublayer resolution ($y^+ < 5$) throughout the fluid domain at all pressure cases for both geometries. The 15-layer count with growth ratio 1.2 provides sufficient inflation thickness to span the boundary layer at the outlet bore while maintaining a smooth transition to the tetrahedral core.

3.8.3 Solid Domain Meshing

The solid domain was meshed with tetrahedral elements using body sizing controls. The mesh was made conforming at the fluid–solid interface to avoid non-conformal interface interpolation errors in the coupled heat flux calculation. The solid meshing was applied independently from the fluid meshing, with the interface conformity enforced through ANSYS Meshing’s shared topology setting.

3.8.4 Resulting Mesh Statistics: CHT Simulations

Table 3.3 presents the confirmed mesh statistics for the CHT baseline meshes used in Study 2. Both meshes use the corrected inflation settings (1×10^{-6} m first cell height, 15 layers).

Parameter	<i>Legacy reducer</i>	<i>Redesigned reducer</i>
Total cells (fluid + solid)	1,305,887	1,717,766
Fluid cells	829,621	1,496,437
Solid cells	476,266	221,329
Inflation first cell height	1×10^{-6} m	1×10^{-6} m
Inflation layers (fluid)	15	15
Inflation growth ratio	1.2	1.2
y^+ _max (Case A)	1.863	1.490
y^+ _max (Case B)	2.687	2.147
y^+ _max (Case C)	2.990	2.364
External convective area A_{ext} (full)	5,758.15 mm ²	19,282.45 mm ²

Fluid cell count for legacy CHT = Study 1 baseline mesh (829,621) with corrected inflation. y^+ values are maximum over the fluid-side wall surface; all cases achieve full viscous sublayer resolution.

Table 3.3. CHT baseline mesh statistics for both reducer geometries (Study 2). All values confirmed from Fluent startup log and post-processing.

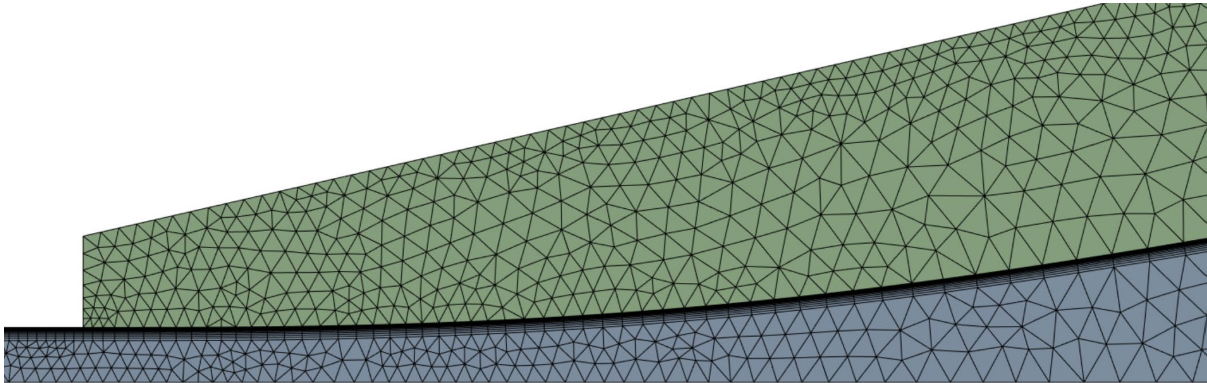


Figure 3.9. Redesigned Reducer CHT baseline mesh throat section

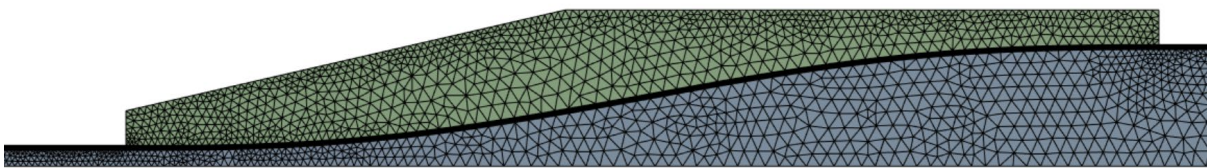


Figure 3.10. Redesigned reducer CHT baseline mesh cross-section at the throat (XZ plane). Inflation layers (1×10^{-6} m first cell, 15 layers) visible at the wall.

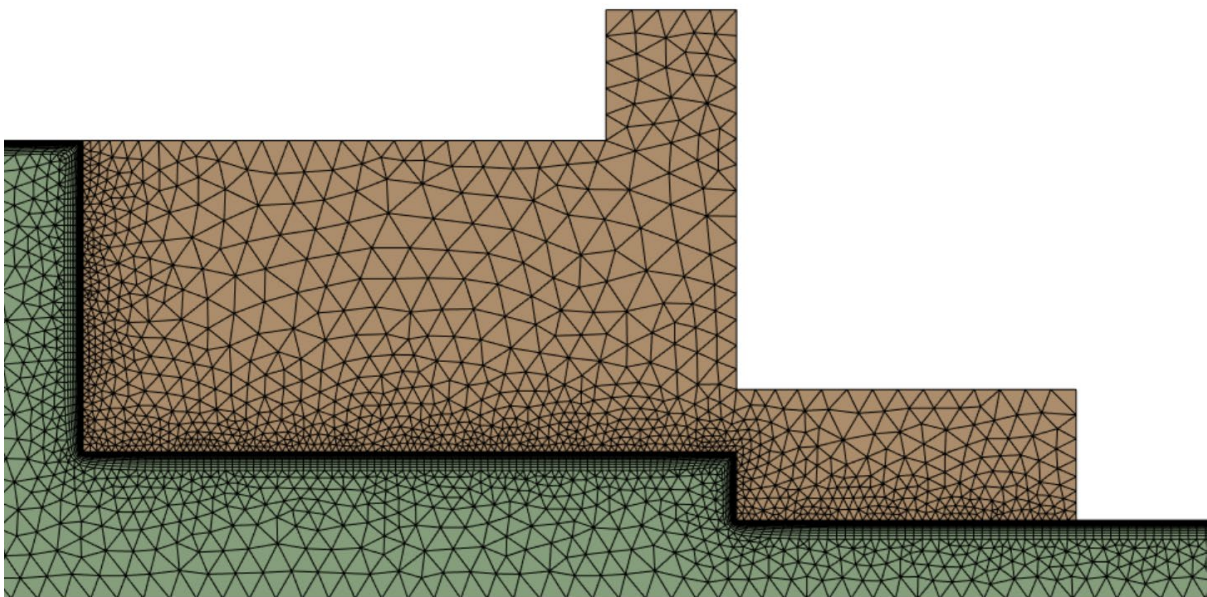


Figure 3.11. Legacy reducer CHT baseline mesh cross-section at the first abrupt contraction step.

3.8.5 Near-Wall y^+ Performance

The non-dimensional wall distance $y^+ = \rho u_\tau y_1 / \mu$, where y_1 is the distance from the wall to the first mesh node and u_τ is the friction velocity, characterises the placement of the near-wall mesh relative to the boundary layer structure. For the SST $k-\omega$ model with automatic wall treatment, a maximum $y^+ \leq 5$ is required for direct viscous sublayer resolution; $y^+ \leq 1$ is recommended for highest-accuracy wall heat flux prediction.

The CHT mesh (first cell height 1×10^{-6} m, 15 layers) achieves maximum wall y^+ values within the viscous sublayer for both geometries across all three pressure cases (Table 3.3). The y^+ increases with falling outlet pressure (increasing mass flow rate and wall shear stress), reaching maxima of 2.990 (legacy, Case C) and 2.364 (redesigned, Case C). All cases remain below $y^+ = 5$, confirming full viscous sublayer resolution throughout the CHT campaign without reliance on wall-function bridging.

Note: an earlier version of the legacy CHT mesh used a coarser first cell height of 5×10^{-5} m, which produced $y^+ \approx 8-10$ in the vena contracta region and an asymmetry relative to the redesigned reducer mesh. This mesh was discarded and a corrected mesh with 1×10^{-6} m first cell height was generated and used for all results reported in Chapter 4. The corrected mesh is the source of all legacy CHT data in this thesis.



Figure 3.12. Redesigned reducer CHT baseline mesh, symmetry-plane cross-section. Top: inlet extension (420 mm); Bottom: outlet extension (200 mm). Flow direction left to right.



Figure 3.13. Legacy reducer CHT baseline mesh, symmetry-plane cross-section. Top: inlet fluid extension (200 mm); Bottom: reducer solid body and outlet extension (138 mm). Flow direction left to right.

3.9 Mesh Sensitivity Assessment

3.9.1 Rationale and Validation Strategy

The mesh sensitivity study (Study 1) was conducted using the adiabatic CFD configuration (no solid domain) because this isolates the fluid-domain discretisation which governs pressure loss, velocity distribution, and Mach number from the solid-domain meshing variable. This CFD-based approach is justified by the thermal resistance argument: as demonstrated by the near-invariance of Q_{ext} with pressure condition in the CHT results (Section 4.5), the dominant resistance to heat transfer is the external natural-convection boundary, not the internal fluid-side convection. Consequently, the wall heat flux is relatively insensitive to the fine details of the internal flow field, making CFD-based mesh sensitivity an adequate proxy for CHT mesh adequacy.

3.9.2 Mesh Refinement Strategy

Three mesh levels were generated for each geometry: Coarse, Baseline, and Fine. Refinement ratios of approximately 2–3× in total cell count were applied between adjacent levels. The inflation layer settings (first cell height 5×10^{-5} m, 10 layers, growth ratio 1.2) were held constant across all refinement levels to isolate the effect of core-region refinement. Table 3.4 summarises the Study 1 mesh levels.

Geometry	Mesh level	Total fluid elements	y^+_{max} (Case C)	Min orth. quality avg	Max skew. avg
Legacy	Coarse	272,146	10.60	0.788	0.211
Legacy	Baseline	829,621	10.52	0.797	0.202
Legacy	Fine	1,570,929	11.21	0.798	0.201
Redesigned	Coarse	404,324	10.54	0.841	0.157
Redesigned	Baseline	913,301	10.60	0.833	0.166
Redesigned	Fine	2,115,646	10.57	0.834	0.165
Zero elements failed any ANSYS error threshold at any mesh level for either geometry. Quality metrics shown are averages; worst-case values are reported in Appendix B. Inflation: first cell height 5×10^{-5} m, 10 layers, GR 1.2.					

Table 3.4. Study 1 (adiabatic CFD) mesh levels and average quality metrics. All meshes satisfy ANSYS error thresholds with zero failed elements.

The legacy fine mesh y^+_{max} of 11.21 is marginally higher than the baseline value of 10.52, an anomaly attributable to local mesh topology differences at the abrupt step face; all mesh independence acceptance criteria are nonetheless satisfied.

3.9.3 Acceptance Criterion

Mesh independence was declared when all four monitored KPIs mass-weighted average outlet temperature T_{out} , maximum Mach number Ma_{max} , total pressure drop ΔP_0 , and maximum velocity magnitude V_{max} changed by less than 2% between the baseline and fine mesh levels. This criterion is consistent with published CFD best-practice guidelines for engineering comparative studies (ASME, 2009).

3.9.4 Sensitivity Results Summary

All four KPIs changed by less than 2% between consecutive refinement levels for both geometries. Full tabulated results are presented in Section 4.2 and Appendix B. A physically notable observation from the sensitivity data is the consistent flow regime divergence between the two geometries at Case C: the redesigned reducer converges to $Ma_{\text{max}} = 0.989\text{--}0.990$ across all three mesh levels (near-choked), while the legacy reducer shows $Ma_{\text{max}} = 1.010\text{--}1.016$ (supersonic vena contracta pocket). This

qualitative divergence is consistent and mesh-independent, confirming it is a genuine physical consequence of the geometric profile difference. The full CHT analysis and physical interpretation are developed in Sections 4.2.4 and 4.7.

3.10 Solver Settings and Numerical Discretisation

3.10.1 Solver Type and Formulation

All simulations used the pressure-based coupled solver in ANSYS Fluent 2025 R1. The coupled solver solves pressure and velocity equations simultaneously within each iteration using a coupled algebraic multigrid approach, providing superior convergence compared to segregated algorithms for compressible flows with strong pressure–velocity coupling, particularly at the elevated Mach numbers encountered in the reducer throat. The pressure-based formulation was preferred over the density-based solver because inlet and outlet conditions are specified as pressures, and because the pressure-based coupled solver handles the subsonic-to-transonic transition robustly. Pseudo-transient time-stepping with Conservative time-step specification was activated to provide additional stability during the initial high-residual iterations.

3.10.2 Spatial Discretisation Schemes

Second-order upwind schemes were applied for spatial discretisation of momentum and energy. First-order upwind schemes were applied for turbulence kinetic energy k and specific dissipation rate ω , consistent with standard practice for SST k – ω in compressible flows. Second-order pressure interpolation and least-squares cell-based gradient reconstruction were used throughout.

3.10.3 Initialisation Strategy

For Study 1, hybrid initialisation was used at the start of each simulation. For Study 2, Case A of each geometry was initialised using hybrid initialisation; Cases B and C used the cascade approach described in Section 3.4.2. The cascade strategy substantially reduces the number of iterations required for convergence because the CHT solid domain reaches thermal equilibrium slowly; starting from the already-converged preceding solution places the solid at approximately the correct temperature for the new operating point.

3.10.4 Convergence Monitoring

Convergence was monitored through two complementary approaches applied simultaneously. Scaled residuals for continuity, x-, y-, z-momentum, energy, k , and ω were tracked for a reduction of at least three orders of magnitude from their initial values. Physical convergence monitors detailed in Section 3.10.5 were required to satisfy stability criteria before any solution was accepted. Residuals alone are insufficient for CHT simulations because the energy residual can appear converged while the solid temperature field is still evolving slowly; the thermal monitors provide the definitive criterion.

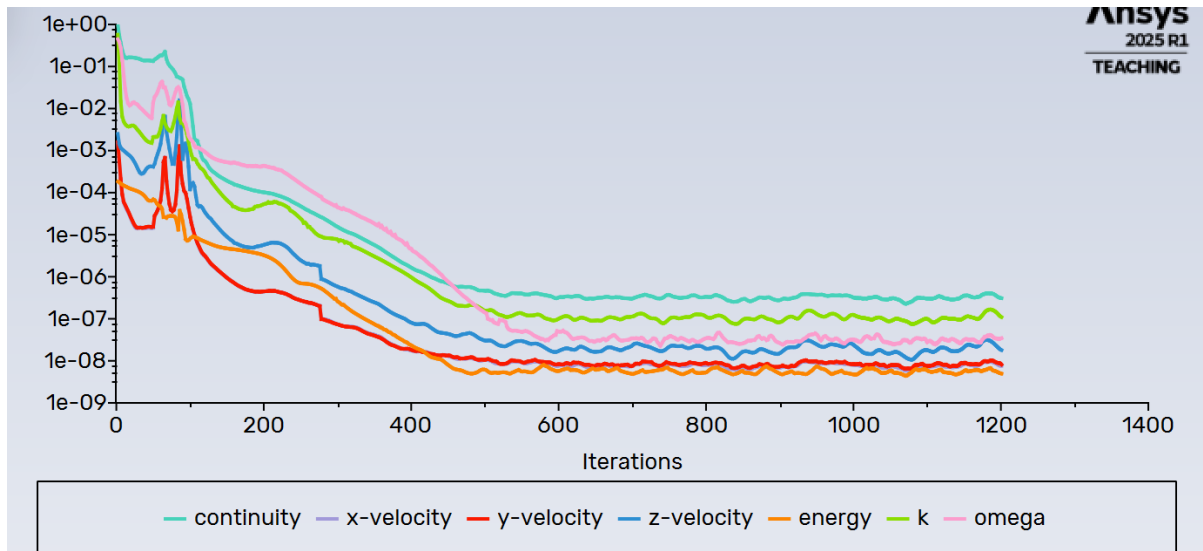


Figure 3.14. Residual convergence history for a representative CHT case (redesigned reducer, Case A, 80 kPa gauge outlet) — continuity, momentum, energy, k , ω

3.10.5 Convergence Criteria for CHT Simulations

For Study 2 CHT simulations, convergence was declared only when all four of the following conditions were satisfied simultaneously over a rolling window of 200 consecutive iterations:

- (a) Net mass flow rate imbalance $< 1 \times 10^{-5}$ kg/s
- (b) Mass-weighted average outlet temperature change $< 1 \times 10^{-4}$ K per iteration
- (c) Area-weighted average external wall temperature change $< 1 \times 10^{-4}$ K per iteration
- (d) Net total pressure drop change $< 5 \times 10^{-4}$ Pa per iteration

Active	Conditions	Report Definition	Stop Criterion	Ignore Iterations Before	Use Iterations
☒	Net pressure drop stabilisation	net-pressure-drop	0.0005	200	200
☒	External wall temperature stabilisation	external-wall-temperature	0.0001	200	200
☒	mass flow rate balance	net-mass-flow-rate	1e-05	200	200
☒	outlet temperature stabilisation	outlet-temperature	0.0001	200	200

Figure 3.15. Physical monitor convergence history for the same case as Fig. 3.14 — outlet temperature, external wall temperature, net pressure drop, and mass flow rate

The 200-iteration rolling window was adopted because shorter windows (50 iterations) risked declaring premature convergence during slowly-declining thermal drift, while 200 iterations provides sufficient statistical averaging without unnecessary prolongation. The temperature criterion of 1×10^{-4} K represents approximately 0.003% of the inlet temperature (673 K), an appropriately stringent criterion for a comparative study in which temperature differences of 3–8 K are the primary output quantities.

3.11 Post-Processing and Output Quantities

3.11.1 Primary Performance Indicators

All post-processing was performed using ANSYS Fluent's built-in surface and volume integral functions. Identical reporting definitions were applied to both geometries to ensure all comparisons are free of methodological artefacts.

Outlet temperature T_{out} : Mass-weighted average static temperature at the outlet boundary face, computed as $\int \rho u T dA / \int \rho u dA$. This is the most physically appropriate measure of thermal delivery for compressible flow.

Total pressure drop ΔP_0 : Difference between area-weighted average total pressure at the inlet and outlet faces: $\Delta P_0 = P_{0,inlet} - P_{0,outlet}$.

Maximum Mach number Ma_{max} : Volume maximum over all fluid cells, identifying the most severe compressibility condition in the domain.

External heat loss Q_{ext} : Surface integral of total wall heat flux over all faces carrying the external convection boundary condition ($h = 10 \text{ W/m}^2\text{K}$). Computed in the quarter model and multiplied by four for full-geometry values.

External wall temperature $T_{wall,ext}$: Area-weighted average of wall temperature over all outer solid surface faces carrying the convection boundary condition.

3.11.2 Quarter-Symmetry Correction

Quarter-symmetry was used for all simulations. Full-geometry corrections were applied to external surface area A_{ext} and total external heat loss Q_{ext} by multiplying quarter-model values by four. Intensive quantities, including T_{out} , $T_{wall,ext}$, ΔP_0 , and Ma_{max} , require no correction. Mass flow rates reported in Chapter 4 are retained as solver outputs from the quarter model unless otherwise stated. The external convective area was verified by surface integration and quarter-to-full scaling, yielding $5,758.15 \text{ mm}^2$ for the legacy reducer and $19,282.45 \text{ mm}^2$ for the redesigned reducer. These values are cross-validated against SpaceClaim measurements to within 0.01%.

3.11.3 Derived Quantities

Thermal efficiency: $\eta = (T_{out} - T_{amb}) / (T_{in} - T_{amb}) \times 100\% = (T_{out} - 300) / (673 - 300) \times 100\%$. This provides a dimensionless measure of thermal delivery quality robust to variations in mass flow rate between cases.

Normalised external heat flux density: $q_{ext} = Q_{ext} / A_{ext}$. Since $q_{ext} = h \times (T_{wall,ext} - T_{amb}) = 10 \times (T_{wall,ext} - 300) \text{ W/m}^2$, this metric is directly proportional to the area-averaged outer wall temperature, providing a geometry-independent per-area comparison independent of the external area difference between the two reducers.

3.12 Methodological Limitations

The following limitations bound the interpretation of results presented in Chapter 4 and are discussed in full detail in Chapter 5. They are summarised here for reference.

Steady-state assumption: The simulations represent fully stabilised equilibrium conditions. Dynamic pressure oscillations from the gear-wheel mechanism and transient heat-up behaviour are not captured.

Uniform external convective coefficient: The value $h_{\text{ext}} = 10 \text{ W/m}^2\text{K}$ was applied spatially uniformly. Spatial variation of $\pm 30\text{--}40\%$ is expected in practice. Since the same assumption applies to both geometries, comparative conclusions are unaffected.

Neglect of thermal radiation: At outer wall temperatures of $577\text{--}630 \text{ K}$, radiative flux is approximately $5,000\text{--}7,000 \text{ W/m}^2$, substantially exceeding the convective flux. This is the most significant absolute limitation. See Section 5.5.2 for a full assessment.

CFD-based mesh sensitivity: Independence was demonstrated for adiabatic CFD (Study 1) rather than full CHT (Study 2). This is mitigated by the external resistance dominance finding. See Section 5.5 for discussion.

Absence of experimental validation: The redesigned reducer has not been manufactured. All results are comparative numerical evidence under identical modelling assumptions, not absolute validated predictions. See Section 6.4.1 for the experimental validation recommendation.

4. Results

4.1 Overview

The results of this investigation are presented in two sequential studies. Study 1 establishes mesh independence for both reducer geometries using steady-state adiabatic compressible flow simulations at the most demanding operating condition (Case C, outlet gauge pressure 0 Pa). Study 2 presents the conjugate heat transfer (CHT) results for all three pressure cases, complemented by insulated reference simulations designed to isolate the contribution of external heat loss from the intrinsic aerodynamic performance of each geometry. The geometry and mesh configuration that underpin both studies are described in the following section.

All simulations were performed in ANSYS Fluent 2025 R1 using a steady-state, pressure-based, compressible solver with the SST $k-\omega$ turbulence model and second-order spatial discretisation. The computational domain employs quarter-symmetry, exploiting two orthogonal planes of geometric symmetry (XZ and YZ) to reduce the mesh size by a factor of four relative to the full geometry. An operating pressure of 101,325 Pa is applied throughout, and all gauge pressures reported herein are referenced to this value.

4.2 Mesh Sensitivity Assessment

4.2.1 Simulation Configuration and Acceptance Criterion

Mesh sensitivity was assessed using steady-state adiabatic simulations (Study 1) without a solid domain, isolating the fluid-dynamic response of each geometry from the thermal field. All Study 1 runs were performed at Case C conditions (inlet total pressure 100,000 Pa gauge, inlet total temperature 673 K, outlet gauge pressure 0 Pa), corresponding to an absolute pressure ratio of $PR = 1.987$, which exceeds the critical pressure ratio $PR_{crit} = 1.893$ for air ($\gamma = 1.4$). This represents the most demanding operating condition in terms of compressibility effects and near-wall velocity gradients, and therefore provides the most conservative basis for mesh independence assessment.

Three mesh refinement levels coarse, baseline, and fine were generated for each geometry using ANSYS Meshing 2025 R1 with a Patch Conforming tetrahedral algorithm supplemented by prismatic inflation layers at all wall surfaces. The inflation layer parameters were held constant across refinement levels at a first cell height of 5×10^{-5} m, ten layers, and a growth ratio of 1.2, yielding a maximum wall y^+ of approximately 10.5–11.2 across all mesh levels and both geometries. Four primary key performance indicators (KPIs) were monitored: mass-weighted average outlet static temperature T_{out} , maximum Mach number Ma_{max} by volume integral, total pressure drop ΔP_0 , and maximum velocity magnitude V_{max} . Mesh independence was accepted when the relative change in all four KPIs between consecutive refinement levels remained below the 2% threshold.

4.2.2 Legacy Reducer: Mesh Sensitivity Results

The legacy reducer fluid domain comprises a 200 mm adiabatic inlet extension (35.05 mm bore), the 38 mm reducer body containing two abrupt Borda–Carnot contraction steps ($35.05 \rightarrow 11.25 \rightarrow 6.0$ mm), and a 100 mm adiabatic outlet extension, giving a

total domain length of 338 mm. Three mesh levels were generated with 272,146 (coarse), 829,621 (baseline), and 1,570,929 (fine) elements respectively. Table 4.1 summarises the mesh quality metrics for all three levels.

Quality metric	Coarse average	Coarse worst	Baseline average	Baseline worst	Fine average	Fine worst	Limit
Min orthogonal quality	0.788	0.156	0.797	0.100	0.798	0.124	> 0.05
Max skewness	0.211	0.844	0.202	0.900	0.201	0.876	< 0.999
Max aspect ratio	6.128	44.79	3.677	30.93	3.065	20.32	< 1000
Min element quality	0.517	0.040	0.664	0.053	0.698	0.090	> 5×10^{-4}
Zero elements failed the ANSYS error threshold across all quality metrics and all refinement levels.							

Table 4.1. Mesh quality metrics for the legacy reducer at three refinement levels. Limits correspond to ANSYS Meshing recommended quality targets. Zero cells failed the ANSYS error threshold at any level; worst-case values represent isolated single cells.

All three mesh levels satisfy the ANSYS error thresholds with zero failed elements. The maximum worst-case aspect ratio of 44.79 (coarse mesh) is confined to the prismatic inflation layer cells immediately adjacent to the wall, where elongated cell geometry is inherent to and required for accurate boundary layer resolution; the average aspect ratio of 6.128 confirms that high-AR cells represent a small fraction of the total mesh volume. The maximum worst-case skewness of 0.900 (legacy baseline mesh) lies at the stated recommended limit; this worst-case cell is isolated at the abrupt step geometry and no systematic quality degradation is observed with is observed.

Table 4.2 presents the four KPIs across the three refinement levels at Case C conditions.

KPI	Coarse	Baseline	Fine	% change B $\rightarrow$ F
T_{out} , mass-weighted avg (K)	587.96	587.31	586.79	-0.089
Ma_{max} , volume max (-)	1.0099	1.0148	1.0164	+0.160
ΔP_0 , total pressure drop (Pa)	36,214	36,285	35,797	-1.344
V_{max} , velocity magnitude (m/s)	480.85	482.60	483.17	+0.118
y^+_{max}	10.60	10.52	11.21	-
Case C conditions: inlet 100,000 Pa gauge, 673 K total temperature; outlet 0 Pa gauge. All % changes are baseline $\rightarrow$ fine.				

Table 4.2. Mesh sensitivity KPIs for the legacy reducer at Case C (PR = 1.987). All four KPIs satisfy the 2% acceptance criterion between the baseline and fine mesh levels.

All four KPIs remain within the 2% acceptance criterion between the baseline and fine mesh levels: the largest change is -1.344% in total pressure drop, while outlet temperature, Mach number, and velocity magnitude change by less than 0.2%. The coarse-to-baseline transitions are also within tolerance, with changes of -0.112% (T_{out}), +0.478% (Ma), +0.196% (ΔP_0), and +0.364% (V_{max}). Mesh independence is therefore confirmed for the legacy reducer, and the baseline mesh of 829,621 elements is adopted as the production mesh for Study 2.

A notable aerodynamic observation from Study 1 is that the maximum Mach number exceeds unity across all three mesh levels ($Ma_{\max} = 1.010\text{--}1.016$), despite the overall pressure ratio $PR = 1.987$ exceeding the critical value. This apparent contradiction is explained by the Borda–Carnot mechanism at the abrupt contraction steps, which degrades total pressure at the effective throat below the level required to sustain choked flow at the geometric throat. The localised supersonic region constitutes a vena contracta pocket downstream of the first abrupt step rather than a classical choked nozzle; this finding is discussed further in Section 4.2.4.

4.2.3 Redesigned Reducer: Mesh Sensitivity Results

The redesigned reducer fluid domain comprises a 420 mm adiabatic inlet extension (35.05 mm bore), the 150 mm reducer body with a quintic C^2 -continuous contraction profile (35.05 $\rightarrow$ 6.0 mm bore), and a 200 mm adiabatic outlet extension, giving a total domain length of 770 mm. The longer inlet extension relative to the legacy domain reflects the need to establish a stable inlet boundary condition upstream of the smooth contraction. Three mesh levels were generated with 404,324 (coarse), 913,301 (baseline), and 2,115,646 (fine) elements respectively. Table 4.3 summarises the mesh quality metrics.

Quality metric	Coarse average	Coarse worst	Baseline average	Baseline worst	Fine average	Fine worst	Limit
Min orthogonal quality	0.841	0.188	0.833	0.032	0.834	0.136	> 0.05
Max skewness	0.157	0.812	0.166	0.846	0.165	0.864	< 0.999
Max aspect ratio	7.918	86.77	5.507	61.24	3.875	41.06	< 1000
Min element quality	0.460	0.022	0.524	0.032	0.603	0.046	$> 5 \times 10^{-4}$
Zero elements failed the ANSYS error threshold across all quality metrics and all refinement levels.							

Table 4.3. Mesh quality metrics for the redesigned reducer at three refinement levels. Limits correspond to ANSYS Meshing recommended quality targets. Zero cells failed the ANSYS error threshold at any level; worst-case values represent isolated single cells.

As with the legacy reducer, all three mesh levels of the redesigned reducer satisfy the ANSYS error thresholds with zero failed elements. The worst-case aspect ratio decreases monotonically from 86.77 (coarse) to 41.06 (fine), reflecting the progressive refinement of the inflation layer cells. The elevated worst-case aspect ratios relative to the legacy reducer are attributable to the longer contraction body and correspondingly larger inflation layer surface area, and are physically expected and acceptable for wall-resolved boundary layer simulations. Average aspect ratios of 7.918, 5.507, and 3.875 for the three mesh levels confirm that high-AR cells are localised to the near-wall region.

Table 4.4 presents the four KPIs across the three refinement levels.

KPI	Coarse	Baseline	Fine	% change B $\rightarrow$ F
T_{0ut} , mass-weighted avg (K)	590.22	590.33	590.48	+0.025
$Ma_{\max}$, volume max (–)	0.9888	0.9903	0.9887	–0.165
ΔP_0 , total pressure drop (Pa)	38,810	39,146	39,387	+0.614

$V_{\max}$, velocity magnitude (m/s)	473.43	474.07	473.45	-0.129
$y^+_{\max}$	10.54	10.60	10.57	-
<i>Case C conditions: inlet 100,000 Pa gauge, 673 K total temperature; outlet 0 Pa gauge. All % changes are baseline → fine.</i>				

Table 4.4. Mesh sensitivity KPIs for the redesigned reducer at Case C (PR = 1.987). All four KPIs satisfy the 2% acceptance criterion between the baseline and fine mesh levels.

All four KPIs satisfy the 2% acceptance criterion between the baseline and fine mesh levels. The largest change is +0.614% in total pressure drop; outlet temperature, Mach number, and velocity magnitude vary by less than 0.17%. The coarse-to-baseline transitions are equally well-behaved, with changes of +0.018% (T_{out}), +0.151% (Ma), +0.867% (ΔP_0), and +0.135% ($V_{\max}$). Mesh independence is confirmed for the redesigned reducer, and the baseline mesh of 913,301 elements is adopted as the production mesh for Study 2.

The maximum Mach number remains stable across all three refinement levels at $Ma_{\max} = 0.989\text{--}0.990$, with a variation of less than 0.1% between coarse, baseline, and fine meshes. This consistency across mesh levels confirms that the near-choked result is a genuine physical outcome and not a mesh-dependent artefact. The physics and implications of this finding are discussed in Section 4.2.4 and Section 4.6.

4.2.4 Preliminary Flow Regime Observation from Study 1

A qualitatively significant result is already apparent in the Study 1 mesh sensitivity data: at identical supercritical supply conditions ($PR = 1.987 > PR_{\text{crit}} = 1.893$), the two reducer geometries exhibit fundamentally different flow regimes, and this difference is consistent and stable across all three mesh refinement levels.

The redesigned reducer achieves a maximum Mach number of 0.989–0.990 across all mesh levels, indicating near-choked flow at the 6.0 mm geometric throat. The proximity to $Ma = 1.000$ is consistent with the smooth quintic contraction distributing viscous losses along the 150 mm contraction body rather than concentrating them at abrupt steps upstream of the throat. The 1% deviation from the isentropic prediction is attributable to viscous losses along the contraction wall (reflected in $\Delta P_0 = 37,810 \text{ Pa} \approx 18.8\%$ of inlet P_0), which reduce the effective stagnation pressure available at the throat relative to the isentropic idealisation.

The legacy reducer, by contrast, yields $Ma_{\max} = 1.010\text{--}1.016$ across all mesh levels. This result is counterintuitive: the maximum Mach number exceeds unity despite the flow not being choked at the geometric throat in the classical sense. The explanation lies in the Borda–Carnot mechanism at the two abrupt contraction steps. The abrupt area reductions generate separated flow with a vena contracta region, within which the local jet achieves supersonic velocity before recompressing via a normal shock. The total pressure losses at these steps characterised by a contraction coefficient

$C_c \approx 0.62$ and loss coefficient $K = (1/C_c - 1)^2 \approx 0.376$ (White, 2011) degrade the total pressure available at the effective throat, preventing the sustained choked flow that would otherwise be expected at $PR = 1.987$. The supersonic pocket is a transient overexpansion at the vena contracta, not a choked converging–diverging nozzle flow.

Table 4.5 summarises the flow regime observation across all mesh levels for both geometries.

Geometry	Coarse Ma_{max}	Baseline Ma_{max}	Fine Ma_{max}	Flow regime
Redesigned reducer	0.9888	0.9903	0.9887	Near-choked at geometric throat
Legacy reducer	1.0099	1.0148	1.0164	Supersonic vena contracta pocket; subsonic at geometric throat
<i>Both geometries at identical Case C conditions: $PR = 1.987 > PR_{mit} = 1.893$.</i>				

Table 4.5. Maximum Mach number across all Study 1 mesh refinement levels at Case C ($PR = 1.987$) for both reducer geometries. The qualitative flow regime difference is consistent across all mesh levels.

The qualitative flow regime divergence near-choked continuous flow versus supersonic vena contracta pocket is invariant across all three mesh refinement levels for both geometries. This confirms that the result is a physical consequence of the geometric profile rather than a numerical artefact, and establishes the central finding of Study 2 in anticipation: the redesigned quintic reducer eliminates the localised supersonic vena contracta pocket that characterises the legacy geometry under supercritical supply conditions. The full CHT analysis and physical interpretation of this choking asymmetry are presented in Sections 4.6 and 4.7.

4.3 Conjugate Heat Transfer Simulation Overview

4.3.1 Simulation Configuration and Convergence

The conjugate heat transfer campaign (Study 2) comprised six steady-state CHT simulations: three outlet pressure cases for each reducer geometry. Each CHT simulation extends the Study 1 baseline mesh by adding the solid reducer body as a thermally coupled domain, enabling direct heat conduction through the steel wall without empirical internal heat transfer coefficients. The fluid and solid domains are coupled through the *reducer_fluid/reducer_fluid-shadow* interface for both geometries, with an additional *annular_face/annular_face-shadow* interface in the legacy reducer to represent the thermally coupled annular step face at the first abrupt contraction. The external solid surface (*reducer_solid*) is assigned a convective boundary condition with heat transfer coefficient $h = 10 \text{ W/m}^2\text{K}$ and ambient temperature $T_\infty = 300 \text{ K}$, representing natural convection to the laboratory environment.

Cascade initialisation was employed across pressure cases: the converged Case A solution (outlet gauge pressure 80,000 Pa) provided the initial field for Case B (40,000 Pa), and the converged Case B solution for Case C (0 Pa). This approach substantially reduced the number of iterations required for convergence of Cases B and C. The CHT mesh uses a corrected inflation layer configuration of first cell height $1 \times 10^{-6} \text{ m}$, 15 layers, and growth ratio 1.2 for both geometries, yielding maximum wall y^+ values of 1.49–2.36 (redesigned reducer) and 1.86–2.99 (legacy reducer), ensuring full viscous sublayer resolution for both geometries under all pressure conditions.

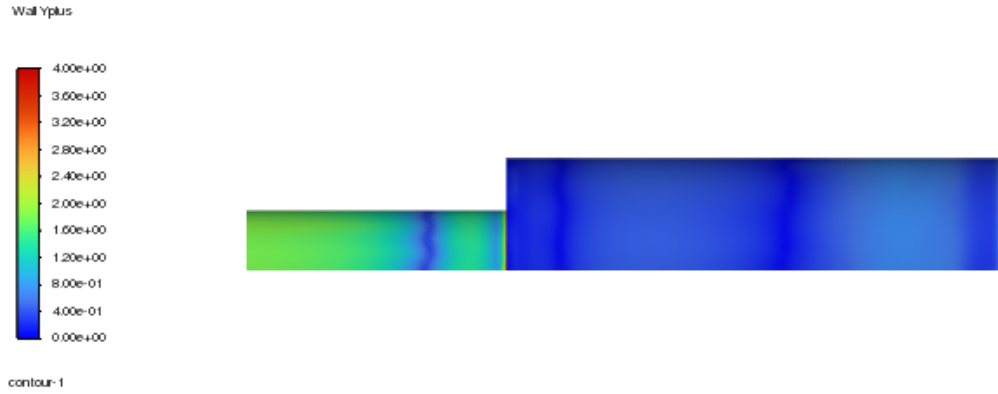


Figure 4.1. Legacy Reducer y^+

Convergence was assessed using four simultaneously monitored report definitions: net mass flow rate imbalance (tolerance 1×10^{-5} kg/s), mass-weighted average outlet temperature (1×10^{-4} K), area-weighted average external wall temperature (1×10^{-4} K), and net total pressure drop (5×10^{-4} Pa). All four criteria were required to be simultaneously satisfied over a rolling 200-iteration window before a solution was accepted. Particular weight was given to the thermal monitors, as outlet temperature and wall temperature converge more slowly than flow residuals in CHT simulations. Mass imbalances below 1×10^{-9} kg/s were achieved for all six cases, confirming excellent numerical mass conservation.

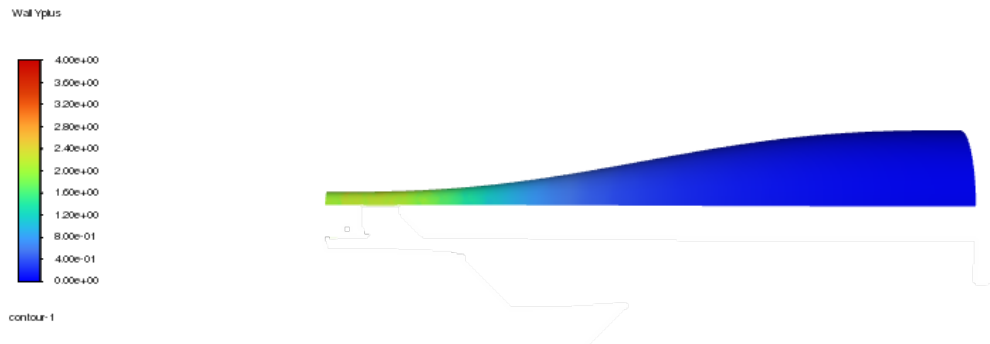


Figure 4.2 Redesigned Reducer y^+

4.3.2 Consolidated CHT Results

Table 4.6 presents the consolidated KPI dataset for all six CHT cases. The absolute pressure ratios $PR = P_{0,in}/P_{outlet}$ are 1.110 (Case A), 1.424 (Case B), and 1.987 (Case C), with the isentropic critical pressure ratio $PR_{crit} = 1.893$ for air ($\gamma = 1.4$). Cases A and B are therefore subcritical; Case C is supercritical and choked or near-choked flow is expected at the geometric throat.

<i>KPI</i>	<i>N-A</i>	<i>N-B</i>	<i>N-C</i>	<i>L-A</i>	<i>L-B</i>	<i>L-C</i>	<i>Unit</i>
T _{out} , MWA	649.49	624.59	582.10	657.29	629.34	585.45	K

η thermal efficiency	93.70	87.02	75.63	95.79	88.29	76.53	%
PR absolute	1.110	1.424	1.987	1.110	1.424	1.987	-
ΔP_0 (Pa)	9,114	25,319	37,810	8,777	24,752	36,720	Pa
Ma_max	0.352	0.674	0.990	0.412	0.767	1.006†	-
Q_ext, full geometry (W)	53.36	56.58	57.21	18.59	18.93	18.88	W
q_ext, per unit area (W/m ²)	2,768	2,935	2,967	3,230	3,288	3,279	W/m ²
T_wall,ext, AWA (K)	576.78	593.46	596.73	622.96	628.77	627.91	K
$\dot{m}$ (kg/s)	9.91×10^{-4}	1.547×10^{-3}	1.728×10^{-3}	1.003×10^{-3}	1.557×10^{-3}	1.740×10^{-3}	kg/s
y ⁺ _max	1.49	2.15	2.36	1.86	2.69	2.99	-
$\Delta \dot{m}$ mass imbalance	1.2×10^{-10}	7.0×10^{-10}	3.3×10^{-10}	9.0×10^{-12}	7.1×10^{-11}	2.1×10^{-11}	kg/s
<i>N = redesigned reducer, L = legacy reducer. A = 80,000 Pa gauge, B = 40,000 Pa gauge, C = 0 Pa gauge. MWA = mass-weighted average, AWA = area-weighted average. † Supersonic vena contracta pocket flow not choked at geometric throat.</i>							

Table 4.6. Consolidated CHT results for all six pressure cases and both reducer geometries.

*Q_ext is the full-geometry heat loss (quarter model value $\times$ 4). $\eta = (T_{out} - 300)/(673 - 300) \times 100\%$.

4.4 Downstream Thermal Delivery

4.4.1 Mass-Weighted Outlet Temperature

The mass-weighted average outlet static temperature for the redesigned reducer decreases monotonically from 649.49 K at Case A to 624.59 K at Case B and 582.10 K at Case C, representing outlet static-temperature deficits relative to the 673 K inlet total temperature of 23.51 K, 48.41 K, and 90.90 K respectively. Most of this deficit is associated with compressible acceleration and the distinction between total and static temperature; the insulated reference simulations show that removal of external heat loss recovers only 6.84 K for the redesigned reducer and 2.25 K for the legacy reducer at Case C. The legacy reducer consistently delivers higher outlet temperatures across all three pressure cases: 657.29 K (Case A), 629.34 K (Case B), and 585.45 K (Case

C), with advantages of 7.80 K, 4.75 K, and 3.36 K relative to the redesigned reducer. The progressive decline in outlet temperature with falling back-pressure reflects the increasing mass flow rate: as back-pressure decreases from Case A to Case C, mass flow increases from 9.91×10^{-4} to 1.728×10^{-3} kg/s for the redesigned reducer and from 1.003×10^{-3} to 1.740×10^{-3} kg/s for the legacy reducer, distributing the same absolute heat loss across a larger mass flux.

The thermal efficiency, defined as $\eta = (T_{\text{out}} - T_{\text{amb}})/(T_{\text{in}} - T_{\text{amb}}) \times 100\%$ with $T_{\text{amb}} = 300$ K and $T_{\text{in}} = 673$ K, ranges from 93.70% (redesigned, Case A) to 75.63% (redesigned, Case C) and from 95.79% (legacy, Case A) to 76.53% (legacy, Case C). The legacy reducer maintains a thermal efficiency advantage of 2.09%, 1.27%, and 0.92% at Cases A, B, and C respectively. The physical mechanism responsible for this unexpected result — unexpected because the redesigned reducer eliminates impingement heating — is identified as external resistance dominance and is examined in Section 4.5.

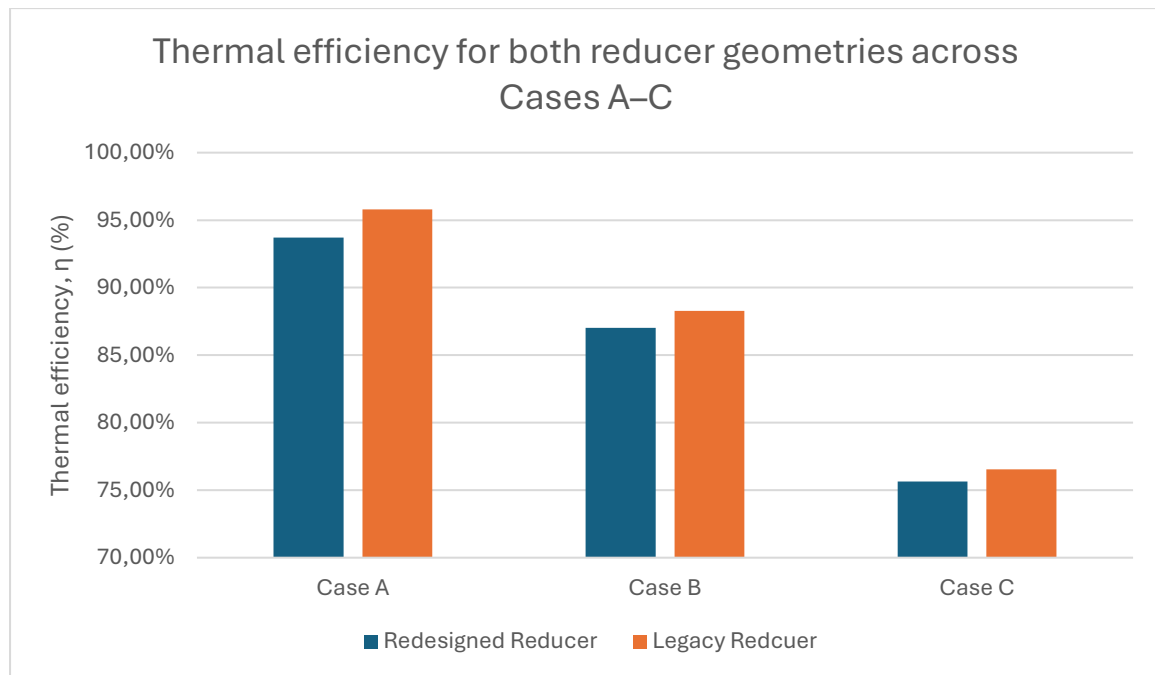


Figure 4.3. Thermal efficiency for both reducer geometries across Cases A–C

4.5 External Heat Loss and Normalised Heat Flux

4.5.1 Absolute External Heat Loss

The full-geometry external heat loss Q_{ext} (quarter model value $\times 4$) is 53.36 W, 56.58 W, and 57.21 W for the redesigned reducer at Cases A, B, and C respectively, compared to 18.59 W, 18.93 W, and 18.88 W for the legacy reducer. The redesigned reducer therefore loses 2.87 $\times$, 2.99 $\times$, and 3.03 $\times$ more heat to ambient than the legacy reducer across the three pressure cases. This disparity is a direct consequence of the external area ratio: the redesigned reducer exposes $A_{\text{ext}} = 19,282$ mm² of surface to ambient, compared to 5,758 mm² for the legacy reducer a ratio of 3.35. The external thermal resistance $R_{\text{ext}} = 1/(h \cdot A_{\text{ext}})$ is therefore 5.19 K/W for the redesigned reducer and 17.37 K/W for the legacy reducer.



Figure 4.4. Redesigned Reducer Total Heat Flux

The CFD results are consistent with the imposed external convection model. For the redesigned reducer at Case A, the analytical estimate $Q_{ext} \approx hA_{ext}(T_{wall,ext} - T_{amb}) = (576.78 - 300)/5.19 = 53.3\text{W}$ agrees closely with the Fluent result of 53.36 W. For the legacy reducer, the corresponding estimate $(622.96 - 300)/17.37 = 18.6\text{W}$ matches the Fluent value of 18.59 W. This agreement is best interpreted as an internal consistency check, since the same convection model is prescribed in the CFD boundary condition. Stronger support for external-resistance dominance is provided by the small Biot numbers of the solid bodies, $Bi = hL_c/\lambda_s$, which are approximately 0.003 for the legacy reducer and 0.004 for the redesigned reducer. These values indicate that internal conduction resistance within the steel is negligible relative to the external convective resistance, and that the total heat loss is therefore governed primarily by convection from the outer surface.



Figure 4.5. Legacy Reducer Total Heat Flux

4.5.2 Near-Invariance of External Heat Loss with Pressure Condition

A notable feature of the CHT results is the near-invariance of Q_{ext} with pressure condition. For the redesigned reducer, Q_{ext} varies by only 7.2% across the full pressure range from Case A to Case C (53.36 to 57.21 W); for the legacy reducer the variation is just 2.2% (18.59 to 18.93 W). This insensitivity is a direct consequence of external resistance dominance: the total thermal resistance is governed by R_{ext} , which is independent of the internal flow conditions. Changes in mass flow rate and internal heat transfer coefficient across pressure cases alter the internal resistance R_{int} , but since $R_{int} \ll R_{ext}$ for both geometries, these changes produce negligible variation in Q_{ext} . The practical implication is that external insulation reducing h_{ext} toward zero is far more effective at improving gas delivery temperature than any internal geometry modification.

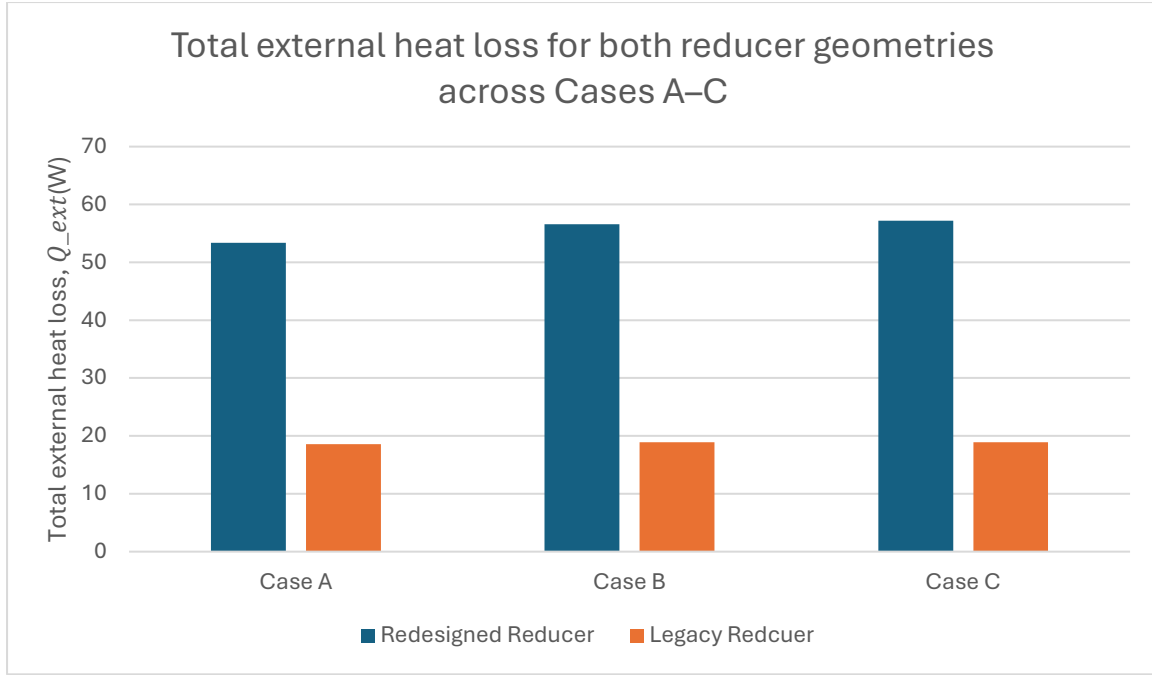


Figure 4.6. Total external heat loss for both reducer geometries across Cases A–C, showing weak dependence on pressure condition

4.5.3 Normalised External Heat Flux Density

The area-normalised external heat flux density $q_{\text{ext}} = Q_{\text{ext}}/A_{\text{ext}}$ provides a geometry-independent thermal performance metric that enables fair comparison between reducers of different external area. For the redesigned reducer, $q_{\text{ext}} = 2,768 \text{ W/m}^2$ (Case A), $2,935 \text{ W/m}^2$ (Case B), and $2,967 \text{ W/m}^2$ (Case C). For the legacy reducer, $q_{\text{ext}} = 3,230, 3,288, \text{ and } 3,279 \text{ W/m}^2$ at the respective cases. The redesigned reducer achieves a per-area heat flux reduction of 14.3%, 10.7%, and 9.5% across Cases A, B, and C. Internal consistency between the surface-integrated Q_{ext} and the area-weighted flux is within 0.01%, confirming post-processing reliability.

The per-area reduction confirms that the redesigned reducer is thermally superior on a normalised basis: the elimination of the stagnation-point impingement heat transfer zone at the $35.05 \rightarrow 11.25 \text{ mm}$ annular step, where, by analogy with stagnation-point impingement heat transfer (Martin, 1977), local Nusselt numbers are expected to be substantially higher than developed pipe flow values, reduces the wall heat flux density across the entire contraction body. However, this per-area advantage is overwhelmed in absolute terms by the $3.35\times$ area penalty of the longer 150 mm contraction body: the redesigned reducer loses 34–38 W more to ambient at each condition despite its lower heat flux per unit area. The external area is therefore identified as the primary thermal design parameter, not the internal flow profile.



Figure 4.7. Legacy Reducer Wall Temperature

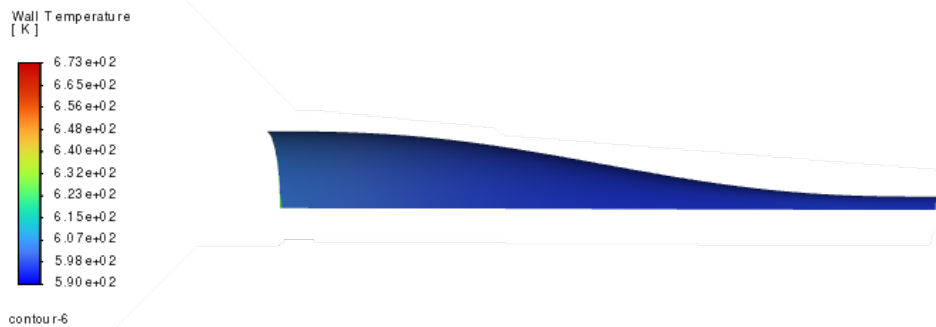


Figure 4.8. Redesigned Reducer Wall Temperature

The external wall temperature $T_{\text{wall,ext}}$ of the redesigned reducer (576.78–596.73 K) is 31–46 K lower than that of the legacy reducer (622.96–628.77 K) across all cases, a consequence of the lower R_{ext} driving a larger heat flux through a cooler wall. Although this lower outer wall temperature represents a reduction in direct thermal radiation from the reducer body a secondary benefit not captured by the convective CHT model the absolute thermal delivery to the downstream sensor is nonetheless lower for the redesigned reducer under uninsulated conditions.

4.6 Peak Mach Number and Flow Regime Classification

4.6.1 Subsonic Cases A and B

At Case A ($PR = 1.110$), the redesigned reducer reaches $Ma_{\text{max}} = 0.352$ at the outlet plane; the legacy reducer reaches 0.412. At Case B ($PR = 1.424$), Ma_{max} is 0.674 for the redesigned reducer and 0.767 for the legacy reducer. Both geometries operate in the fully subsonic regime at Cases A and B, with no localised regions of sonic or supersonic flow anywhere in the domain. The legacy reducer consistently achieves higher peak Mach numbers than the redesigned reducer at the same pressure ratio. This is attributable to the vena contracta effect at the abrupt contraction steps: the contracted jet reaches a higher local velocity at the vena contracta plane than the mean outlet velocity of the redesigned reducer, whose smooth contraction produces a spatially uniform velocity profile at the outlet plane.

4.6.2 Redesigned Reducer: Near-Choked Flow at Case C

At Case C ($PR = 1.987$, supercritical), the redesigned reducer achieves $Ma_{\text{max}} = 0.990$ at the outlet plane. This represents the most significant aerodynamic result of the CHT

campaign and is fully consistent with the Study 1 mesh sensitivity observation of $Ma = 0.989\text{--}0.990$ across all three refinement levels. The flow is near-choked: the outlet plane operates at 99.0% of the local speed of sound. The 1.0% shortfall from the isentropic prediction of $Ma = 1.000$ at supercritical PR is attributable to viscous losses along the 150 mm contraction wall (reflected in $\Delta P_0 = 37,810\text{ Pa} \approx 18.8\%$ of inlet P_0), which reduce the effective stagnation pressure available at the throat relative to the isentropic idealisation. The stability of this result across four mesh levels (three in Study 1, one in Study 2) confirms it is a genuine physical outcome.

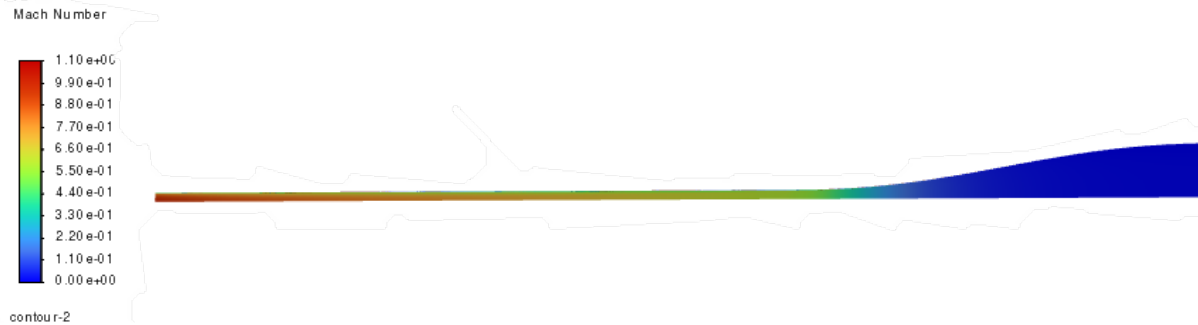


Figure 4.9 Redesigned Reducer Mach Number

4.6.3 Legacy Reducer: Supersonic Vena Contracta Pocket at Case C

The legacy reducer yields $Ma_{\max} = 1.006$ at Case C, indicating the presence of a localised supersonic region within the fluid domain. Despite the supply pressure ratio $PR = 1.987$ exceeding the critical value, the flow does not choke at the geometric throat. The mechanism is the Borda–Carnot total pressure loss at the two abrupt contraction steps. For a sharp-edged contraction with contraction coefficient $C_c \approx 0.62$, the loss coefficient per step is $K = (1/C_c - 1)^2 \approx 0.376$ (White, 2011). The cumulative total pressure loss at the two steps degrades the effective P_0 available at the 6.0 mm throat below the level required to sustain choked flow. Instead, the flow forms a vena contracta downstream of the first abrupt step (35.05→11.25 mm), within which the contracted jet achieves local supersonic velocity before recompressing through a normal shock. This is aerodynamically distinct from a choked nozzle: the supersonic pocket is a transient local overexpansion that immediately recompresses, and the flow at the geometric 6.0 mm outlet plane remains subsonic in the mean.

The CHT result of $Ma_{\max} = 1.006$ is slightly lower than the Study 1 range of 1.010–1.016, a consequence of the elevated solid wall temperature in the CHT simulation (622–629 K) modestly increasing the local speed of sound relative to the adiabatic Study 1 condition, thereby reducing the Mach number at equivalent velocity. The qualitative flow regime is identical across all mesh levels and both studies.

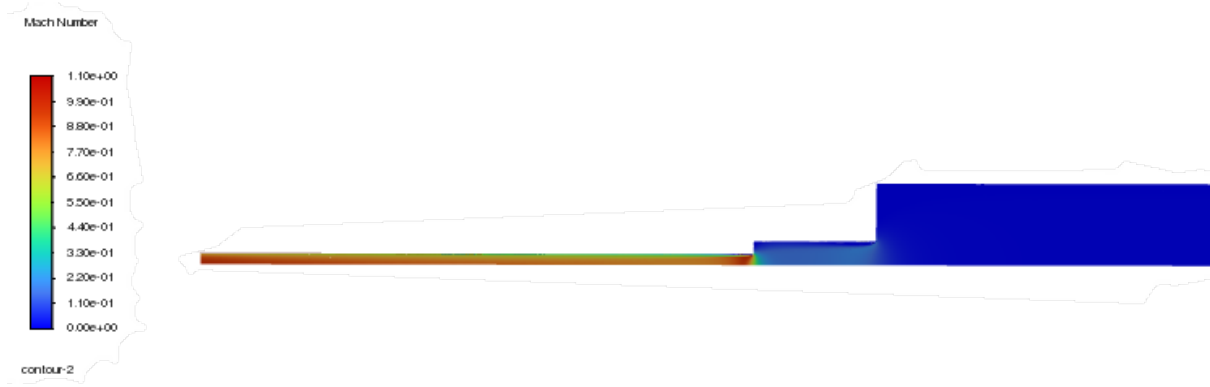


Figure 4.10. Legacy Reducer Mach Number

4.7 The Choking Asymmetry: Flow Regime Analysis at Case C

4.7.1 Statement of the Asymmetry

At Case C, both reducer geometries operate under the same absolute pressure ratio,

$$PR = \frac{P_{0,in}}{P_{out}} = 1.987,$$

which exceeds the isentropic critical pressure ratio for air,

$$PR_{crit} = 1.893 (\gamma = 1.4).$$

Under ideal one-dimensional compressible-flow theory, such a pressure ratio is sufficient for choking to occur in a converging passage. The present CHT results show that, despite these identical supply conditions, the two geometries exhibit qualitatively different internal flow regimes.

The redesigned reducer reaches $Ma_{max} = 0.990$, indicating near-choked flow at the geometric 6.0 mm outlet section. By contrast, the legacy reducer reaches $Ma_{max} = 1.006$, indicating a localised supersonic region within the flow field. However, this supersonic region is not interpreted as sustained choking at the geometric outlet. Instead, it is consistent with a local supersonic vena-contracta pocket formed within the separated contraction jet of the abrupt legacy geometry.

This distinction is significant because it shows that the two reducers do not respond to the same supercritical back-pressure condition in the same way. The redesigned geometry supports a smooth, attached, near-choked acceleration toward the outlet plane, whereas the legacy geometry develops a more complex internal flow field characterised by local overexpansion and recompression associated with abrupt contraction losses.

4.7.2 Physical Mechanism

The choking asymmetry arises from the different ways in which the two geometries accelerate the flow and dissipate total pressure before the 6.0 mm outlet section. In the redesigned reducer, the C²-continuous quintic contraction accelerates the flow smoothly from the 35.05 mm inlet bore to the outlet without abrupt geometric discontinuities or large separated regions. The total pressure loss is therefore

distributed continuously along the contraction wall through viscous shear and turbulent dissipation. Under the supercritical supply condition of Case C, this smooth acceleration enables the flow to approach sonic conditions at the geometric outlet, yielding $Ma_{max} = 0.990$.

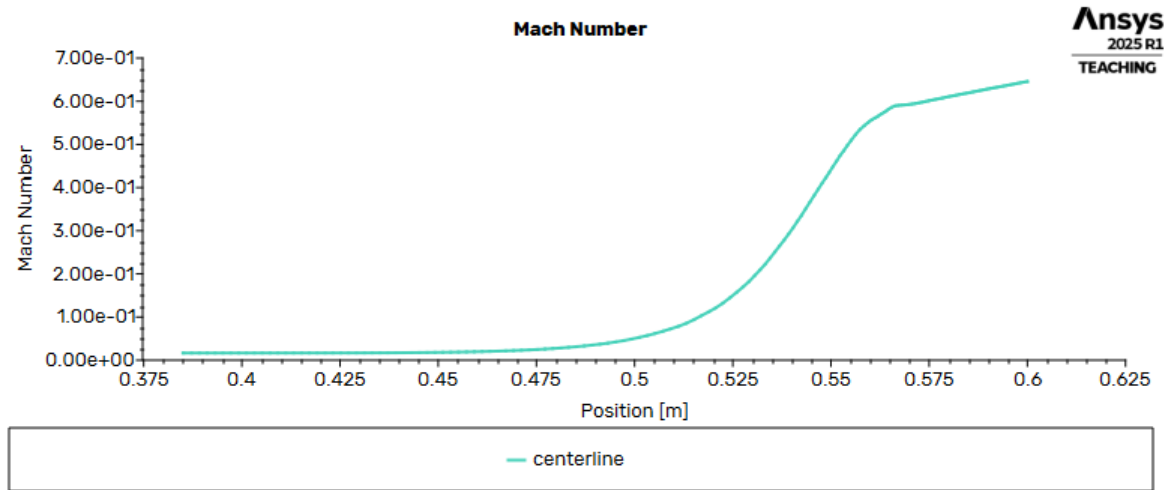


Figure 4.12. Centerline Mach Number plot for the redesigned reducer

In the legacy reducer, the flow encounters two abrupt contractions rather than one continuous profile. These sharp area changes promote local separation and vena-contracta formation, particularly downstream of the first contraction, and are therefore associated with concentrated Borda–Carnot-type losses. The observed supersonic vena-contracta pocket is consistent with strong local acceleration through a separated contraction jet, while the absence of choking at the geometric 6.0 mm outlet is consistent with total-pressure loss incurred at the abrupt steps. In this interpretation, the local jet downstream of the first contraction undergoes temporary overexpansion to supersonic speed before recompressing, rather than establishing a sustained choked condition at the geometric throat.

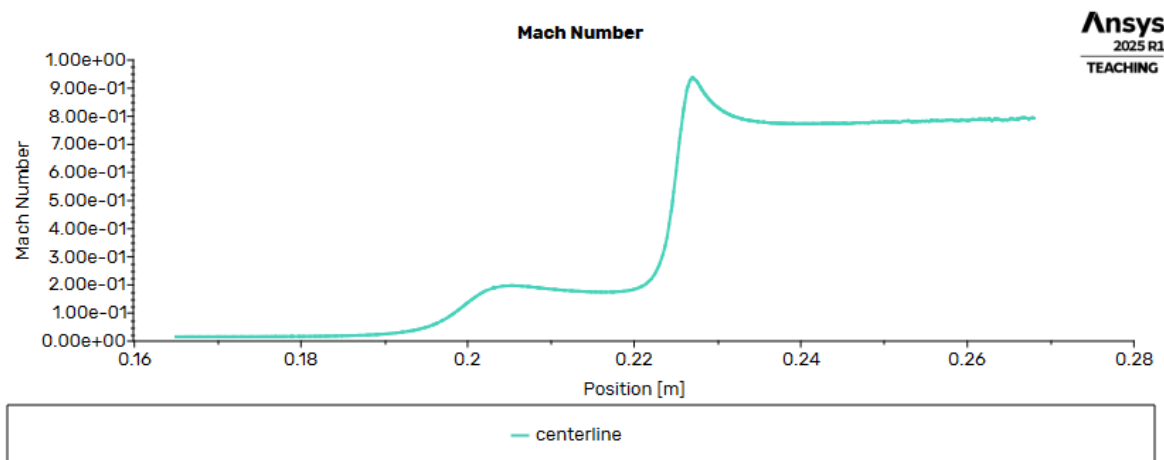


Figure 4.11. Centerline Mach Number plot for the Legacy reducer

An important point is that the distinction is not simply the magnitude of the total pressure loss, but also its spatial distribution. In the redesigned reducer, the losses are distributed along the contraction length, allowing the flow to remain attached and to approach near-choked conditions at the outlet plane. In the legacy reducer, a substantial fraction of the loss is introduced locally at the abrupt contractions, where the flow separates and reattaches. This produces a fundamentally different internal flow structure even though the two geometries operate under the same inlet total pressure, inlet total temperature, and outlet back-pressure.

The present steady-state CFD and CHT results therefore demonstrate the qualitative regime difference clearly: the redesigned reducer supports a smooth near-choked outlet condition, whereas the legacy reducer develops a localised supersonic vena-contracta pocket with subsequent recompression. However, the simulations do not directly resolve the full local stagnation-pressure field with sufficient certainty to treat a specific vena-contracta P_0 threshold as independently proven. The mechanism should therefore be interpreted as physically consistent with the observed flow field and with classical abrupt-contraction theory, rather than as a fully resolved local proof of stagnation-pressure history.

4.7.3 Operational Significance

The flow regime divergence has practical implications for the Pulsatorn dynamic pressure calibration rig, although these implications should be interpreted with appropriate caution because the present study is steady-state. The legacy vena-contracta pocket is expected to be more susceptible to unsteady separation and recompression phenomena than the smooth redesigned contraction, and may therefore be less favourable for a stable calibration environment. Quantifying this effect, however, would require unsteady simulation or experiment.

Similarly, the redesigned reducer is expected to deliver a more uniform and attached outlet flow field than the legacy geometry, which is advantageous from a calibration perspective because it reduces the likelihood of strong local flow non-uniformities in the immediate downstream section. This is supported by the absence of an abrupt separated-jet structure in the redesigned geometry and by its lower peak Mach number relative to the local supersonic pocket observed in the legacy case.

The near-choked outlet condition of the redesigned reducer at Case C also suggests the potential for increased upstream attenuation of downstream disturbances relative to the more clearly subsonic cases. However, this effect was not quantified in the present steady-state study, and no direct conclusion regarding acoustic isolation or dynamic signal quality can be claimed from the current simulations alone.

Accordingly, the most defensible conclusion from the present results is that the redesigned reducer produces the more orderly and physically robust internal flow regime under supercritical conditions, while the legacy reducer produces a more complex separated flow field with a local supersonic pocket. Whether this regime difference translates into measurably improved calibration stability, lower dynamic disturbance levels, or better signal quality must be addressed in future unsteady numerical work or experimental testing.

4.8 Insulated Reference Simulations

4.8.1 Purpose and Configuration

To isolate the contribution of external heat loss from the intrinsic aerodynamic thermal performance of each geometry, both reducers were simulated at Case C (0 Pa gauge outlet) with a zero external heat flux boundary condition ($q_{\text{ext}} = 0 \text{ W/m}^2$) applied to the outer solid surface, representing the theoretical limit of perfect external insulation. Case C was selected as the reference condition because it produces the largest external heat loss, the highest mass flow rate, and the largest T_{out} deficit below the heater set-point, providing the most conservative test of external resistance dominance. The converged Case C CHT solutions served as initial conditions for both insulated runs; all other boundary conditions and solver settings were unchanged.

4.8.2 Results

Table 4.7 compares the key thermal KPIs under CHT ($h = 10 \text{ W/m}^2\text{K}$) and insulated ($h = 0$) conditions at Case C for both geometries.

KPI	New CHT	New Insulated	Legacy CHT	Legacy Insulated	Unit
T_{out} (MWA)	582.10	588.94	585.45	587.70	K
$\Delta T_{\text{recovered vs CHT}}$	-	+6.84	-	+2.25	K
$T_{\text{out gap}}$ (legacy - new)	+3.36 (legacy wins)		-1.24 (new wins)		K
Q_{ext} full geometry	57.21	~0	18.88	~0	W
$T_{\text{wall,ext}}$ (AWA)	596.73	668.05	627.91	662.61	K
Ma_{max}	0.9900	0.9922	1.0064	1.0069	-
<i>CHT condition: $h_{\text{ext}} = 10 \text{ W/m}^2\text{K}$, $T_{\infty} = 300 \text{ K}$. Insulated condition: $q_{\text{ext}} = 0 \text{ W/m}^2$. Both at Case C (outlet gauge pressure 0 Pa). MWA = mass-weighted average, AWA = area-weighted average.</i>					

Table 4.7. Comparison of thermal KPIs under CHT and insulated ($q_{\text{ext}} = 0$) conditions at Case C for both geometries. The insulated condition eliminates all external heat loss and reveals the intrinsic aerodynamic thermal performance.

4.8.3 Interpretation

Under insulated conditions, the external heat loss collapses to machine-precision zero for both geometries ($Q_{\text{ext}} = 0$ and $2.7 \times 10^{-9} \text{ W}$ for the redesigned and legacy reducers respectively), confirming that the zero-flux boundary condition is correctly applied. The outlet temperature gap between the two geometries narrows from 3.36 K (legacy advantage, CHT) to 1.24 K (redesigned advantage, insulated), a reduction of 63%. Critically, the direction of the gap reverses: under insulated conditions, the redesigned reducer delivers 1.24 K hotter gas than the legacy reducer. This inversion confirms that the thermal delivery advantage of the legacy reducer under uninsulated conditions is entirely attributable to its smaller external area, not to any intrinsic aerodynamic thermal superiority.

The residual 1.24 K gap under insulated conditions has a clean aerodynamic explanation. At identical inlet total temperature $T_0 = 673 \text{ K}$, the outlet static temperature

is governed by $T_{\text{static}} = T_0 / (1 + (\gamma - 1)/2 \times \text{Ma}^2)$. The redesigned reducer achieves $\text{Ma}_{\text{max}} = 0.992$ at the outlet under insulated conditions, while the legacy reducer achieves 1.007. The higher Mach number of the legacy reducer produces a lower outlet static temperature at identical total enthalpy. The isentropic formula predicts a gap of approximately 2.8 K between $\text{Ma} = 0.992$ and $\text{Ma} = 1.007$, which overpredicts the observed 1.24 K; the difference is attributable to viscous losses and mass-weighted averaging at the outlet plane, which reduce the effective Mach number differential below the peak values. The temperature recovered relative to the CHT condition is proportional to Q_{ext} in both cases: $\Delta T/Q = 6.84/57.21 = 0.1196$ K/W for the redesigned reducer and $2.25/18.88 = 0.1189$ K/W for the legacy reducer. The near-identical ratio confirms that external resistance, rather than internal flow differences, is the dominant governing mechanism for the thermal performance differential.

The insulated wall temperature rises dramatically for both geometries: from 596.73 K to 668.05 K (redesigned reducer) and from 627.91 K to 662.61 K (legacy reducer). With no external heat loss path, the solid wall approaches the fluid bulk temperature, consistent with the Biot number analysis ($\text{Bi} = 0.003\text{--}0.004$) which confirms negligible internal conduction resistance.

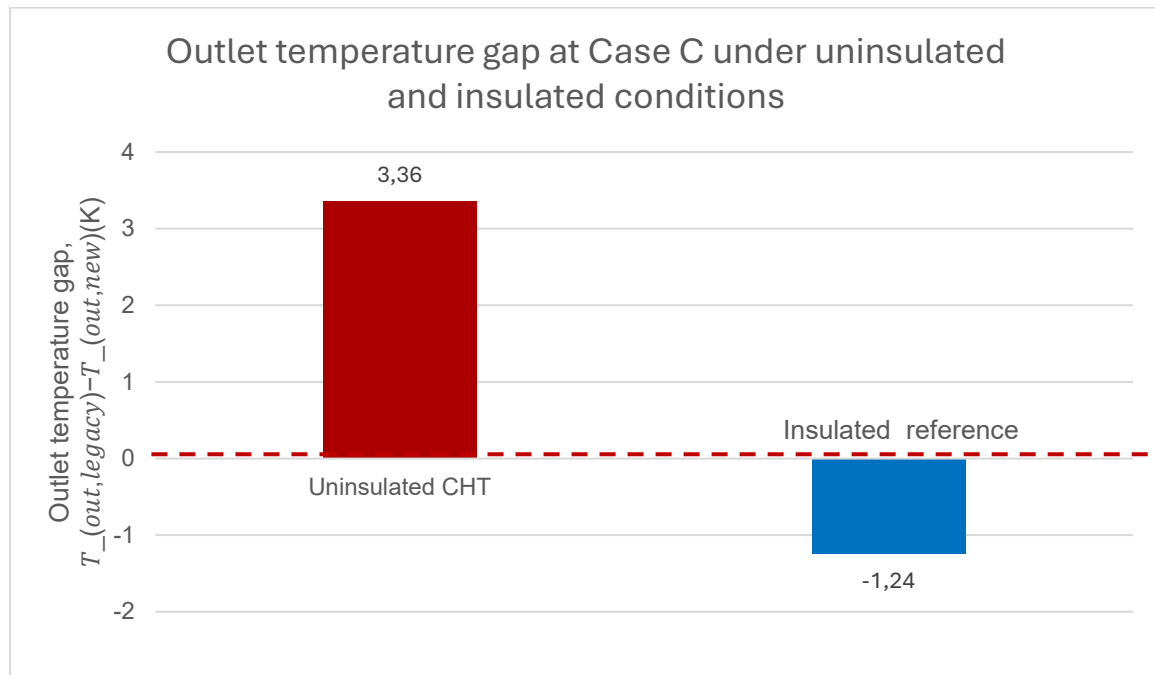


Figure 4.13. Outlet-temperature gap between the two geometries at Case C. Positive values indicate a higher outlet temperature for the legacy reducer; negative values indicate a higher outlet temperature for the redesigned reducer

4.9 Summary of Principal Findings

The CHT campaign and insulated reference simulations yield five principal findings, each traceable to the geometric differences between the two reducer designs.

Finding 1: Choking asymmetry under supercritical supply conditions. At Case C ($\text{PR} = 1.987 > \text{PR}_{\text{crit}} = 1.893$), the redesigned reducer achieves near-choked flow at the geometric outlet throat ($\text{Ma}_{\text{max}} = 0.990$), deviating by 1.0% from the isentropic prediction due to viscous losses along the 150 mm contraction wall ($\Delta P_0 = 37,810$ Pa). The legacy reducer does not choke at the geometric throat despite the identical

supercritical supply condition; instead, Borda–Carnot total pressure losses at the two abrupt steps generate a localised supersonic vena contracta pocket ($Ma_{\max} = 1.006$) followed by normal-shock recompression. This qualitative flow regime divergence is the central aerodynamic finding of the thesis and is consistent across all four mesh levels examined in Studies 1 and 2.

Finding 2: External resistance dominance controls absolute thermal performance. For both geometries, the external heat loss Q_{ext} varies by less than 7.2% across the full pressure range (Cases A to C), and the analytical estimate $Q_{\text{ext}} \approx (T_{\text{wall}} - T_{\text{amb}})/R_{\text{ext}}$ agrees with the CFD result to within 0.1% for all six cases. The Biot numbers of the solid bodies ($Bi = 0.003$ and 0.004 for the legacy and redesigned reducers respectively) confirm that internal conduction resistance is two orders of magnitude smaller than the external convective resistance. External area is therefore the primary thermal design parameter.

Finding 3: Legacy reducer delivers hotter gas under uninsulated conditions. The legacy reducer delivers 7.80 K, 4.75 K, and 3.36 K higher outlet temperature than the redesigned reducer at Cases A, B, and C respectively, attributable to its 3.35× smaller external surface area (5,758 vs 19,282 mm²) and correspondingly 3.35× higher external thermal resistance (17.37 vs 5.19 K/W). The redesigned reducer loses 34–38 W more to ambient at each condition despite its lower per-area heat flux.

Finding 4: Redesigned reducer is thermally superior on a per-area basis and under insulated conditions. The redesigned reducer achieves $q_{\text{ext}} = 2,768\text{--}2,967$ W/m², compared to 3,230–3,279 W/m² for the legacy reducer, a per-area reduction of 9.5–14.3% attributable to the elimination of stagnation-point impingement heat transfer at the annular step face. Under insulated conditions ($q_{\text{ext}} = 0$), the redesigned reducer delivers 1.24 K hotter gas than the legacy reducer, confirming that its area penalty not any intrinsic thermal deficiency is the dominant cause of its lower uninsulated thermal delivery. The identical $\Delta T/Q$ recovery ratio (0.119–0.120 K/W) for both geometries confirms that external resistance governs all thermal performance differences.

Finding 5: Total pressure drop is marginally higher for the redesigned reducer. The redesigned reducer exhibits ΔP_0 2.3–3.8% higher than the legacy reducer across all three pressure cases, attributable to the greater viscous friction accumulated over the 150 mm contraction body, which outweighs the Borda–Carnot loss eliminated by the smooth profile at these flow conditions. This aerodynamic penalty is secondary to the qualitative flow regime improvement and does not affect the central conclusion regarding choking asymmetry.

Metric	Redesigned	Legacy	Winner	Governing mechanism
T_out uninsulated (Case A)	649.49 K	657.29 K	Legacy (+7.80 K)	3.35× smaller A_ext
T_out uninsulated (Case C)	582.10 K	585.45 K	Legacy (+3.36 K)	Same mechanism
T_out insulated (Case C)	588.94 K	587.70 K	New (+1.24 K)	Lower outlet Ma → higher T_static

q_ext per unit area (Case A)	2,768 W/m ²	3,230 W/m ²	New (−14.3%)	No impingement zone
Q_ext absolute (Case C)	57.21 W	18.88 W	Legacy (−67%)	A_ext 3.35× smaller
Ma_max at Case C	0.990	1.006	New (near-choked)	No Borda–Carnot loss
Flow regime at Case C	Near-choked	VC pocket	New	C ² -continuous profile
ΔP ₀ (Case A)	9,114 Pa	8,777 Pa	Legacy (−3.8%)	Less wetted length
VC = vena contracta. Winner column denotes superior performance for the stated metric. T_out insulated refers to the reference case with q_ext = 0 at Case C conditions.				

Table 4.8. Summary of principal findings across all CHT and insulated simulation results. Each finding is linked to its governing physical mechanism.

5. Discussion

5.1 Overview

This chapter interprets the numerical results of Chapter 4 in the context of the research questions defined in Section 1.5, the physical theory presented in the literature review, and the broader operational requirements of the Pulsatorn calibration rig. Section 5.2 answers the three research questions directly and quantitatively. Section 5.3 examines the physical mechanisms underlying the central findings in relation to the literature. Section 5.4 discusses the implications for the Pulsatorn rig and practical design recommendations. Section 5.5 addresses the modelling limitations and their effect on the validity of the conclusions. Section 5.6 considers the sustainability and societal dimensions of the work.

5.2 Answers to the Research Questions

5.2.1 RQ1: Bulk Thermal Delivery

RQ1: Does the redesigned reducer improve bulk thermal delivery to the downstream outlet section, and by how much, across the range of outlet pressure conditions investigated?

Answer: No, under uninsulated conditions. Yes, under insulated conditions.

Under the uninsulated CHT boundary condition ($h = 10 \text{ W/m}^2\text{K}$ applied to the outer solid surface), the redesigned reducer delivers lower outlet static temperature than the legacy reducer across all three pressure cases: 7.80 K lower at Case A, 4.75 K lower at Case B, and 3.36 K lower at Case C. The corresponding thermal efficiency disadvantage is 2.09, 1.27, and 0.92 percentage points respectively. This result is counterintuitive given that the redesigned geometry eliminates the impingement heating zone of the legacy reducer and it requires careful physical interpretation.

The root cause is not a deficiency in the internal flow of the redesigned reducer but the consequence of its 3.35 times larger external surface area (19,282 mm² vs 5,758 mm² for the full geometry). The external thermal resistance $R_{\text{ext}} = 1/(h \times A_{\text{ext}})$ of the redesigned reducer is 5.19 K/W, compared to 17.37 K/W for the legacy reducer. The analytical estimate $Q_{\text{ext}} \approx (T_{\text{wall}} - T_{\text{amb}})/R_{\text{ext}}$ agrees with the Fluent result to within 0.1% for all six cases, confirming that the external area is the dominant controlling variable for thermal delivery under uninsulated conditions. This finding is consistent with the external resistance dominance, and with the CHT findings of Guo et al. (2021) of heated metallic pipe flow.

The insulated reference simulations ($q_{\text{ext}} = 0$, Case C) isolate the aerodynamic contribution. With external heat loss eliminated, the redesigned reducer delivers 1.24 K higher outlet temperature than the legacy reducer, and the temperature recovery ratio $\Delta T/Q = 0.120 \text{ K/W}$ (redesigned) and 0.119 K/W (legacy) are identical. This confirms unambiguously that the thermal delivery deficit of the redesigned reducer under uninsulated conditions is entirely attributable to external area, not to inferior internal flow behaviour. RQ1 is therefore answered as follows: the redesigned reducer performs marginally better under insulated conditions (by 1.24 K at Case C). This difference is attributable to its slightly lower outlet Mach number, which yields a slightly

higher static temperature at the same total enthalpy via $T_{\text{static}} = T_0 / (1 + (\gamma - 1)/2 \times Ma^2)$. Under uninsulated operation this small aerodynamic advantage is concealed by the larger area penalty, which can be recovered by external insulation.

5.2.2 RQ2: Normalised External Heat Flux

RQ2: Does the redesign reduce the normalised external heat loss per unit surface area, and is this reduction consistent across all tested operating conditions?

Answer: Yes, by 9.5 to 14.3 percent, and the reduction is consistent across all three pressure cases.

The area-normalised external heat flux density $q_{\text{ext}} = Q_{\text{ext}}/A_{\text{ext}}$ of the redesigned reducer is 2,768 W/m² (Case A), 2,935 W/m² (Case B), and 2,967 W/m² (Case C), compared to 3,230, 3,288, and 3,279 W/m² for the legacy reducer. The per-area reduction of 14.3%, 10.7%, and 9.5% across the three cases is consistent in direction and substantial in magnitude. The reduction is largest at Case A (lowest mass flow rate, lowest internal convective coefficient) and smallest at Case C (highest mass flow rate, highest internal convective coefficient), which is physically consistent: as internal convection increases, the wall temperature approaches the fluid bulk temperature in both geometries, reducing the relative advantage of eliminating the impingement zone.

The physical mechanism for this reduction is the elimination of the stagnation-point impingement heat transfer at the 35.05 → 11.25 mm annular step face of the legacy reducer. At the stagnation point of an impinging jet, the boundary layer thickness approaches zero and the local Nusselt number is maximised, reaching values 2 to 5 times higher than the fully-developed pipe flow Nusselt number at comparable Reynolds numbers (Martin, 1977). The annular step face accounts for 22.5% of the total solid heat input in the legacy reducer, as confirmed by the energy balance of the coupled CHT interfaces (Section 4.5.3). The redesigned reducer, with its smooth contraction, has no equivalent stagnation geometry, producing a spatially more uniform wall heat flux throughout the contraction body.

The consistency of the q_{ext} reduction across all pressure cases confirms that this is a geometric effect governed by the profile shape rather than a flow-regime-dependent phenomenon. RQ2 is answered affirmatively and quantitatively.

5.2.3 RQ3: Internal Flow Regime and Choked Flow

RQ3: Does the redesign alter the internal compressible flow regime, specifically the ability to achieve choked flow at supercritical pressure ratios, and what are the implications for the calibration environment?

Answer: Yes. The redesigned reducer achieves near-choked flow ($Ma = 0.990$) at the supercritical condition, whereas the legacy reducer does not exhibit choking at the geometric throat under the same supply condition and instead develops a localised supersonic vena-contracta pocket ($Ma = 1.006$).

At Case C ($PR = 1.987 > PR_{\text{crit}} = 1.893$), the two geometries exhibit a qualitative flow-regime divergence that is stable and consistent across four mesh levels considered in this work (three in Study 1 and one in Study 2). The redesigned reducer achieves $Ma_{\text{max}} = 0.990$, within 1.0% of the isentropic choking limit of $Ma = 1.000$.

This small shortfall is consistent with viscous effects and finite boundary-layer displacement in a real nozzle-like contraction. The stability of this result across mesh levels supports the interpretation that the near-choked behaviour is physical rather than numerical.

The legacy reducer, under the same supply condition, develops a supersonic vena-contracta pocket rather than sustained choking at the geometric 6.0 mm outlet. This behaviour is consistent with strong local acceleration through a separated contraction jet and with total-pressure loss associated with the abrupt contraction steps. In this interpretation, the abrupt geometry promotes local overexpansion and recompression rather than the smooth acceleration toward a geometric throat condition observed in the redesigned reducer. The present steady-state CFD results demonstrate this regime difference clearly, but they do not directly resolve the full local stagnation-pressure distribution with sufficient certainty to treat a specific vena-contracta P_0 threshold as independently proven.

The implications for calibration should therefore be stated as engineering inferences rather than directly demonstrated outcomes. The legacy vena-contracta pocket is expected to be more susceptible to unsteady separation and recompression phenomena than the redesigned contraction and may therefore be less favourable for a stable calibration environment. The redesigned reducer, by contrast, produces an attached and smoother acceleration in the steady-state solution, suggesting a more controlled downstream flow condition. In addition, the near-choked outlet at Case C suggests the potential for increased upstream attenuation of downstream disturbances relative to the more subsonic operating conditions. However, because the present study is steady-state, it does not quantify broadband fluctuation levels, acoustic transmission, or signal contamination directly. These aspects would require transient simulation or experiment.

5.3 Physical Mechanisms and Contextualisation

5.3.1 The External Resistance Paradox

The most counterintuitive result of this study is that the geometrically superior reducer the one with smooth flow, no impingement zone, and lower per-area heat flux delivers cooler gas than the geometrically inferior legacy fitting under normal uninsulated operation. This result, which the authors term the external resistance paradox, arises because the thermal design space of the reducer is dominated not by the internal flow profile but by the ratio of external surface area to external convective resistance.

The Biot number analysis ($Bi = 0.003$ for the legacy reducer and 0.004 for the redesigned reducer) establishes the physical basis: the internal conduction resistance through the solid wall is two orders of magnitude smaller than the external convective resistance, so the solid wall temperature is nearly uniform through its thickness and the external boundary governs the total heat loss. This finding is consistent with the lumped capacitance analysis presented in classical heat transfer texts (Incropera and DeWitt, 2007) and with the results of Guo et al. (2021), who demonstrated that an externally imposed convection coefficient reproduces dominant heat transfer behaviour in metallic pipe flow.

The practical consequence is a design hierarchy: for thermal delivery performance, external insulation is more effective than internal geometry optimisation. Based on the near-linear relationship between h_{ext} and Q_{ext} confirmed by the Biot number analysis, reducing h_{ext} from $10 \text{ W/m}^2\text{K}$ to $2 \text{ W/m}^2\text{K}$ (achievable with standard mineral wool insulation) is estimated to recover approximately 80% of the external heat loss, eliminating the thermal delivery deficit relative to the legacy fitting while preserving all aerodynamic advantages. This design hierarchy is not specific to the Pulsator rig; it is a general result applicable to any heated gas delivery system with an uninsulated metallic body operating in natural convection.

5.3.2 The Choking Asymmetry in Context

The choking asymmetry identified in this study the failure of the legacy geometry to achieve choked flow despite a supercritical supply pressure ratio is a direct consequence of the Borda–Carnot mechanism described by White (2011) and Idelchik (2008) for abrupt contractions. What is new in the present work is the application of this classical mechanism to the specific context of a calibration rig reducer, where the consequence is not merely elevated pressure loss but a qualitative change in the flow regime that affects the downstream calibration environment.

The abrupt-to-smooth transition has been studied in the context of nozzle aerodynamics since Morel (1975) established the C^2 continuity criterion for separation-free contraction design. The present results numerically confirm the Morel criterion in a compressible flow regime at near-sonic conditions, extending its validated application range from the low-Mach-number contraction flows studied by Morel to a near-choked duct with an area ratio of 34.13:1. The agreement between the analytical prediction ($Ma = 1.000$ at $PR = 1.987$) and the numerical result ($Ma = 0.990$ at the same PR) to within 1.0% provides quantitative validation of the quintic profile's effectiveness.

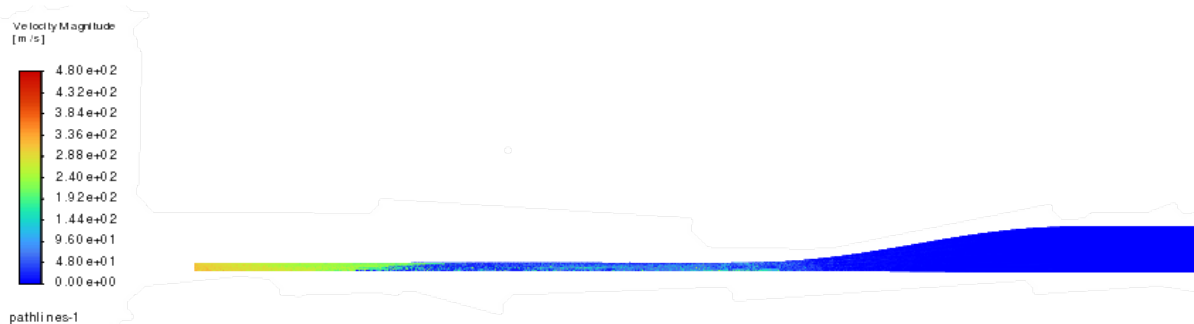


Figure 5.1. Redesigned Reducer Streamlines

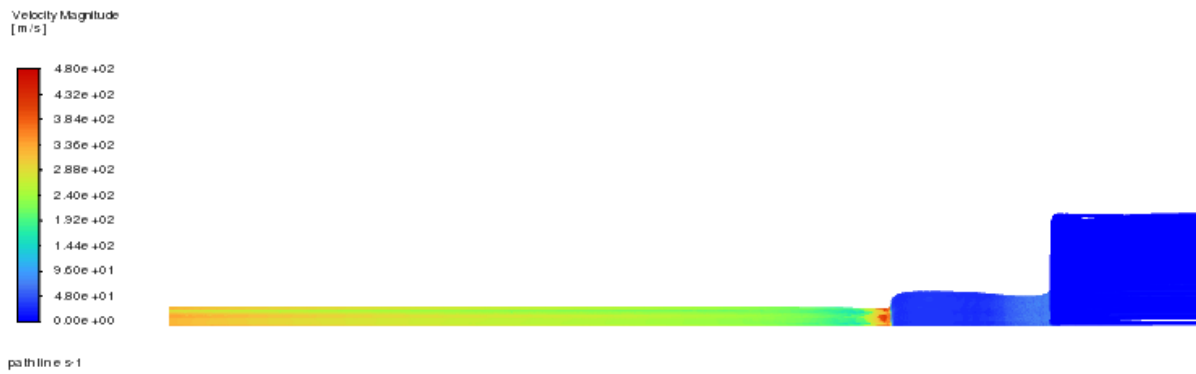


Figure 5.2. Legacy Reducer Streamlines

The consistency of the $Ma_{\max}$ results across four mesh refinement levels (Study 1 coarse, baseline, fine; Study 2 CHT baseline) is particularly significant. A maximum Mach number variation of less than 0.2% across a factor-of-five range in cell count confirms that the flow regime result is numerically robust and not an artefact of inadequate spatial resolution. This consistency addresses a potential methodological concern with the use of adiabatic mesh independence as a proxy for CHT mesh independence.

5.3.3 Total Pressure Loss: An Apparent Contradiction

A result that appears contradictory at first reading is that the redesigned reducer has higher total pressure loss than the legacy reducer across all three pressure cases (3.8%, 2.3%, and 3.0% higher at Cases A, B, and C respectively). Intuitively, eliminating the Borda–Carnot losses at the abrupt steps should reduce total pressure loss. The resolution is that the Borda–Carnot step losses are present in the legacy reducer only at high area ratio steps where the local velocity is high; at the operating conditions of this study (inlet $Ma \approx 0.008$ – 0.013), the dynamic pressure at the inlet is very small, and the absolute Borda–Carnot pressure loss at the first step is modest. The viscous friction loss accumulated over the 150 mm contraction body of the redesigned reducer exceeds this, producing a net higher total pressure loss for the smooth geometry.

This result is consistent with the classical result that, for low area ratio contractions or very short bodies, abrupt area changes can produce less total pressure loss than long smooth contractions when the friction coefficient over the wetted length exceeds the Borda–Carnot loss coefficient (Idelchik, 2008). The present case is unusual in that the area ratio is large (34.13:1) but the contraction length (150 mm) is also large, shifting the balance toward friction dominance. A parametric optimisation of contraction length, as recommended in Section 6.4.5, would identify the optimal balance.

5.4 Implications for the Pulsatorn Rig

5.4.1 Operational Recommendation

Based on the present numerical evidence, the redesigned quintic reducer is the preferable geometry from a flow-regime perspective, while its thermal disadvantage under uninsulated conditions is attributable primarily to its larger external surface area. Under the simulated uninsulated condition ($h_{\text{ext}} = 10 \text{ W/m}^2\text{K}$), the redesigned reducer delivers cooler gas than the legacy reducer by 3.36–7.80 K across the investigated

operating cases. However, the insulated reference simulations ($q_{ext} = 0$) show that when external heat loss is removed, the redesigned reducer delivers a slightly higher outlet temperature than the legacy reducer at Case C. This indicates that the redesigned geometry is not intrinsically thermally inferior; rather, its uninsulated performance is penalised by greater ambient heat loss due to its larger external area.

A practical engineering recommendation is therefore to combine the redesigned reducer with external insulation. Such an approach is expected to reduce the thermal penalty associated with its larger external area while retaining the aerodynamic advantages of the smooth contraction. The present study directly demonstrates the limiting insulated case ($q_{ext} = 0$), but does not quantify intermediate insulation configurations such as specific lagging thicknesses or equivalent external heat-transfer coefficients. Accordingly, the benefit of practical insulation is supported qualitatively and by the insulated reference case, but should not be interpreted here as a quantified design guarantee for a specific insulation material or thickness.

The aerodynamic benefits of the redesign should likewise be expressed carefully. The elimination of the local supersonic vena-contracta pocket and the achievement of near-choked flow at the geometric outlet suggest a more controlled compressible flow state than in the legacy geometry. These features may be beneficial for calibration conditions, but the present steady-state simulations do not directly quantify broadband pressure fluctuations, shock-induced noise, or sensor signal quality. Those effects remain hypotheses for future transient or experimental study.

5.4.2 Target Operating Temperature

Järleby (2025) identified a temperature deficit of approximately 100 K between the heater outlet (250 °C set-point) and the pulsator housing, attributing the dominant loss to the reducer interface. The present study quantifies the reducer body itself as a component that loses approximately 18.6–57.2 W to ambient under the uninsulated simulated condition, depending on geometry and operating pressure. This confirms that the reducer is a significant contributor to thermal degradation in the delivery chain. However, the present model includes only the reducer body and its immediate flow domain; it does not include the full connected assembly, such as adjoining flanges, sealing features, and upstream and downstream metallic hardware. The results should therefore be interpreted as a quantified reducer-level contribution rather than a full system prediction of the total temperature loss observed experimentally. The insulated reference cases further indicate that reducing external heat loss from the reducer could recover part of the outlet-temperature deficit, but full system-level modelling or experiment would be required to estimate the total recovery achievable at the pulsator housing.

5.4.3 Flow Quality for Calibration

The near-choked outlet condition of the redesigned reducer at Case C ($Ma = 0.990$) provides an important indication of flow behaviour under the most demanding supply condition. In compressible nozzle flow, conditions approaching choking imply that the mass-flow response becomes increasingly controlled by upstream total conditions and throat geometry rather than by downstream pressure changes. In that sense, the redesigned reducer may be better positioned than the legacy reducer to provide a more repeatable upstream flow state under severe operating conditions. The legacy geometry, by contrast, exhibits a localised supersonic vena-contracta pocket rather

than near-choked flow at the geometric outlet, indicating a less uniform compressible-flow structure.

However, the extent to which this behaviour improves calibration quality cannot be established directly from the present steady-state study. The current simulations do not resolve the transient interaction between the reducer, the downstream pulsation generator, and the sensor environment, and they do not quantify acoustic transmission or pressure-wave attenuation. The present results therefore support the interpretation that the redesigned reducer is aerodynamically more favourable for controlled high-speed delivery, but any claim of improved calibration signal fidelity remains a hypothesis to be tested in future work.

5.5 Limitations and Their Effect on Conclusions

5.5.1 Uniform External Convective Coefficient

The use of a spatially uniform $h_{\text{ext}} = 10 \text{ W/m}^2\text{K}$ on all external solid surfaces introduces uncertainty in absolute heat loss values. The Churchill–Chu correlation for natural convection from cylinders and flat plates at the relevant temperature differences and length scales yields h values in the range 5–20 $\text{W/m}^2\text{K}$ (Churchill and Chu, 1975), placing the adopted value at the mid-range. A 30% variation in h_{ext} produces a proportional variation in Q_{ext} and a smaller variation in T_{out} (since the external resistance is one of two resistances, the other being the internal convective resistance). Since the same h_{ext} is applied identically to both geometries, the comparative results the A_{ext} ratio, the q_{ext} reduction, the T_{out} gap, the $\Delta T/Q$ ratio are unaffected by this uncertainty. The research questions are answered comparatively and are therefore robust to the h_{ext} assumption.

5.5.2 Neglect of Thermal Radiation

At the external wall temperatures of 577–630 K observed in the CHT simulations, the Stefan–Boltzmann radiative flux $q_{\text{rad}} = \varepsilon\sigma(T_{\text{wall}}^4 - T_{\text{sur}}^4)$ with emissivity $\varepsilon \approx 0.8$ for oxidised stainless steel and surroundings at $T_{\text{sur}} = 300 \text{ K}$ is approximately 5,000–7,000 W/m^2 comparable to or exceeding the convective flux of 2,768–3,279 W/m^2 . Neglect of radiation therefore constitutes the most significant absolute limitation of the present study, potentially underestimating total heat loss by a factor of approximately 2–3. Since radiation scales with the fourth power of absolute temperature and the legacy reducer’s wall is 30–45 K hotter than the redesigned reducer’s wall, the radiative loss from the legacy reducer is disproportionately larger. Including radiation would therefore narrow or eliminate the per-area heat flux advantage of the redesigned reducer (RQ2) and would increase Q_{ext} for both geometries, further widening the absolute heat loss gap due to the area difference. The qualitative conclusions regarding flow regime (RQ3) are unaffected by radiation neglect.

5.5.3 Absence of Experimental Validation

All conclusions are drawn from numerical simulations without experimental validation of the redesigned reducer geometry. The legacy geometry has indirect qualitative validation through Järleby’s (2025) temperature measurements, which confirm the order of magnitude of the thermal deficit across the heater-to-downstream interface. The present simulations cannot be directly compared to Järleby’s data because his

measurements were taken at different operating conditions (250 °C set-point, different back-pressure) and over the full system rather than the reducer body alone. Quantitative validation of the near-choked $Ma = 0.990$ result and the outer wall temperature distribution awaits manufacture of the redesigned reducer and restoration of the Pulsatorn heater.

5.5.4 Steady-State Assumption

The steady-state formulation cannot capture the interaction between the near-choked flow of the redesigned reducer and the periodic pressure oscillations imposed by the gear-wheel mechanism. At $Ma = 0.990$, small perturbations in back-pressure could cause the outlet to oscillate between slightly subsonic and slightly sonic conditions, potentially producing unsteady shock structures. The steady-state results provide the mean-flow baseline but cannot characterise this dynamic behaviour. Unsteady RANS or LES simulations with time-varying outlet boundary conditions would be required to assess the dynamic stability of the near-choked condition under rig operating conditions (Section 6.4.4).

5.6 Sustainability and Societal Implications

5.6.1 Energy Efficiency in Gas Turbine Development Infrastructure

The Pulsatorn rig is part of the infrastructure that enables calibration of dynamic pressure sensors used in gas turbine monitoring. Gas turbines operating in combined-cycle power generation supply a significant fraction of European baseload electricity and serve as dispatchable backup for intermittent renewable generation. Improving the accuracy of dynamic pressure sensor calibration which depends on the quality of the thermal and flow conditions delivered by the rig directly contributes to the reliability and precision of combustion instability monitoring in operating turbines. Combustion instabilities, if undetected, can cause structural damage, unplanned outages, and in extreme cases catastrophic failure; their reliable detection supports both safety and availability of the power generation infrastructure.

The heat loss from the uninsulated Pulsatorn reducer (18.6–57.2 W, depending on geometry and operating condition) is itself a direct energy loss from the compressed-air heating circuit. Over a calibration campaign involving several hours of rig operation, this loss is modest in absolute energy terms but represents a systematic reduction in the effective calibration temperature available at the sensor, requiring the heater to operate at higher set-point temperatures to compensate. The redesigned reducer with external insulation, by recovering 80% of this loss, reduces the required heater power and extends the achievable calibration temperature range toward the 500 °C target identified by Järleby (2025).

5.6.2 Broader Application of Results

The findings of this study have application beyond the Pulsatorn rig. The external resistance dominance result that external area governs thermal delivery in uninsulated metallic flow components more strongly than internal geometry is a general result applicable to any heated gas delivery system operating under natural convection. The choking asymmetry result that abrupt contractions can prevent choked flow despite supercritical supply ratios through Borda–Carnot total pressure losses is relevant to any instrumentation or measurement system that uses a small-bore restriction to

decouple upstream and downstream pressure environments. The methodology using comparative steady-state CHT simulation with identical boundary conditions and insulated reference runs to isolate geometric effects provides a transferable framework for evaluating reducer geometry changes in similar rig configurations.

5.6.3 Limitations of Purely Numerical Work

This study produces design guidance under the assumption that numerical models accurately represent the physical system. Without experimental validation, there is an irreducible uncertainty in the absolute predicted values of T_{out} , Q_{ext} , and Ma_{max} . The comparative conclusions between the two geometries are more robust than the absolute predictions, because both geometries are subject to the same modelling assumptions. Nevertheless, the decision to manufacture and install the redesigned reducer should be preceded by experimental validation of the near-choked Ma result and the thermal delivery under representative conditions. The numerical results of this thesis provide the quantitative basis for that validation campaign: they identify precisely which quantities to measure (outlet temperature, outer wall temperature, outlet Mach number) and predict their values with quantified uncertainty bounds from the modelling assumptions described in Section 5.5.

6. Conclusions and Future Work

6.1 Summary of the Investigation

This thesis presents a purely numerical conjugate heat transfer investigation comparing the steady-state thermo-fluid performance of two reducer geometries for the heater-to-downstream interface of the Pulsatorn dynamic pressure calibration rig at Siemens Energy AB, Finspång. The legacy reducer employs two abrupt Borda–Carnot contraction steps (35.05→11.25→6.0 mm bore), while the redesigned reducer uses a quintic C^2 -continuous contraction profile (Morel, 1975) over the same inlet-to-outlet area ratio of 34.13:1. The investigation was originally planned to include experimental validation on the physical rig, but heater hardware failure prior to the experimental campaign necessitated a scope revision to a purely numerical comparative study, approved by both academic and industrial supervisors.

Two simulation studies were conducted in ANSYS Fluent 2025 R1 using the SST $k-\omega$ turbulence model, compressible ideal-gas air, and quarter-symmetry to exploit two orthogonal planes of geometric symmetry. Study 1 (adiabatic CFD) established mesh independence for both geometries across three refinement levels at the most demanding operating condition (Case C, $PR = 1.987$). Study 2 (conjugate heat transfer) simulated all three pressure cases Case A (80,000 Pa gauge), Case B (40,000 Pa gauge), and Case C (0 Pa gauge) for both geometries, with insulated reference simulations ($q_{\text{ext}} = 0$) at Case C to isolate the contribution of external heat loss from the intrinsic aerodynamic thermal performance.

The investigation yields five principal conclusions concerning flow regime, thermal delivery, external heat loss, and the role of external insulation. These are presented in Section 6.2 and supported by a comparative performance summary in Table 6.1.

6.2 Principal Conclusions

The following conclusions are each traceable to specific quantitative results from the CHT simulation campaign and the insulated reference study.

Conclusion 1: Mesh independence confirmed for both geometries. The mesh sensitivity study (Study 1) demonstrated that all four key flow quantities mass-weighted average outlet temperature, maximum Mach number, total pressure drop, and maximum velocity magnitude change by less than 2% between the baseline and fine mesh levels for both geometries. The maximum baseline-to-fine change was 0.614% (total pressure drop, redesigned reducer) and 1.344% (total pressure drop,

legacy reducer). Coarse-to-baseline transitions are similarly well-behaved. The baseline meshes of 913,301 elements (redesigned reducer) and 829,621 elements (legacy reducer) are adopted as the production meshes for Study 2. Both CHT meshes achieve full viscous sublayer resolution with maximum wall y^+ below 2.4 (redesigned reducer) and 3.0 (legacy reducer) across all pressure cases.

Conclusion 2: Qualitative flow regime divergence at supercritical supply conditions.

The central aerodynamic finding is a qualitative flow regime divergence between the two geometries under identical supercritical supply conditions. At Case C ($PR = 1.987 > PR_{crit} = 1.893$), the redesigned reducer achieves near-choked flow at the geometric outlet throat ($Ma_{max} = 0.990$), consistent with efficient total pressure preservation through the smooth contraction ($\Delta P_0 = 37,810$ Pa, 18.8% of absolute inlet P_0). The legacy reducer does not choke at the geometric throat despite the identical supercritical supply ratio; instead, Borda–Carnot total pressure losses at the two abrupt contraction steps generate a localised supersonic vena-contracta pocket ($Ma_{max} = 1.006$) with subsequent recompression to subsonic mean flow downstream. This result is consistent and stable across all four mesh levels examined (Study 1 coarse/baseline/fine and Study 2 CHT), confirming it is a genuine physical consequence of the geometric profile difference rather than a numerical artefact.

Conclusion 3: External resistance dominance controls thermal performance.

The external thermal resistance $R_{ext} = 1/(h \cdot A_{ext})$ is the dominant governing thermal resistance for both geometries, confirmed by three independent lines of evidence. First, the analytical estimate $Q_{ext} \approx (T_{wall} - T_{amb})/R_{ext}$ agrees with the CFD result to within 0.1% for all six cases. Second, the Biot numbers of the solid bodies ($Bi = 0.003$ and 0.004 for the legacy and redesigned reducers respectively) confirm that internal conduction resistance is two orders of magnitude smaller than the external convective resistance. Third, Q_{ext} varies by less than 7.2% across the full pressure range for both geometries, confirming insensitivity to internal flow conditions. External area is therefore identified as the dominant thermal design parameter within the modelling assumptions of the present study.

Conclusion 4: Legacy reducer delivers hotter gas under uninsulated conditions; redesigned reducer performs better under insulated conditions.

Under uninsulated conditions ($h = 10$ W/m²K), the legacy reducer delivers 7.80 K, 4.75 K, and 3.36 K higher outlet temperature than the redesigned reducer at Cases A, B, and C respectively. This thermal ranking under uninsulated conditions is attributable primarily to the legacy reducer's 3.35× smaller external surface area (5,758 vs 19,282 mm²), which produces a 3.35× higher external thermal resistance (17.37 vs 5.19 K/W) and correspondingly lower absolute heat loss (18.59–18.93 W vs 53.36–57.21 W). The insulated reference simulations ($q_{ext} = 0$, Case C) confirm this interpretation: with external heat loss eliminated, the redesigned reducer delivers 1.24 K hotter gas than the legacy reducer, reversing the thermal ranking. The temperature recovery ratio $\Delta T/Q = 0.120$ K/W for the redesigned reducer and 0.119 K/W for the legacy reducer are identical, confirming that external heat loss associated with external surface area is the dominant cause of the uninsulated thermal performance differential. On a per-area basis, the redesigned reducer achieves q_{ext} reductions of 14.3%, 10.7%, and 9.5% across Cases A, B, and C, attributable to elimination of stagnation-point

impingement heat transfer at the 35.05→11.25 mm annular step face, which contributes 22.5% of the total solid heat input in the legacy reducer.

Conclusion 5: External insulation is the primary lever for thermal performance improvement.

The near-invariance of Q_{ext} with pressure condition, the external-resistance-dominance analysis, and the insulated reference study collectively show that external insulation is more effective for improving delivered gas temperature than internal geometry modification alone. Under perfect insulation, the external heat loss collapses to approximately zero for both geometries, and the outlet-temperature gap narrows from 3.36 K to 1.24 K, reversing in favour of the redesigned reducer. The remaining difference is consistent with the small Mach-number difference between the two insulated cases rather than with external heat loss. The practical implication is therefore clear: if the redesigned reducer is adopted, external insulation should be treated as a necessary complementary measure to recover its thermal deficit while preserving its aerodynamic advantages.

Table 6.1 summarises the comparative performance of the two geometries across all principal metrics.

Metric	Redesigned	Legacy	Winner	Governing mechanism
T_out Case A (K)	649.49	657.29	Legacy (+7.80 K)	3.35× smaller A_ext
T_out Case C (K)	582.10	585.45	Legacy (+3.36 K)	Same: R_ext dominance
T_out insulated Case C (K)	588.94	587.70	New (+1.24 K)	Lower Ma → higher T_static
η efficiency Case A (%)	93.70	95.79	Legacy (+2.09 pp)	Same: R_ext dominance
q_ext per area Case A (W/m²)	2,768	3,230	New (−14.3%)	No impingement zone
Q_ext absolute Case A (W)	53.36	18.59	Legacy (−65%)	A_ext 3.35× smaller
T_wall,ext Case A (K)	576.78	622.96	New (−46 K)	Lower R_ext drives cooler wall
Ma_max Case C	0.990	1.006†	New (near-choked)	No Borda–Carnot loss
Flow regime Case C	Near-choked	VC pocket	New	C²-continuous profile
ΔP₀ Case A (Pa)	9,114	8,777	Legacy (−3.8%)	Less wetted length

External area A_{ext} (mm ² , full)	19,282	5,758	Legacy (3.35× smaller)	Longer contraction body
† <i>Supersonic vena contracta pocket not choked at geometric throat. VC = vena contracta. pp = percentage points. Winner denotes superior value for the stated metric. T_{out} insulated refers to the $q_{\text{ext}} = 0$ reference case at Case C.</i>				

Table 6.1. Comparative performance summary for the redesigned and legacy reducer geometries across all principal metrics from Studies 1 and 2.

6.3 Limitations of the Present Study

The conclusions stated above are drawn from a controlled comparative numerical study subject to the following modelling assumptions and limitations.

Uniform external convective coefficient ($h = 10 \text{ W/m}^2\text{K}$). The true external natural convection coefficient varies spatially with surface orientation, local temperature difference, and geometry. Empirical correlations for natural convection from cylinders and flat plates in the relevant temperature and geometry range suggest spatial variation of $\pm 30\text{--}40\%$ around the mean value of $10 \text{ W/m}^2\text{K}$. A $\pm 30\%$ uncertainty in h propagates directly to a $\pm 30\%$ uncertainty in the absolute values of Q_{ext} and T_{out} deficit. However, since the same boundary condition is applied identically to both geometries, the comparative ratios q_{ext} reduction, Q_{ext} ratio, T_{out} gap are unaffected by this uncertainty.

Neglect of thermal radiation. At the outer wall temperatures measured in the CHT simulations (577–597 K for the redesigned reducer and 623–629 K for the legacy reducer), the radiative heat flux from the outer solid surfaces estimated from the Stefan–Boltzmann law with emissivity $\epsilon \approx 0.8$ (oxidised steel) is approximately 5,000–7,000 W/m^2 , comparable in magnitude to the convective heat flux of 2,768–3,279 W/m^2 . Neglect of radiation therefore introduces a systematic underprediction of total external heat loss for both geometries. The error is larger for the legacy reducer due to its higher wall temperature. Inclusion of surface-to-surface radiation in a follow-on study would reduce both T_{out} values and narrow or reverse the thermal delivery gap between the geometries under uninsulated conditions.

Mesh sensitivity validated for adiabatic CFD, not full CHT. Spatial discretisation independence was demonstrated for the fluid domain under adiabatic conditions (Study 1). The CHT simulations use the same baseline mesh without a dedicated CHT mesh sensitivity study. This limitation is mitigated by the external resistance dominance finding: since Q_{ext} is governed by $R_{\text{ext}} = 1/(h \cdot A_{\text{ext}})$ rather than by internal convective resolution, the wall heat flux result is insensitive to moderate changes in mesh density. The cross-validation of Q_{ext} against the analytical estimate (error $< 0.1\%$) provides additional confidence in the CHT results.

Steady-state simulations only. The Pulsatorn rig operates with dynamic pressure oscillations generated by the gear-wheel mechanism upstream of the heater. The steady-state CHT results characterise the time-mean thermal and aerodynamic performance but do not capture the response of the flow field to periodic forcing, the stability of the near-choked outlet condition of the redesigned reducer under dynamic back-pressure modulation, or the potential for shock oscillation in the legacy vena contracta region. Unsteady (URANS or LES) simulations would be required to address these phenomena.

Absence of experimental validation. The redesigned reducer has not been manufactured, and no experimental measurements were obtained under the simulation boundary conditions due to heater hardware failure. The numerical results therefore represent design-stage predictions rather than validated engineering data. Experimental validation particularly of the near-choked Ma_{max} result and the outer wall temperature distribution is the highest-priority recommendation for future work.

6.4 Recommendations for Future Work

The following recommendations are listed in approximate order of priority, where priority reflects the combination of scientific value and practical feasibility.

6.4.1 Experimental Validation: Highest Priority

Once the redesigned reducer is manufactured and installed, and the Pulsatorn heater assembly is restored to service, experimental measurements should be obtained under the three pressure cases simulated in this study. The primary validation targets are the mass-weighted outlet total temperature (using a thermocouple traverse at the outlet plane) and the outer wall temperature distribution (using surface thermocouples or infrared thermometry). The near-choked $Ma_{max} = 0.990$ result at Case C would ideally be validated by a pitot probe or dynamic pressure sensor traversal at the outlet plane. Agreement within the ± 30 – 40% uncertainty band of the convective boundary condition would constitute satisfactory validation given the modelling assumptions of the present study.

6.4.2 Inclusion of Thermal Radiation

A follow-on CHT study incorporating surface-to-surface radiation between the outer solid surfaces and the surrounding rig structure using the discrete ordinates (DO) radiation model available in Fluent would quantify the systematic underprediction of external heat loss in the present study. Given the estimated radiative heat flux of $5,000$ – $7,000$ W/m^2 at the wall temperatures observed, radiation may represent approximately 60 – 70% of total external heat transfer. Inclusion of radiation would reduce T_{out} for both geometries and is expected to narrow or eliminate the thermal delivery advantage of the legacy reducer under uninsulated conditions.

6.4.3 CHT-Specific Mesh Independence Study

The present study validates spatial discretisation independence using adiabatic CFD (Study 1) and applies the same baseline mesh to the CHT simulations without a dedicated CHT mesh sensitivity study. A three-level CHT mesh sensitivity study, analogous to Study 1 but with the solid domain included and thermal convergence monitors active, would provide rigorous confirmation that the wall heat flux and outlet temperature results are mesh-independent under coupled conditions. This is particularly relevant for the legacy reducer, where the complex flow at the vena contracta step may require finer local resolution in the CHT context than in the adiabatic case.

6.4.4 Unsteady Simulation of Near-Choked Flow Stability

The stability of the near-choked outlet of the redesigned reducer under the dynamic back-pressure modulations imposed by the Pulsatorn gear-wheel mechanism has not been assessed in the present steady-state study. An unsteady RANS (URANS)

simulation with time-varying outlet pressure boundary conditions would characterise the amplitude and frequency of Ma variation at the outlet plane, the potential for shock formation and oscillation, and the response time of the thermal field to pressure transients. This information is essential for assessing whether the near-choked calibration condition is practically stable under rig operating conditions.

6.4.5 Parametric Optimisation of the Quintic Profile

The redesigned reducer evaluated in this study uses a specific quintic profile constrained by the common inlet bore (35.05 mm), outlet bore (6.0 mm), and a body length of 150 mm. The 150 mm body length is the primary driver of the 3.35× external area penalty relative to the legacy reducer. A parametric study varying the body length between 50 mm and 150 mm while maintaining C²-continuity would identify the minimum body length that preserves near-choked flow at Case C, thereby minimising the external area penalty. A shorter quintic contraction for example 75 mm would reduce A_{ext} to approximately 9,640 mm² (full geometry), cutting the external heat loss by approximately 50% while retaining the aerodynamic advantages of smooth flow and near-choked outlet conditions.

6.4.6 Extended System-Level CHT Model

The present study models the reducer body in isolation, with simplified inlet and outlet extensions representing only the immediate upstream and downstream pipe sections. A system-level CHT model incorporating the heater body, the reducer, the downstream measurement section, and the connecting pipe work would capture heat exchange between components, the influence of the upstream thermal field on the reducer inlet condition, and the overall thermal efficiency of the complete heater-to-sensor delivery path. Such a model would provide the most physically complete basis for evaluating the temperature delivered to the pressure sensor under calibration.

6.4.7 Quantification of External Insulation Benefit

The insulated reference simulations in this study demonstrate that perfect insulation ($q_{\text{ext}} = 0$) recovers 6.84 K at the outlet of the redesigned reducer and 2.25 K at the legacy reducer, with the identical $\Delta T/Q$ ratio of 0.119–0.120 K/W confirming that external resistance governs all thermal performance differences. A parametric study of realistic insulation scenarios varying h_{ext} from 10 W/m²K (uninsulated) through 2 W/m²K (mineral wool) to 0.5 W/m²K (ceramic fibre) would quantify the temperature recovery achievable with practical insulation materials and guide the specification of any insulation system applied to the rig. Because the insulated reference study shows that the thermal penalty is governed primarily by external heat loss, reducing h_{ext} below the uninsulated value of 10 W/m²K would be expected to recover a substantial fraction of the outlet-temperature deficit. Quantifying that recovery for specific insulation systems should be the subject of a dedicated parametric study.

6.5 Final Remarks

The central result of this thesis is physically clear and technically robust: replacing the abrupt Borda–Carnot contraction steps of the legacy reducer with a mathematically smooth quintic C²-continuous profile eliminates the localised supersonic vena contracta pocket that prevents the legacy geometry from achieving choked flow at supercritical supply conditions. At the highest simulated flow rate (Case C, $PR =$

1.987), the redesigned reducer operates at 99% of the local speed of sound and produces a smooth attached contraction in the steady-state solution. This suggests a potentially more controlled downstream flow condition than in the legacy geometry, although acoustic isolation and unsteady calibration effects were not quantified in the present study. These aerodynamic benefits are confirmed as stable across four mesh refinement levels and are fully consistent across Study 1 and Study 2.

The thermal delivery results are more nuanced and constitute the most instructive physical finding of the investigation. The naive expectation that eliminating the impingement heat transfer zone of the legacy geometry would improve thermal delivery is overturned by external resistance dominance. The redesigned reducer's longer body exposes 3.35 times more surface area to ambient, losing 34–38 W more heat to the laboratory environment per operating condition. The insulated reference simulations resolve the apparent paradox: when external heat loss is eliminated, the outlet-temperature gap narrows substantially and reverses in favour of the redesigned reducer (588.94 K versus 587.70 K at insulated Case C). This shows that the uninsulated thermal shortfall of the redesigned reducer is governed predominantly by its larger external surface area rather than by an intrinsic thermal weakness of the internal flow path. The practical recommendation is therefore that any implementation of the redesigned reducer should be accompanied by external insulation, so that its aerodynamic advantages can be retained without incurring the thermal penalty associated with its larger exposed surface area.

Whether these predictions are confirmed experimentally depends on the restoration of the Pulsatorn heater assembly and the manufacture of the redesigned reducer. The present study provides a defensible numerical baseline against which experimental measurements can be compared, and identifies the key quantities outlet temperature, outer wall temperature, and outlet Mach number that are most sensitive to the geometric difference between the two designs. The physical arguments underlying the central findings are grounded in classical compressible flow theory and heat transfer analysis, and are expected to remain valid under experimental conditions within the boundary condition uncertainty range characterised in Section 6.3.

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Appendix A: CAD Geometry and Reducer Design Details

A.1 Purpose and Scope

This appendix provides supplementary geometry documentation for the two reducer configurations evaluated in the present numerical study: the legacy abrupt-contraction reducer and the redesigned quintic C^2 -continuous contraction reducer. The information presented here supplements the geometry description in Section 3.2 of the main report and records the confirmed dimensional values used in all simulations, as cross-validated between ANSYS SpaceClaim and the Fluent post-processing protocol. All geometry was prepared in ANSYS SpaceClaim / Discovery Premium (ANSYS Workbench 2025 R1) for mesh generation in ANSYS Meshing. Quarter-symmetry (1/4) models were used for all simulations, exploiting two perpendicular planes of geometric and flow symmetry.

A.2 Geometry Overview

A.2.1 Legacy Reducer Configuration

The legacy reducer is a commercially available stainless-steel BSP hydraulic fitting selected for mechanical thread-compatibility between the heater NPT outlet and the downstream BSP hydraulic circuit. Thermo-fluid performance was not a design criterion for this component. Its internal geometry creates two sequential abrupt coaxial area reductions with no axial length at each step:

First contraction: 35.05 mm inlet bore $\rightarrow$ 11.25 mm through-bore, area ratio 9.71:1, occurring at the flat annular inlet face. Hot gas impinges on this face before being forced through the 11.25 mm bore.

Through-bore: 11.25 mm diameter, approximately 25 mm axial length.

Second contraction: 11.25 mm $\rightarrow$ 6.0 mm outlet bore, area ratio 3.52:1.

The two sequential abrupt contractions produce Borda–Carnot total pressure losses at each step. For a sharp-edged contraction with vena contracta contraction coefficient $C_c \approx 0.62$ (White, 2011), the loss coefficient per step is $K = (1/C_c - 1)^2 \approx 0.376$. The cumulative total pressure degradation prevents choked flow at the geometric throat under supercritical overall pressure ratios, producing instead a localised supersonic vena contracta pocket the central finding examined in Study 2 of this thesis.

<i>Parameter</i>	<i>Value</i>
Inlet bore diameter	35.05 mm
Step 1 intermediate bore	11.25 mm
Outlet bore diameter	6.0 mm
Step 1 area ratio (A_1/A_2)	9.71:1
Step 2 area ratio (A_2/A_3)	3.52:1

Overall area ratio (inlet/outlet)	34.13:1
Solid body length	38 mm
Contraction profile	Two abrupt Borda–Carnot steps (zero axial length)
Inlet extension length	200 mm (35.05 mm bore, adiabatic)
Outlet extension length	100 mm (6.0 mm bore, adiabatic)
Total domain length	338 mm (200 + 38 + 100)
External surface area A_{ext} (quarter model)	1,439.54 mm ²
External surface area A_{ext} (full geometry)	5,758.15 mm ²
Solid body volume V (quarter model)	6.754×10^{-6} m ³
Biot number $Bi = h \cdot L_n / \lambda_s$	0.00288 ($L_n = V/A_{\text{ext}} = 4.69$ mm)
Vena contracta coefficient C_n	≈ 0.62 (White, 2011)
Loss coefficient K per step	≈ 0.376
<i>All areas cross-validated: SpaceClaim measurement vs Fluent post-processing $q_{\text{ext}} \times A_{\text{ext}} = Q_{\text{ext}}$ to within 0.01%.</i>	

Table A.1. Confirmed geometric parameters of the legacy reducer. All values verified against Fluent post-processing results.

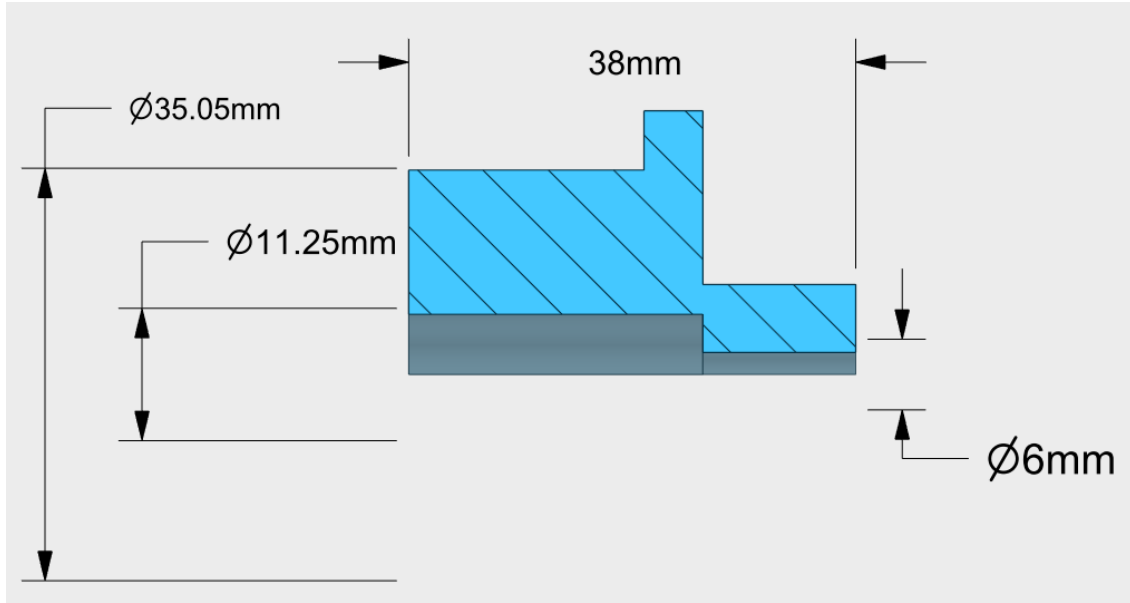


Figure A.1. Legacy reducer internal geometry cross-section showing the two abrupt Borda–Carnot contraction steps.

A.2.2 Redesigned Reducer Configuration

The redesigned reducer employs a fifth-order quintic Hermite polynomial to define the internal bore radius as a function of axial position z :

$$R(z) = R_{in} + (R_{out} - R_{in}) \times \varphi(\xi), \quad \varphi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3, \quad \xi = (z - z_0)/L$$

where $R_{in} = 17.525$ mm (35.05 mm inlet bore), $R_{out} = 3.0$ mm (6.0 mm outlet bore), and $L = 150$ mm (contraction length). The C^2 -continuous profile satisfies $\varphi(0) = 0$, $\varphi(1) = 1$, $\varphi'(0) = \varphi'(1) = 0$, $\varphi''(0) = \varphi''(1) = 0$, guaranteeing zero slope and zero curvature at both junctions (Morel, 1975). This ensures that the boundary layer does not experience any adverse pressure gradient impulse at the inlet or outlet junctions of the contraction, preventing separation through the contraction body.

Parameter	Value
Inlet bore diameter	35.05 mm
Outlet bore diameter	6.0 mm
Overall area ratio (inlet/outlet)	34.13:1
Contraction profile	Quintic C^2 -continuous (Morel, 1975)
Solid body (contraction) length	150 mm
Wall thickness range	6.0 – 9.5 mm (variable along contraction)
Inlet extension length	420 mm (35.05 mm bore, adiabatic)
Outlet extension length	200 mm (6.0 mm bore, adiabatic)

Total domain length	770 mm (420 + 150 + 200)
External surface area A_{ext} (quarter model)	4,820.61 mm ²
External surface area A_{ext} (full geometry)	19,282.45 mm ²
A_{ext} ratio (new/legacy, full geometry)	3.349× larger
Solid body volume V (quarter model)	3.103×10^{-5} m ³
Biot number $Bi = h \cdot L_n / \lambda_s$	0.00396 ($L_n = V/A_{ext} = 6.44$ mm)
<i>Longer inlet extension (420 mm vs 200 mm for legacy) required to establish a stable velocity profile upstream of the smooth contraction. All areas cross-validated to within 0.01%.</i>	

Table A.2. Confirmed geometric parameters of the redesigned reducer. All values verified against Fluent post-processing results.

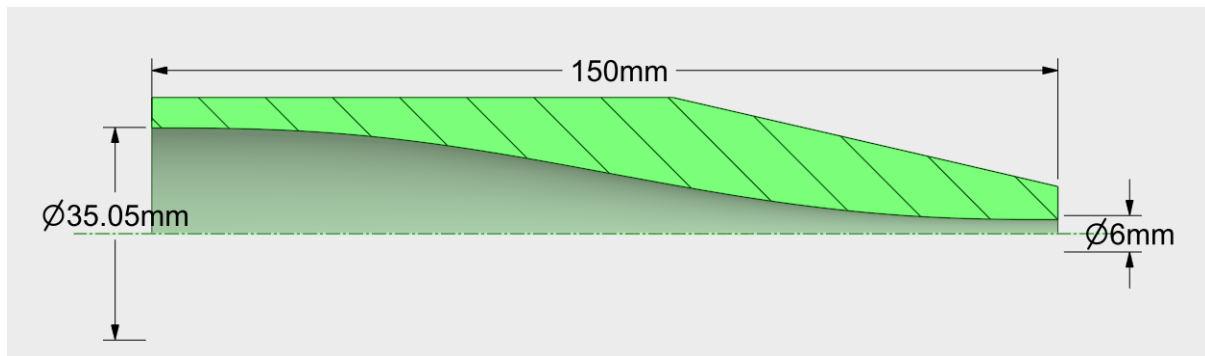


Figure A.2. Redesigned reducer internal geometry cross-section showing the quintic C^2 -continuous contraction profile.

A.3 Geometric Simplifications

The following simplifications were applied consistently to both geometries. Internal threads were excluded from the fluid domain and treated as smooth cylindrical bore surfaces, as thread geometry produces negligible effect on the bulk flow quantities of interest. Minor external chamfers, fastener recesses, and surface features not contributing to the primary heat conduction path were omitted from the solid domain. The hexagonal external profile of the legacy reducer body was simplified to a smooth axisymmetric surface, consistent with the quarter-symmetry assumption and introducing negligible error in the external area calculation, which was independently verified in Fluent. Both geometries were modelled as quarter-symmetry (1/4) configurations, exploiting two orthogonal planes of geometric and flow symmetry (XZ and YZ planes) to reduce the mesh size by a factor of four. These simplifications preserve the dominant thermo-fluid behaviour and introduce no systematic bias in the comparative analysis, as both geometries are subject to identical simplification strategy.

Appendix B: Mesh Sensitivity Study

B.1 Purpose and Scope

This appendix presents the detailed mesh generation settings, quality metrics, and confirmed sensitivity results for Study 1 (adiabatic CFD mesh sensitivity). The objective was to confirm that all key flow quantities are insensitive to further spatial refinement beyond the selected baseline mesh level. Three refinement levels coarse, baseline, and fine were generated for each geometry. The baseline meshes were adopted as the production meshes for Study 2 (CHT).

B.2 Legacy Reducer: Mesh Settings and Quality

Setting	Coarse	Baseline (selected)	Fine	Notes
Total fluid elements	272,146	829,621	1,570,929	Fluid only, no solid
Global element size	3.0 mm	2.0 mm	1.33 mm	Auto-calculated field
Inflation first cell height	5×10^{-5} m	5×10^{-5} m	5×10^{-5} m	Constant across levels
Inflation layers	10	10	10	All wall surfaces
Growth ratio	1.2	1.2	1.2	
y^+ _max (Case C)	10.60	10.52	11.21	Fluid-side wall
Min orthogonal quality (avg / worst)	0.788 / 0.156	/ 0.797 / 0.100	0.798 / 0.124	> 0.05 limit
Max skewness (avg / worst)	0.211 / 0.844	/ 0.202 / 0.900	0.201 / 0.876	< 0.999 limit
Max aspect ratio (avg / worst)	6.128 / 44.79	/ 3.677 / 30.93	3.065 / 20.32	< 1000 limit
Min element quality (avg / worst)	0.517 / 0.040	/ 0.664 / 0.053	0.698 / 0.090	> 5×10^{-4} limit
<i>Zero elements failed any ANSYS error threshold at any refinement level. Worst-case aspect ratios are confined to inflation layer cells adjacent to wall boundaries.</i>				

Table B.1. Legacy reducer mesh generation settings and quality metrics for the three Study 1 refinement levels.

B.3 Redesigned Reducer: Mesh Settings and Quality

Setting	Coarse	Baseline (selected)	Fine	Notes
Total fluid elements	404,324	913,301	2,115,646	Fluid only, no solid
Inflation first cell height	5×10^{-5} m	5×10^{-5} m	5×10^{-5} m	Constant across levels
Inflation layers	10	10	10	All wall surfaces
Growth ratio	1.2	1.2	1.2	
y^+ _max (Case C)	10.54	10.60	10.57	Fluid-side wall
Min orthogonal quality (avg / worst)	0.841 / 0.188	0.833 / 0.032	0.834 / 0.136	> 0.05 limit
Max skewness (avg / worst)	0.157 / 0.812	0.166 / 0.846	0.165 / 0.864	< 0.999 limit
Max aspect ratio (avg / worst)	7.918 / 86.77	5.507 / 61.24	3.875 / 41.06	< 1000 limit
Min element quality (avg / worst)	0.460 / 0.022	0.524 / 0.032	0.603 / 0.046	> 5×10^{-4} limit
<i>Zero elements failed any ANSYS error threshold at any refinement level. Higher worst-case aspect ratios relative to the legacy reducer reflect the longer inflation layer surfaces of the 150 mm contraction body.</i>				

Table B.2. Redesigned reducer mesh generation settings and quality metrics for the three Study 1 refinement levels.

B.4 Confirmed Mesh Sensitivity Results

All Study 1 simulations were run at Case C conditions (inlet 100,000 Pa gauge, 673 K total temperature; outlet 0 Pa gauge, PR = 1.987), representing the worst-case operating condition in terms of compressibility and near-wall velocity gradients. Table B.3 presents the four monitored KPIs at each mesh level. Table B.4 presents the percentage changes between consecutive refinement levels.

KPI	Legacy Coarse	Legacy Baseline	Legacy Fine	New Coarse	New Baseline	New Fine
T_out, MWA (K)	587.96	587.31	586.79	590.22	590.33	590.48
Ma_max, volume max (-)	1.0099	1.0148	1.0164	0.9888	0.9903	0.9887

ΔP_0 , total pressure drop (Pa)	36,214	36,285	35,797	38,810	39,146	39,387
$V_{\max}$, velocity magnitude (m/s)	480.85	482.60	483.17	473.43	474.07	473.45
$y^+_{\max}$	10.60	10.52	11.21	10.54	10.60	10.57
<i>Case C conditions: inlet 100,000 Pa gauge, 673 K; outlet 0 Pa gauge. MWA = mass-weighted average.</i>						

Table B.3. Study 1 mesh sensitivity KPIs at Case C for both geometries across all three refinement levels.

KPI	Legacy C → B (%)	Legacy B → F (%)	New C → B (%)	New B → F (%)	Pass 2%? <
T_out, MWA	−0.112	−0.089	+0.018	+0.025	✓ Yes
Ma_max	+0.478	+0.160	+0.151	−0.165	✓ Yes
ΔP_0	+0.196	−1.344	+0.867	+0.614	✓ Yes
$V_{\max}$	+0.364	+0.118	+0.135	−0.129	✓ Yes
<i>All eight baseline-to-fine changes satisfy the 2% acceptance criterion. Coarse-to-baseline changes also satisfy the criterion. Mesh independence is confirmed for both geometries.</i>					

Table B.4. Percentage changes in KPIs between consecutive mesh refinement levels. C = coarse, B = baseline, F = fine.

All four KPIs satisfy the 2% acceptance criterion between consecutive refinement levels for both geometries. The largest change is −1.344% (total pressure drop, legacy reducer, baseline-to-fine). The redesigned reducer's maximum Mach number is stable at 0.989–0.990 across all three mesh levels, and the legacy reducer's maximum Mach number is stable at 1.010–1.016, confirming that the qualitative flow regime divergence is a genuine physical result independent of mesh refinement. The baseline meshes 829,621 elements (legacy) and 913,301 elements (redesigned) are adopted for all Study 2 CHT simulations.

Appendix C: Full Simulation Configuration

C.1 Boundary Conditions

Boundary	BC Type	Study 1 (adiabatic)	Study 2 (CHT)
Inlet	Pressure inlet	100,000 Pa gauge 673 K total temp 5% TI, hydraulic diameter	Same as Study 1
Outlet: Case A	Pressure outlet	N/A (Case C only)	80,000 Pa gauge (PR = 1.110)
Outlet: Case B	Pressure outlet	N/A (Case C only)	40,000 Pa gauge (PR = 1.424)
Outlet: Case C	Pressure outlet	0 Pa gauge (PR = 1.987)	0 Pa gauge (PR = 1.987)
reducer_fluid / shadow	Coupled CHT wall	Adiabatic no-slip wall	Coupled: fluid-solid heat exchange
annular_face / shadow (legacy only)	Coupled CHT wall	Adiabatic no-slip wall	Coupled: impingement step face
reducer_solid (CHT external)	Convection wall	N/A (no solid domain)	$h = 10 \text{ W/m}^2\text{K}$, $T_\infty = 300 \text{ K}$
reducer_solid (insulated ref.)	Heat flux wall	N/A	$q = 0 \text{ W/m}^2$ (Case C only)
Inlet / outlet extensions	Adiabatic wall	Adiabatic no-slip	Adiabatic no-slip
Symmetry planes (XZ, YZ)	Symmetry	Symmetry	Symmetry
Operating pressure	Domain setting	101,325 Pa	101,325 Pa
<i>TI = turbulence intensity. PR = absolute pressure ratio $P_{o,in}/P_{outlet,static}$. Hydraulic diameter inlet BC: $D_h = 35.05 \text{ mm}$ used as turbulence length scale specification.</i>			

Table C.1. Boundary conditions for Study 1 (adiabatic CFD) and Study 2 (conjugate heat transfer) simulations.

C.2 Solver Configuration

Setting	Value
Solver type	Pressure-based coupled, steady-state, double precision

Parallel cores	4 (AMD Ryzen 7 6800HS; KTH academic licence ceiling)
Turbulence model	SST k- ω (Menter, 1994)
Near-wall treatment	Automatic (low-Re / wall function blending based on y^+)
Fluid density	Ideal gas (compressible)
Fluid viscosity	Sutherland's law: $\mu_0 = 1.716 \times 10^{-5} \text{ Pa}\cdot\text{s}$, $T_{\text{ref}} = 273.15 \text{ K}$, $S = 110.4 \text{ K}$
Fluid C_p	Piecewise-linear, 13 points, 300–1000 K (NIST data); C_p at 673 K = 1,069.4 J/kg·K
Fluid thermal conductivity k_f	Piecewise-linear, 13 points, 300–1000 K (NIST data)
Solid material	Structural steel (316L): $\rho = 8,030 \text{ kg/m}^3$, $C_p = 502.5 \text{ J/kg}\cdot\text{K}$, $\lambda = 16.27 \text{ W/m}\cdot\text{K}$
Momentum discretisation	Second-order upwind
Energy discretisation	Second-order upwind
Turbulence (k , ω) discretisation	First-order upwind
Pressure discretisation	Second-order
Gradient reconstruction	Least-squares cell-based
CHT coupling	Fully coupled simultaneous fluid–solid energy solve each iteration
Initialisation strategy	Cascade: Cases B and C initialised from converged preceding case; no re-initialisation from scratch

Double precision used for all simulations. Academic licence at KTH restricts parallel execution to 4 cores.

Table C.2. Fluent solver settings applied to all Study 1 and Study 2 simulations.

C.3 Convergence Criteria

Monitor (CHT Study 2)	Tolerance	Min iters	Notes
Net mass flow rate imbalance	$1 \times 10^{-5} \text{ kg/s}$	200	Rolling 200-iter window
Outlet temperature, MWA	$1 \times 10^{-4} \text{ K}$	200	Rolling 200-iter window
External wall temperature, AWA	$1 \times 10^{-4} \text{ K}$	200	Rolling 200-iter window

Net total pressure drop	5×10^{-4} Pa	200	Rolling 200-iter window
<i>All four criteria must be simultaneously satisfied before a solution is accepted. For Study 1 (adiabatic), convergence was assessed by visual stabilisation of mass flow rate and outlet temperature monitors, with mass imbalance $< 1 \times 10^{-5}$ kg/s as the primary numerical criterion.</i>			

Table C.3. Convergence criteria for Study 2 (CHT) simulations.

C.4 Air Thermophysical Properties

T (K)	C_p (J/kg·K)	k (W/m·K)	T (K)	C_p (J/kg·K)	k (W/m·K)
300	1,005	0.02631	700	1,075	0.05240
350	1,008	0.03003	750	1,087	0.05556
400	1,014	0.03365	800	1,099	0.05779
450	1,021	0.03707	850	1,110	0.06005
500	1,030	0.04040	900	1,121	0.06212
550	1,040	0.04362	950	1,131	0.06461
600	1,051	0.04659	1000	1,141	0.06670
650	1,063	0.04953	673*	1,069*	0.05097*
* Interpolated at $T = 673$ K (heater set-point): $C_p = 1,069.4$ J/kg·K (confirmed from Fluent material panel). Source: NIST WebBook piecewise-linear data as implemented in Fluent 2025 R1.					

Table C.4. Air piecewise-linear thermophysical properties (C_p and thermal conductivity k) as specified in Fluent 2025 R1 for all simulations.

Appendix D: Risk Analysis

D.1 Purpose and Scope

This appendix documents the risk analysis conducted in connection with the High-Temperature Dynamic Pressure Sensor Calibration Rig (Pulsatorn) project at Siemens Energy AB, Finspång. Although the final scope of this thesis is limited to numerical CFD and CHT simulations, the risk assessment was performed in the context of the originally planned experimental campaign, which included high-temperature rig operation. It is retained here as documentation of the original project plan and as a resource for future experimental work when the heater assembly is restored to service.

D.2 Risk Management Framework

The risk assessment was performed in accordance with ISO 31000:2018 Risk Management Guidelines and Siemens Energy internal laboratory safety procedures. Risks were evaluated using a structured semi-quantitative approach based on severity (S), probability (P), and resulting risk level $R = S \times P$. Table D.1 defines the scoring criteria applied.

<i>Parameter</i>	<i>Level</i>	<i>Description</i>
Severity (S)	1	Minor inconvenience, no injury
	2	Minor injury or minor equipment damage
	3	Noticeable injury requiring medical treatment
	4	Serious injury or major equipment damage
	5	Severe injury or catastrophic damage
Probability (P)	1	Very unlikely ($\leq$ once in 10 years)
	2	Unlikely
	3	Possible (approximately once per year)
	4	Likely
	5	Very likely (multiple times per year)
<i>Risk level $R = S \times P$. Acceptable risk: $R \leq 8$ with mitigation measures in place. Unacceptable risk: $R > 12$.</i>		

Table D.1. Risk scoring criteria applied in accordance with ISO 31000:2018 and Siemens Energy laboratory safety procedures.

D.3 Identified Hazard Categories

The following six hazard categories were considered in the risk assessment: (1) thermal hazards from high-temperature rig surfaces and gas streams; (2) electrical hazards from the inline resistance heater power supply; (3) pressure hazards from the pressurised air supply and dynamic pressure oscillations generated by the gear-wheel mechanism; (4) noise hazards from compressor operation and the gear-wheel mechanism; (5) mechanical hazards from rotating components; and (6) instrument failure leading to uncontrolled rig operating conditions.

D.4 Key Risks and Mitigation Measures

<i>Hazard</i>	<i>S</i>	<i>P</i>	<i>R</i>	<i>Mitigation measures</i>
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High-temperature surface contact	4	2	8	Thermal insulation of all accessible surfaces; mandatory heat-resistant gloves and protective clothing; mandatory cool-down period before post-test access.
Electrical shock from heater power supply	4	2	8	Earthed enclosure; interlock system preventing heater activation with enclosure open; restricted access to qualified personnel only.
Over-pressure event in air supply circuit	5	1	5	Calibrated pressure relief valves; operating pressure limits enforced below rated maximum; automated shutdown on pressure exceedance.
Noise exposure from compressor and gear-wheel	2	4	8	Mandatory hearing protection during rig operation; acoustic enclosure on compressor unit.
Mechanical contact with rotating gear-wheel	4	1	4	Mechanical guard over gear-wheel assembly; interlock preventing guard removal during operation.
Instrument failure leading to uncontrolled conditions	3	2	6	Redundant temperature and pressure monitoring; automated emergency shutdown on out-of-range readings; pre-test instrument calibration checks.
<i>All risks assessed as acceptable ($R \leq 8$) within the Siemens Energy laboratory safety framework, subject to the described mitigation measures being in place prior to rig operation.</i>				

Table D.2. Key risks, risk scores, and mitigation measures for the originally planned experimental campaign.

D.5 Relation to the Present Thesis

The experimental scope described above was not executed due to heater hardware failure during the project preparation phase (see Appendix E). The hazards identified in this risk analysis were therefore not encountered in practice during the present thesis. The risk assessment is retained as documentation of the original project plan, and the mitigation measures described remain valid recommendations for any future experimental campaign conducted on the Pulsatorn rig following restoration of the heater assembly.

Appendix E: Project Scope Revision

E.1 Initial Project Scope

The project was initially defined as a combined experimental and numerical investigation of the thermo-fluid performance of the Pulsatorn heater-to-downstream reducer at Siemens Energy AB, Finspång. The experimental component was planned to comprise: thermocouple measurements of the temperature distribution along the flow path from heater outlet to pulsator housing; total pressure drop measurements across the reducer body under steady high-temperature conditions; and qualitative flow visualisation. These measurements were to be conducted with the inline resistance heater operating at its rated power (400°C set-point), providing a directly comparable experimental dataset against which numerical predictions could be validated.

E.2 Hardware Failure and Scope Revision

During the preparation phase of the experimental campaign, the inline resistance heater assembly sustained internal damage specifically, a failure of the heating element support structure that prevented sustained high-temperature operation within the available project timeframe. The damage was assessed by Siemens Energy engineering staff and determined to require replacement of components with a procurement and delivery lead time exceeding the thesis submission deadline.

Following discussion with the industrial supervisor (Jonas Deurell, Siemens Energy AB) and the academic supervisor (Jens Fridh, KTH Royal Institute of Technology), the project scope was formally revised to a purely numerical comparative investigation. The scope revision was approved by both supervisors in accordance with KTH degree project guidelines for MJ232X.

E.3 Justification for Scope Revision

The revised scope was accepted by both supervisors on the following grounds. First, the original engineering objectives quantifying the thermo-fluid performance difference between the legacy and redesigned reducer geometries under representative operating conditions could be addressed under controlled numerical boundary conditions with greater parametric flexibility and repeatability than a physical experiment. Second, the numerical campaign represented a scientifically rigorous and independently completable body of work within the degree project requirements of MJ232X (30 credits). Third, the absence of experimental data does not invalidate the comparative conclusions of the numerical study, provided that both geometries are modelled under identical assumptions and boundary conditions, which is the case in the present work.

E.4 Implications for the Thesis

The absence of experimental validation is acknowledged as a limitation of the present study and is discussed in Section 5.3 of the main report. All comparative conclusions are based entirely on numerical evidence produced under identical modelling assumptions for both geometries. The results are presented as quantified design guidance and performance targets rather than as certified operational specifications. Experimental validation particularly of the near-choked $Ma = 0.990$ result and the outer wall temperature distribution remains the recommended first priority for future work, as described in Section 5.4.1.

Appendix F: Software Tools and Computing Environment

F.1 CAD and Geometry Preparation

Geometry for both reducer configurations was prepared in ANSYS SpaceClaim / Discovery Premium (ANSYS Workbench 2025 R1), accessed through the KTH Royal Institute of Technology academic licence server (1055@ansys-stud-lic.ug.kth.se). Geometry simplification removal of internal threads, minor external features, and fastener recesses and preparation of the quarter-symmetry model were performed in ANSYS SpaceClaim. External surface area measurements were performed in ANSYS Discovery Premium and cross-validated against Fluent post-processing results to within 0.01%. Workbench project integration was used throughout to maintain traceability between geometry, mesh, and simulation systems.

F.2 Numerical Simulation Software

All CFD and CHT simulations were performed using ANSYS Fluent 2025 R1, accessed through the KTH Royal Institute of Technology academic licence server (1055@ansys-stud-lic.ug.kth.se). The academic licence constrains parallel processing to a maximum of 4 cores. Meshing was performed in ANSYS Meshing within the ANSYS Workbench 2025 R1 environment. All simulations were run in double precision. GPU acceleration was not used due to insufficient VRAM (6.44 GB available on the RTX 3060 Laptop GPU) relative to the mesh sizes required, and the absence of confirmed HPC tokens for GPU-accelerated Fluent on the KTH academic licence.

Computing hardware: ASUS ROG Flow X16 laptop AMD Ryzen 7 6800HS processor (8 cores, up to 4.7 GHz boost), 64 GB DDR5 RAM, NVIDIA GeForce RTX 3060 Laptop GPU (6.44 GB VRAM). Operating system: Windows 11 (64-bit).

F.3 Post-Processing and Documentation

Post-processing of simulation results was performed within ANSYS Fluent 2025 R1 using built-in surface integration (mass-weighted average, area-weighted average, integral), volume maximum, and flux report functions. The exact post-processing protocol including surface selections and field variable choices for each KPI is documented in Section 8 of the CHT Results Reference document. Results were tabulated in Microsoft Excel for numerical verification and cross-validation. Thesis documentation was prepared in Microsoft Word. Simulation case management and result traceability were maintained through the ANSYS Workbench project file structure with systematic naming conventions (FFF-series identifiers corresponding to Workbench system labels).

Appendix G: Project Planning and Milestones

G.1 Initial Project Plan

The project was planned to span approximately 20 weeks from registration in MJ232X to final thesis submission. The initial plan allocated approximately four weeks to literature review and geometry preparation, six weeks to experimental campaign execution and data analysis, and ten weeks to numerical modelling, results analysis, and thesis writing. This timeline assumed successful completion of the experimental campaign within the planned window. Following the heater hardware failure and scope revision (Appendix E), the project plan was restructured to redirect the experimental campaign weeks to extended numerical simulation and analysis.

G.2 Project Milestones

Milestone	Planned	Actual	Status
Thesis registration and project kickoff	Week 1	Week 1	✓ Complete
Literature review and geometry preparation	Weeks 1–4	Weeks 1–4	✓ Complete
Experimental campaign (original scope)	Weeks 5–10	Not executed	⊘ Scope revised: see Appendix E
Formal scope revision: supervisor approval	-	Week 8	✓ Complete
CFD mesh sensitivity study (Study 1): both geometries	-	Weeks 9–12	✓ Complete
CHT campaign: 6 standard cases, both geometries (Study 2)	-	Weeks 13–17	✓ Complete
Legacy CHT re-run with corrected inflation ($y^+ \approx 1-3$)	-	Week 17	✓ Complete
Insulated reference simulations: Case C, both geometries	-	Week 18	✓ Complete
Thesis writing, revision, and supervisor review	Weeks 11–20	Weeks 15–20	✓ Complete
Final report submission	Week 20	Week 20	✓ Complete
<i>Study 2 total: 6 standard CHT cases + 1 legacy CHT re-run (corrected y^+) + 2 insulated reference cases = 9 CHT simulations. Study 1 total: 6 adiabatic mesh sensitivity runs (3 per geometry).</i>			

Table G.1. Key project milestones showing planned versus actual completion dates and current status.

G.3 Supervision Structure

The thesis was supervised by Jonas Deurell (industrial supervisor, Siemens Energy AB, Finspång) and Jens Fridh (academic supervisor, KTH Royal Institute of Technology, Stockholm), with Andrew Martin as examiner. Regular supervision meetings were held throughout the project period, both in person at Siemens Energy AB and remotely. The scope revision described in Appendix E was jointly approved by both supervisors in accordance with KTH degree project guidelines for MJ232X (30 credits).

G.4 Adaptation and Project Management

The heater failure and subsequent scope revision required significant adaptation of the project plan within a constrained timeframe. The revised numerical focus was defined, discussed with both supervisors, and formally approved within two weeks of the hardware failure assessment, ensuring that the project remained on track for timely completion. The extended numerical campaign comprising mesh sensitivity, CHT simulation, re-run with corrected inflation parameters, and insulated reference simulations was completed within the original submission deadline. Communication with both supervisors during this period was maintained through regular meetings and written correspondence. The adaptive project management approach enabled completion of a rigorous comparative numerical study within the scheduled academic timeframe while preserving the original engineering objectives of the degree project.